DevOps Project

Problem Statement:

Create an end-to-end CI/CD pipeline in AWS platform using Jenkins as the orchestration tool, GitHub as scm, maven as the build tool, deploy in a docker instance and create a docker image, store the docker image in ECR, Kubernetes deployment using ECR image. Build sample java web app using maven.

CI/CD Pipeline Setup Approach:

Requirements:

- Git (Local Version Control System)
- GitHub (Distributed Version Control System)
- Jenkins (Continuous Integration Tool)
- Maven (Build Tool)
- Docker (Containerization)
- Kubernetes (Container Orchestration)

Step 1: CI/CD with GitHub, Jenkins, Maven & Tomcat

1. Install & Configure Jenkins
2. Install & Configure Maven and Git
3. Set Up Apache Tomcat Server
4. Integrate Jenkins with GitHub, Maven, and Tomcat
5. Create CI/CD Jobs in Jenkins
    - CI Job: Build code using Maven
    - CD Job: Deploy to Tomcat
6. Test Deployment on Tomcat

Step 2: CI/CD with GitHub, Jenkins, Maven & Docker

1. Set Up Docker Environment
2. Create Docker Image and Run Container
3. Integrate Docker with Jenkins

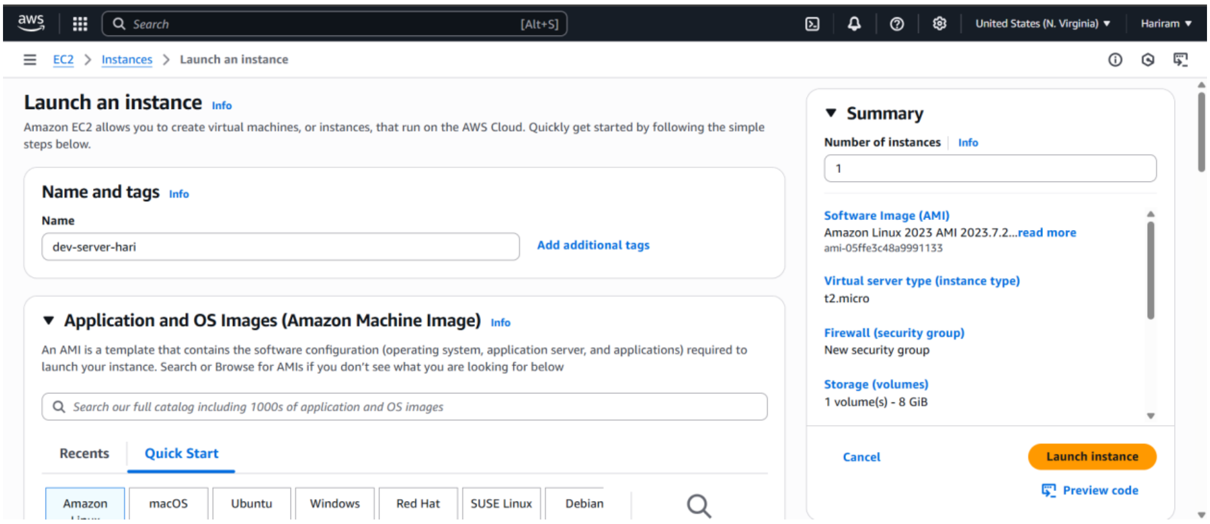
4. Create Jenkins CI/CD Job
    - CI Job: Build code with Maven
    - CD Job: Package into Docker image and deploy to container

Step 3: CI/CD with GitHub, Jenkins, Maven & Kubernetes

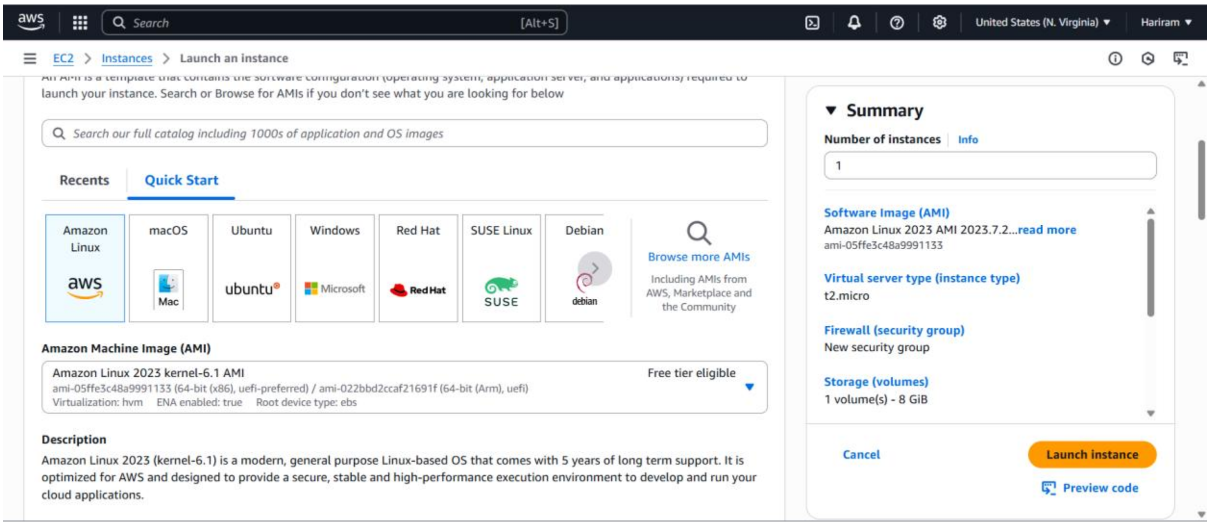
1. **Set Up Kubernetes Cluster (e.g., EKS)**
2. **Write Kubernetes Manifest Files**
    - Deployment
    - Service
    - Pod specifications
3. **Create Jenkins CI/CD Job**
    - CI Job: Build code with Maven
    - CD Job: Deploy to Kubernetes using kubectl

Step 4: Final Deployment & Validation

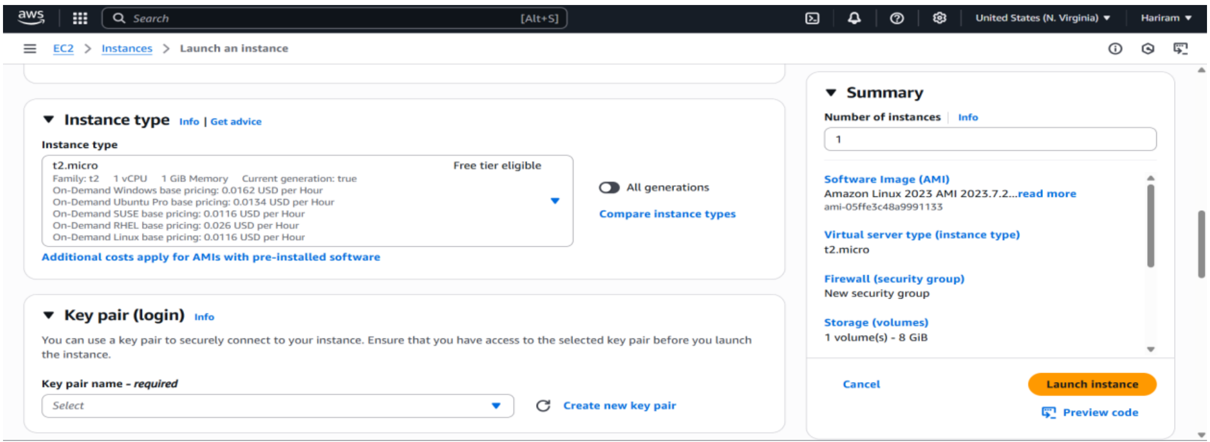
1. Deploy Artifacts to Kubernetes
2. Ensure Docker and Kubernetes Artifacts Contain Required Code
3. Trigger Jenkins Build
4. Verify Pod Creation in Kubernetes
5. Access Application via Kubernetes Service IP
    - Copy service IP
    - Paste in browser to validate output



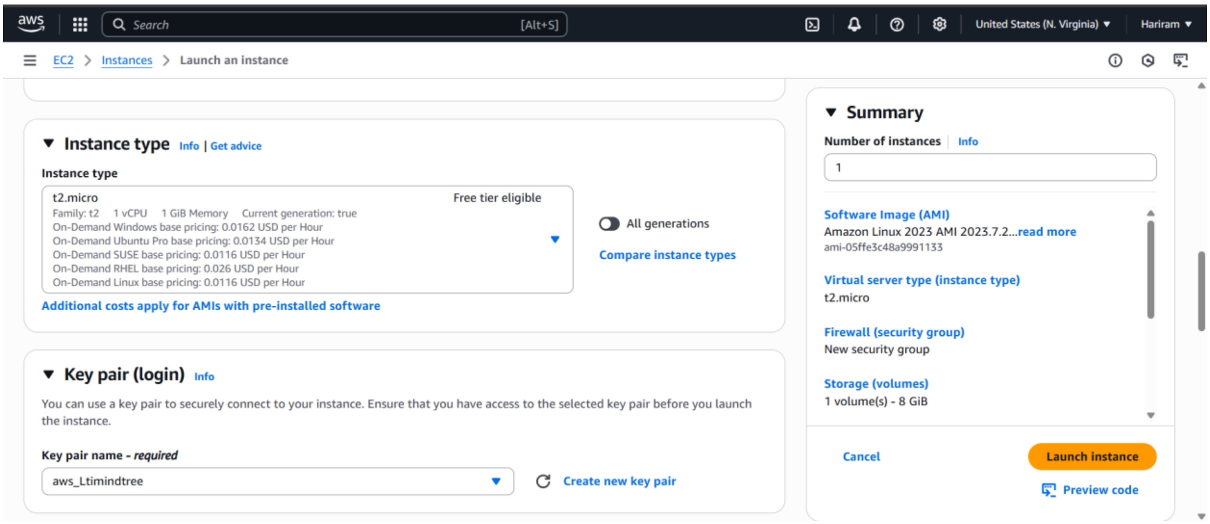
Creating Dev-server instance .



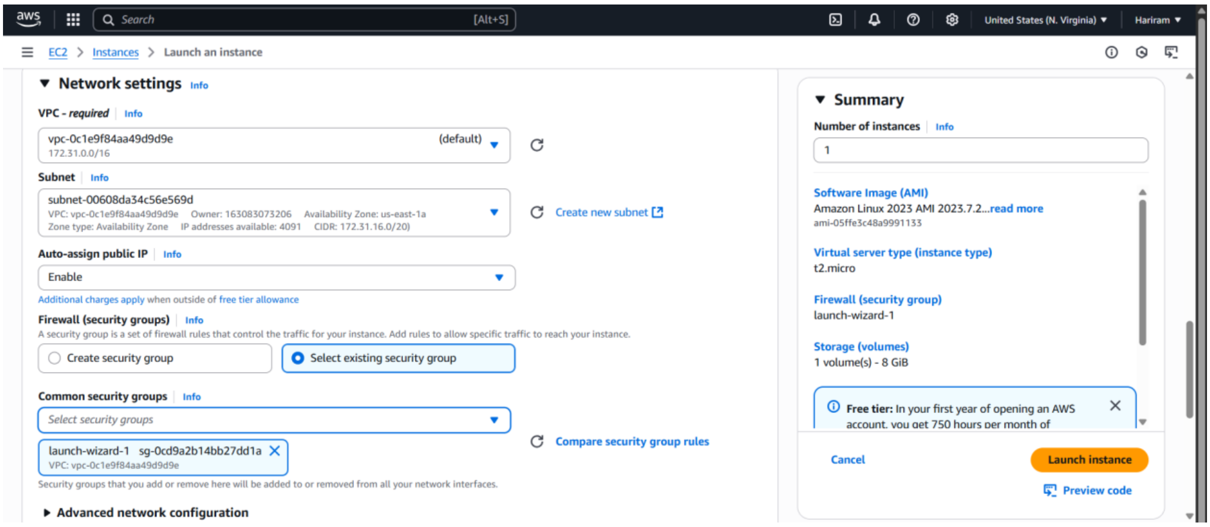
Selected amazon linux.



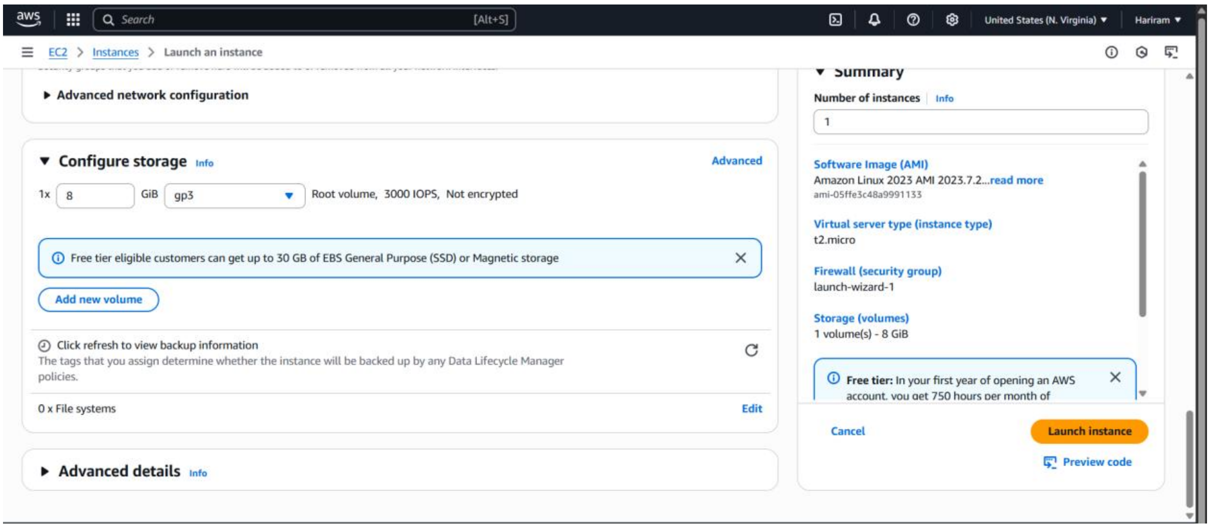
T2.micro is selected.



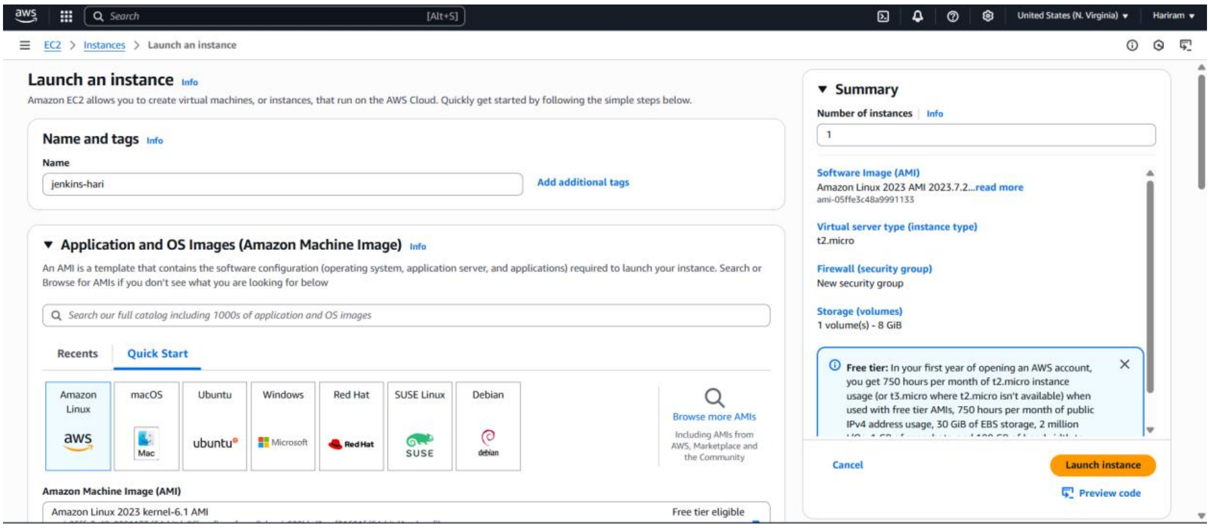
Existing key pair is selected.



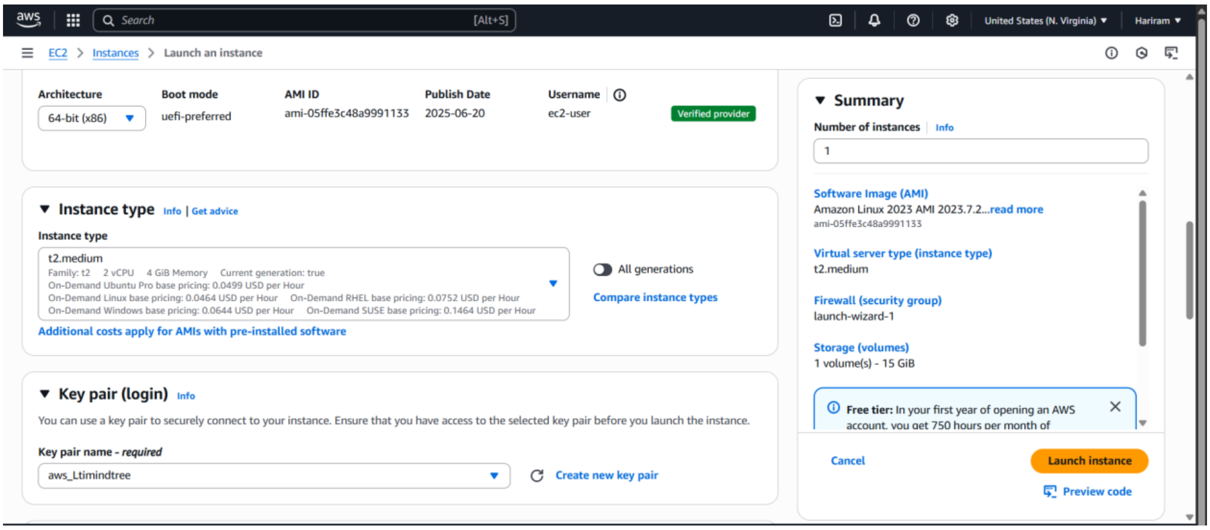
Existing security group is selected with required ports allowed.



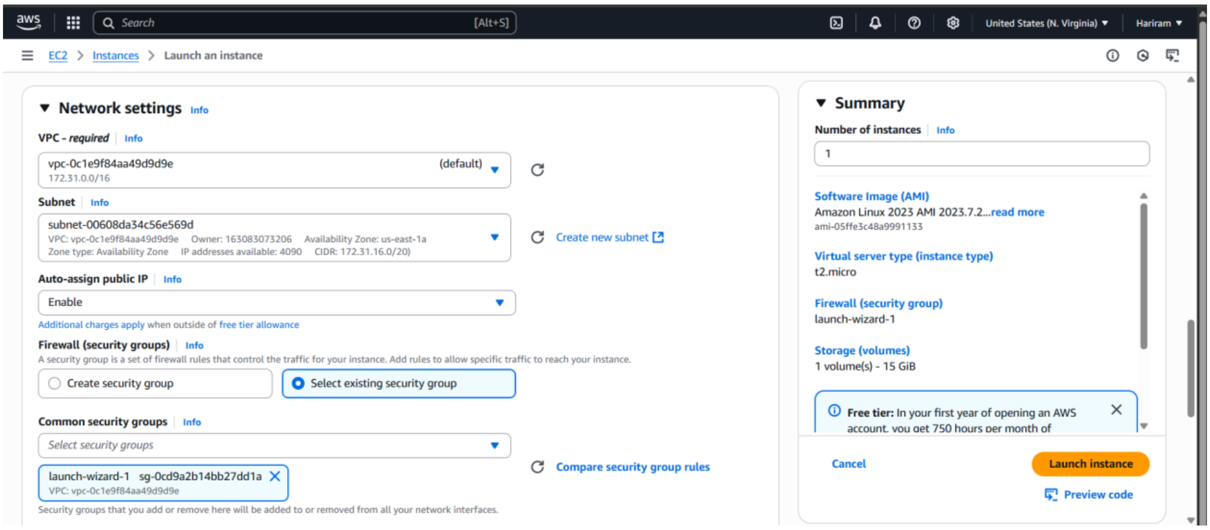
8GB storage and gp3 is selected.



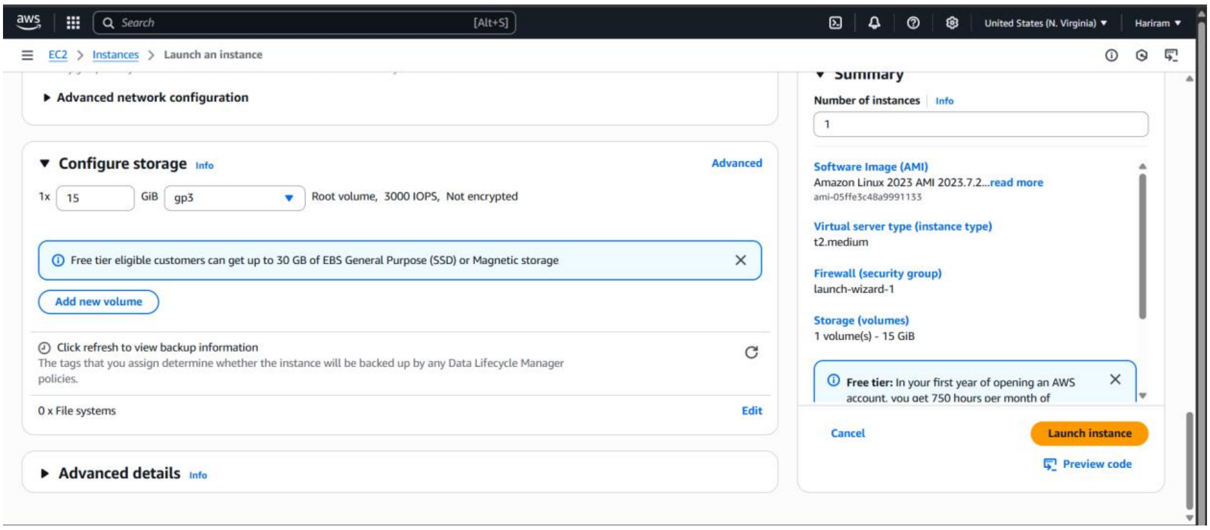
Jenkins server is created with amazon linux.



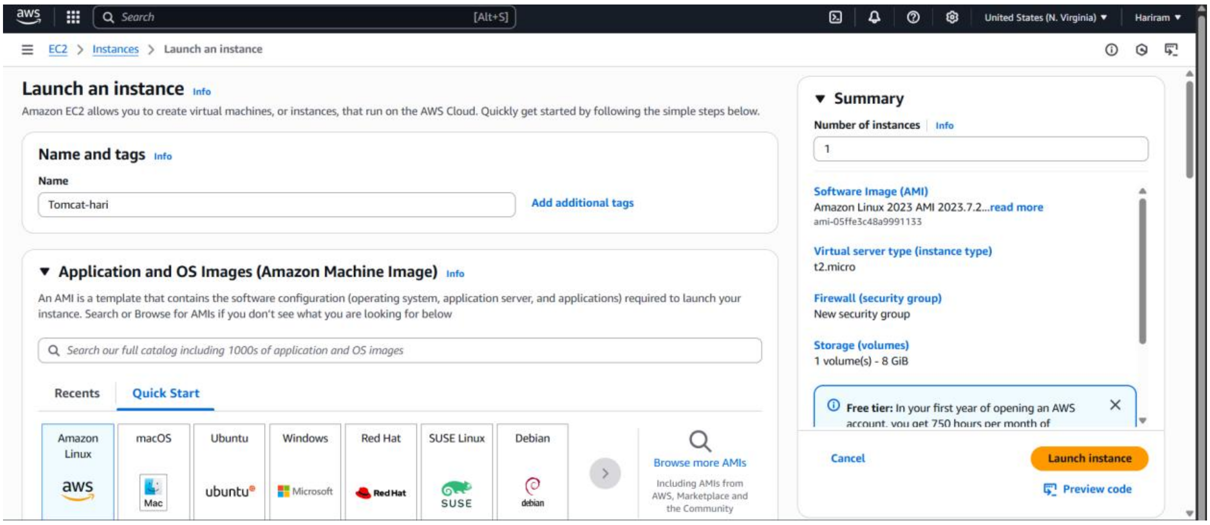
T2.medium is selected as a requirement for Jenkins.



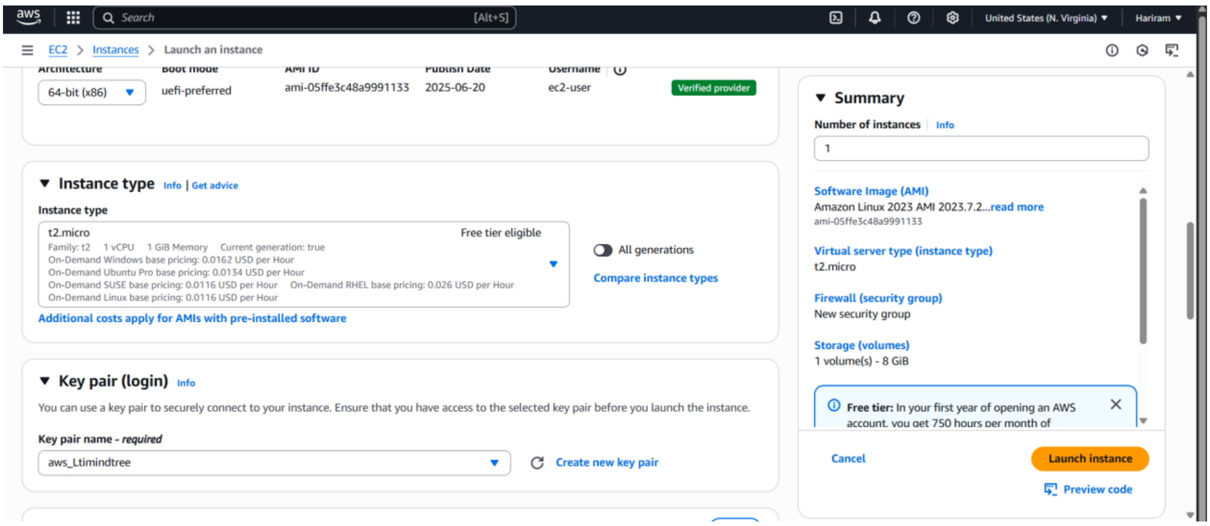
Existing security group is selected.



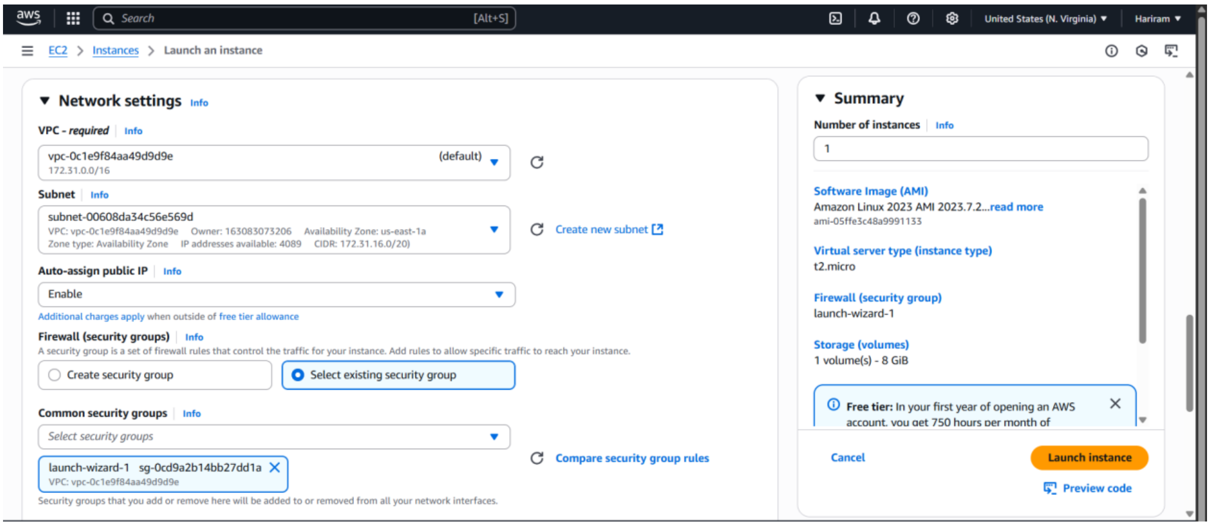
15GB storage and gp3 is selected.



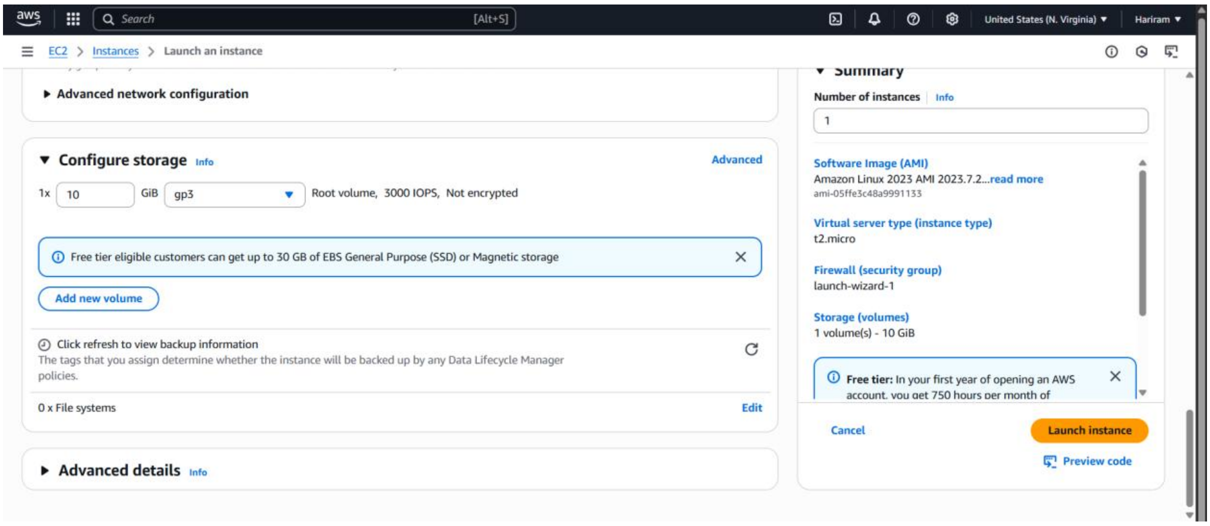
Tomcat server is created with amazon linux.



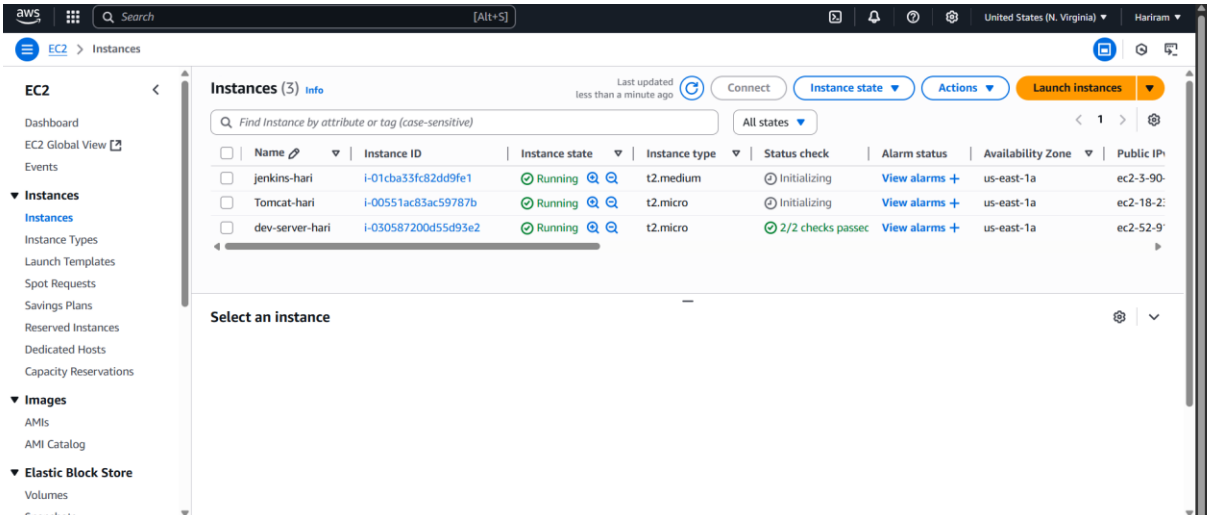
T2.micro is selected with existing key apir.



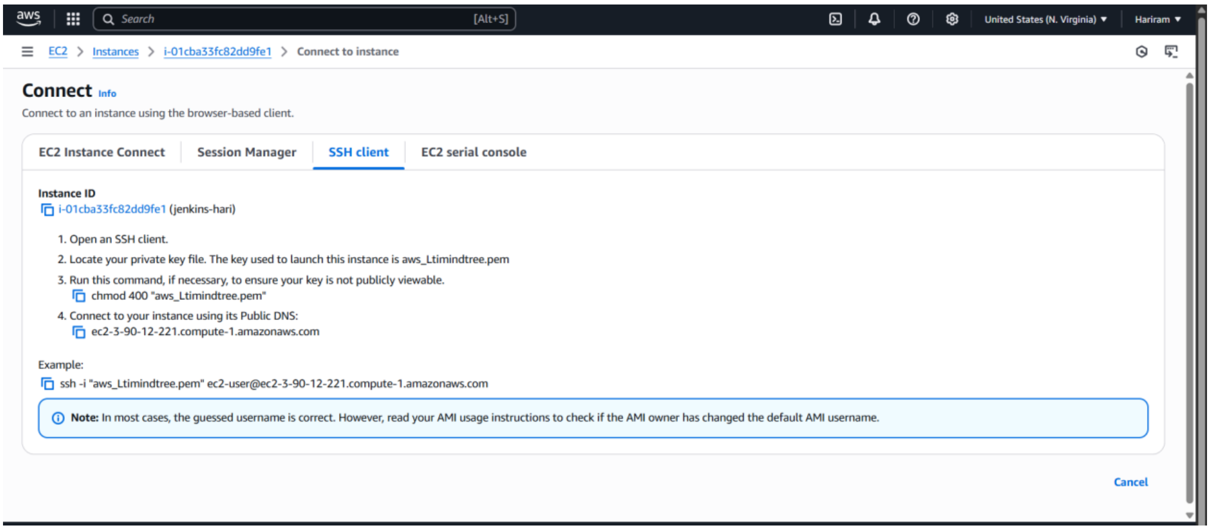
Existing security group is selected.



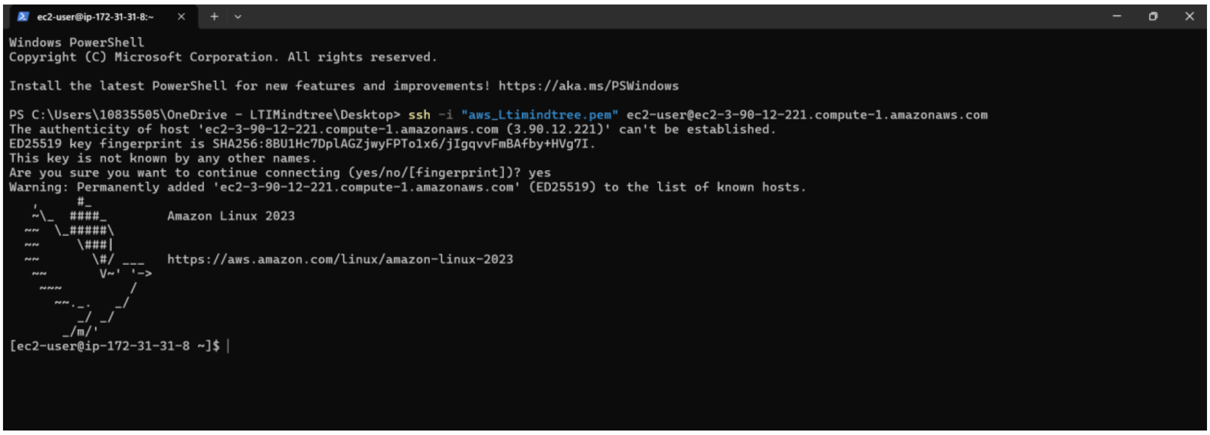
10GB storage and gp3 is selected.



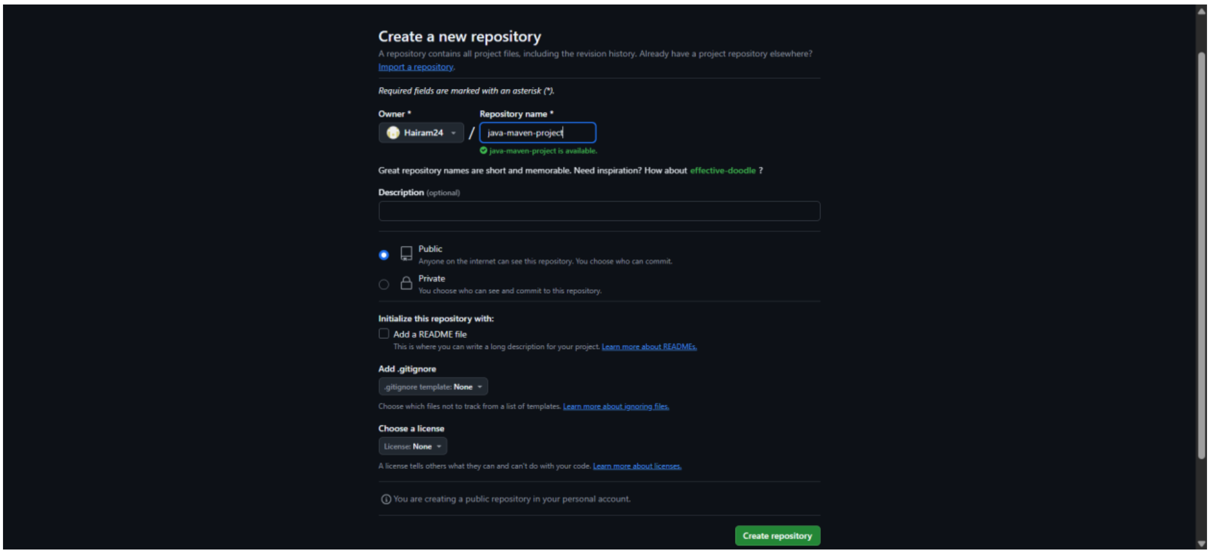
All the three servers are ready to run.



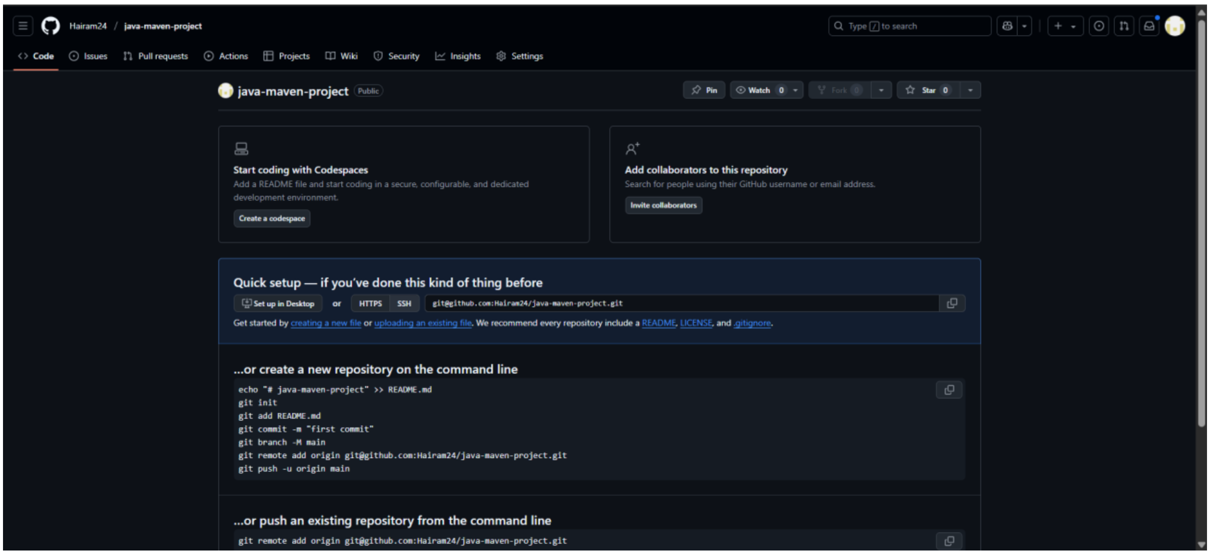
Instance is connected vie SSH client.



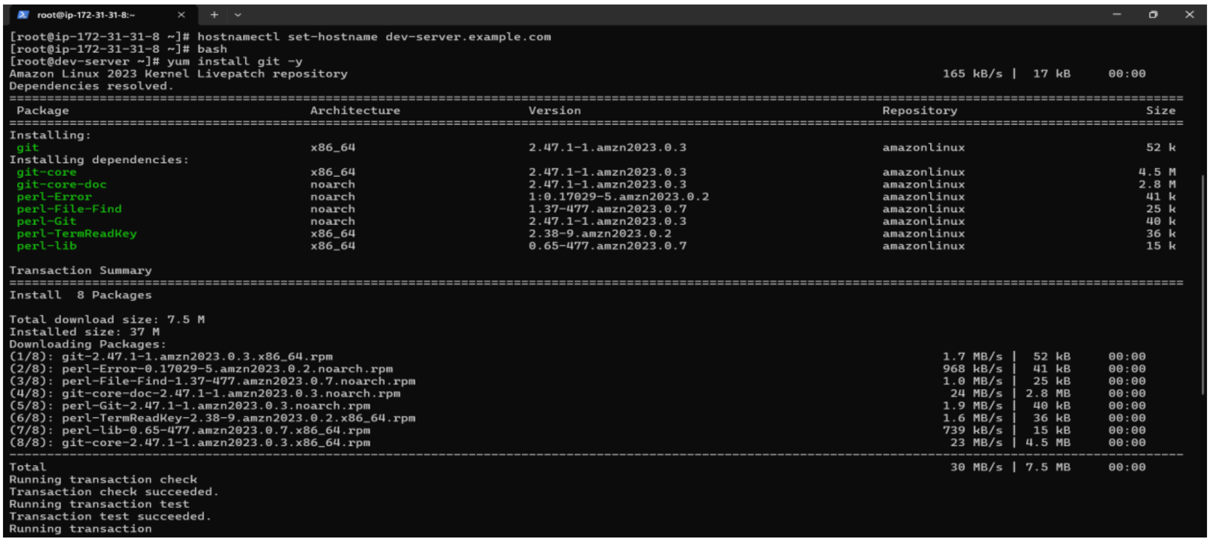
Instance is successfully connected.



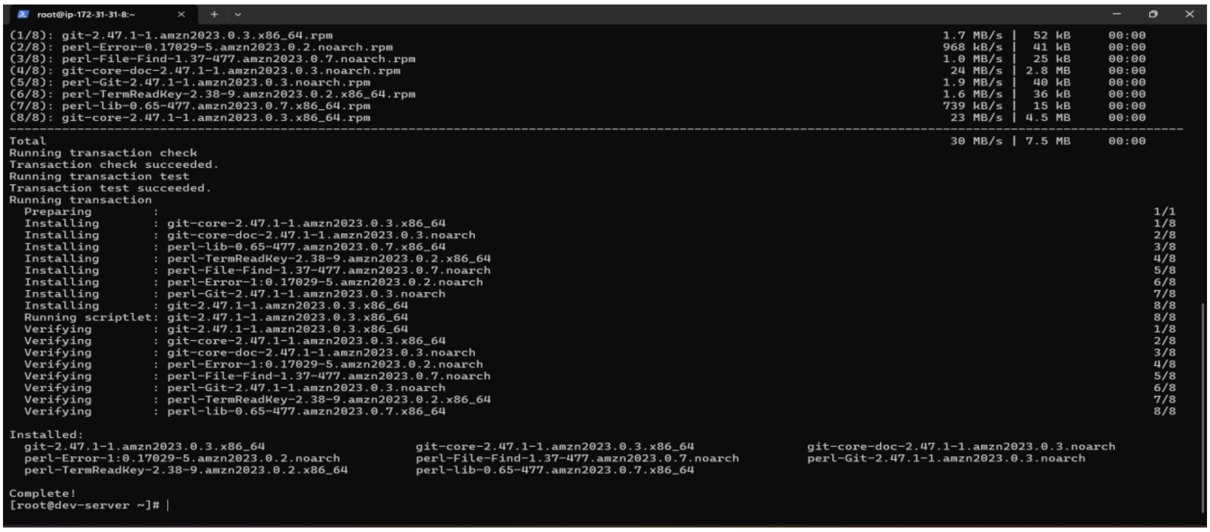
New public repository is created as java-maven project.



Public repository is created successfully.



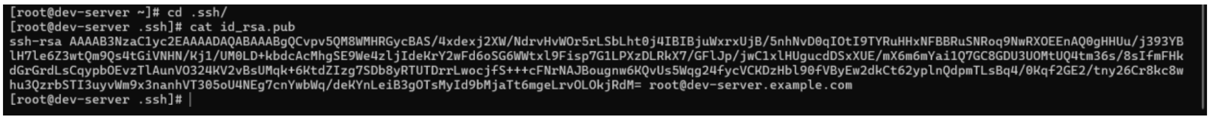
Hostname is set and git package is installed.



Git package is successfully installed.

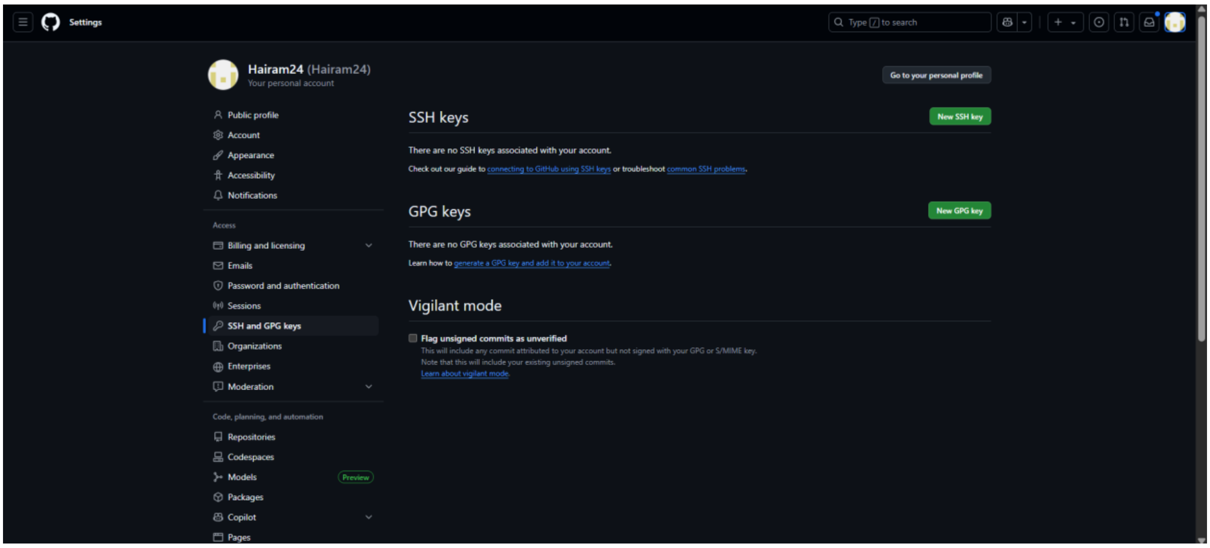
```
[root@dev-server ~]#  ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:60RtT/+Vmve741l4RKlbu7xYzinz8dZS2Kfa64vysR8 root@dev-server.example.com
The key's randomart image is:
+---[RSA 3072]----+
|                 |
|                .|
|               ..|
|         .    .. |
|        S o .. +.|
|       . o o .+o*|
|        o   o.EB*|
|       o  .  B@*@|
|        .  o=B%^X|
+----[SHA256]-----+
[root@dev-server ~]# |
```

SSH key is generated.

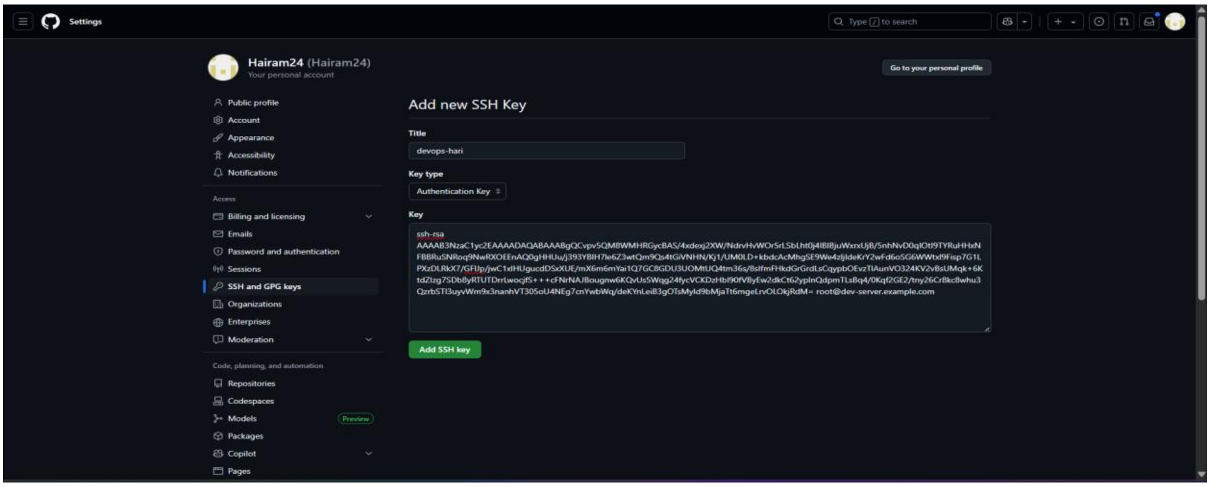


```
[root@dev-server ~]# cd .ssh/
[root@dev-server .ssh]# cat id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCvpv5QM8WMHRGycBAS/4xdexj2XW/NdrvHvWOr5rLSbLht0j4IBIBjuWxrxUjB/5nhNvD0qIOtI9TYRuHHxNFBBRuSNRoq9NwRXOEEnAQ0gHHUu/j393YB
lH7le6Z3wtQm9Qs4tGiVNHN/Kj1/UM0LD+kbdcAcMhgSE9We4zljIdeKrY2wFd6oSG6WWtxl9Fisp7G1LPXzDLRkX7/GFlJp/jwC1xlHUgucdDSxXUE/mX6m6mYai1Q7GC8GDU3UOMtUQ4tm36s/8sIfmFHk
dGrGrdLsCqypbOEvzTlAunVO324KV2vBsUMqk+6KtdZIzg7SDb8yRTUTDrrLwocjfS+++cFNrNAJBougnw6KQvUs5Wqg24fycVCKDzHbl90fVByEw2dkCt62yplnQdpmTLsBq4/0Kqf2GE2/tny26Cr8kc8w
hu3QzrbSTI3uyvWm9x3nanhVT305oU4NEg7cnYwbWq/deKYnLeiB3gOTsMyId9bMjaTt6mgeLrvOLOkjRdM= root@dev-server.example.com
[root@dev-server .ssh]# |
```

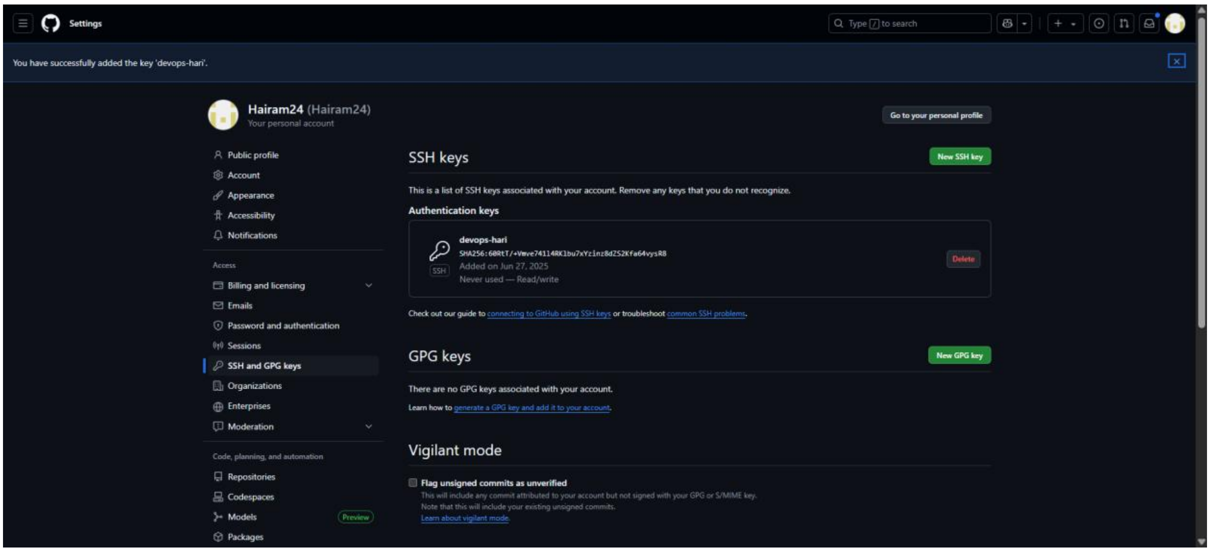
Public key is copied.



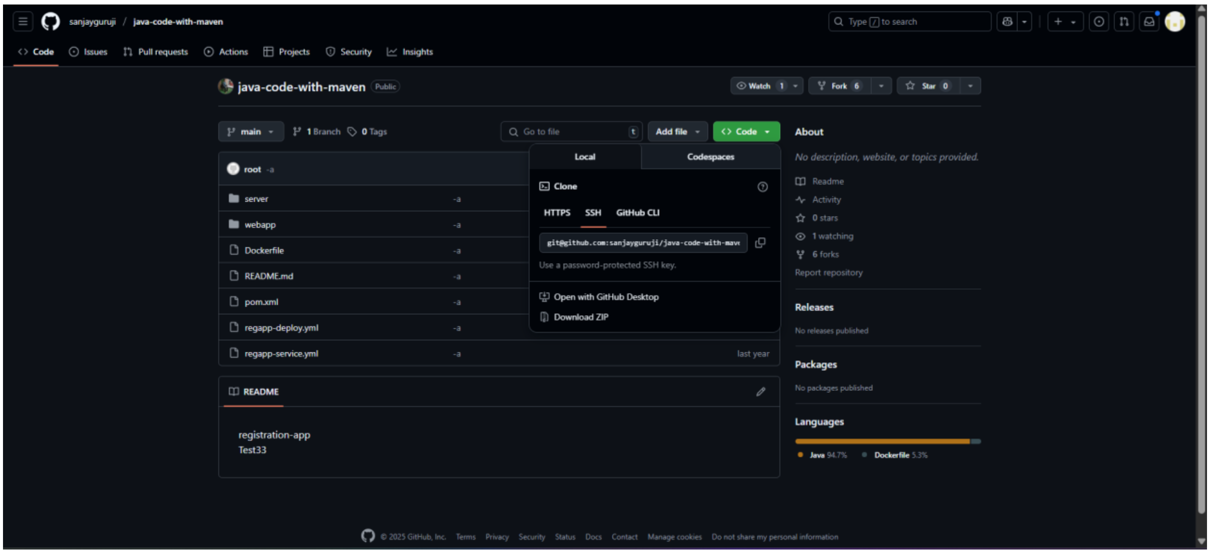
To establish connection to git through ssh key.



SSH key is added.



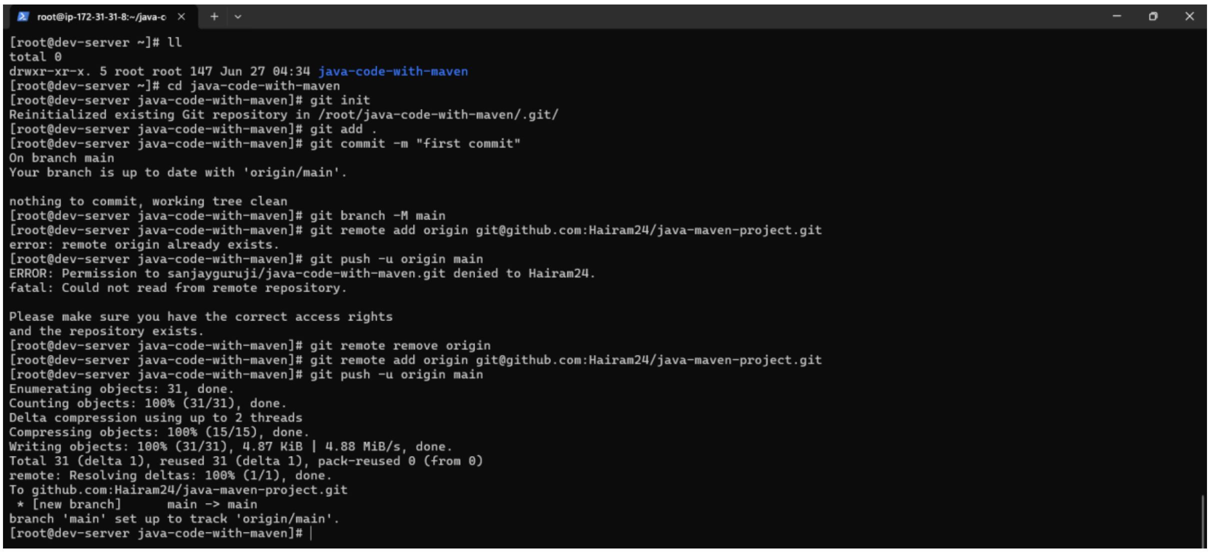
SSH key added successfully.



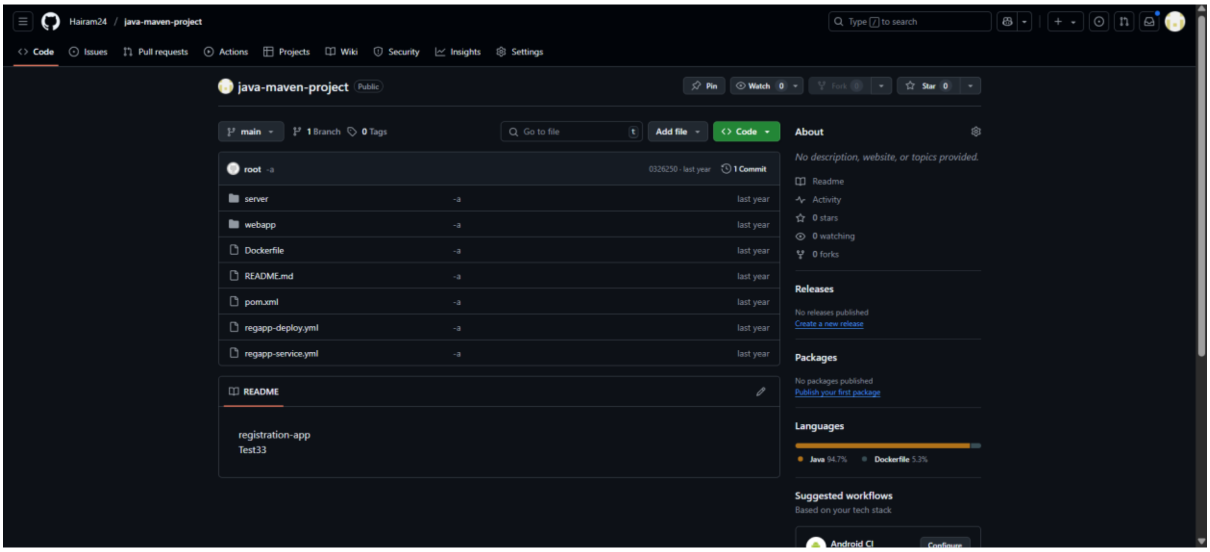
Copied the repo ssh link to clone it.

```
[root@dev-server .ssh]# cd
[root@dev-server ~]# git clone git@github.com:sanjayguruji/java-code-with-maven.git
Cloning into 'java-code-with-maven'...
The authenticity of host 'github.com (140.82.112.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 31, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 31 (delta 0), reused 0 (delta 0), pack-reused 29 (from 2)
Receiving objects: 100% (31/31), 4.87 KiB | 4.88 MiB/s, done.
Resolving deltas: 100% (1/1), done.
[root@dev-server ~]# |
```

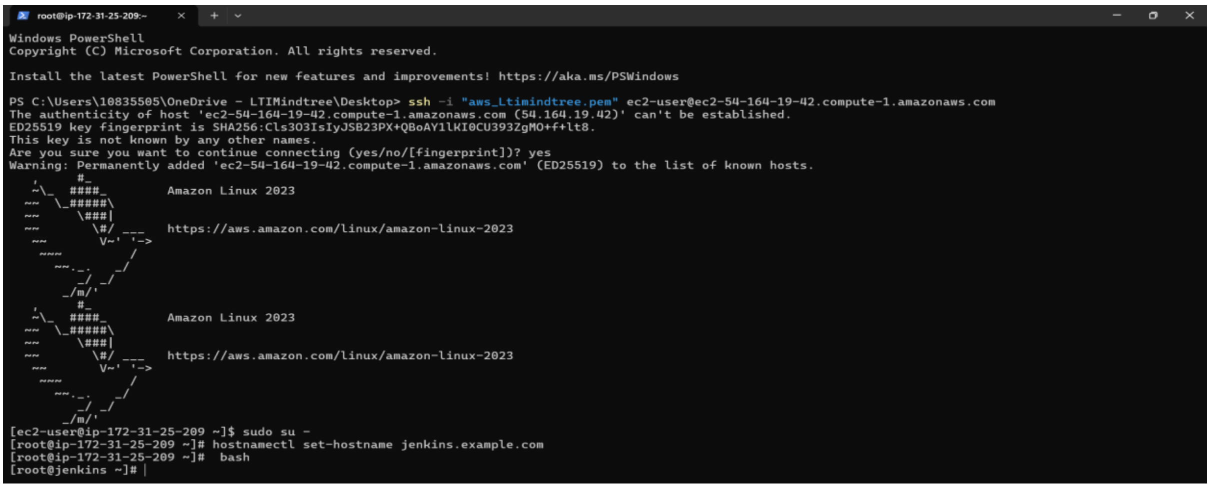
Git repository is cloned successfully.



Cloned repo is copied to the java-maven project repo.



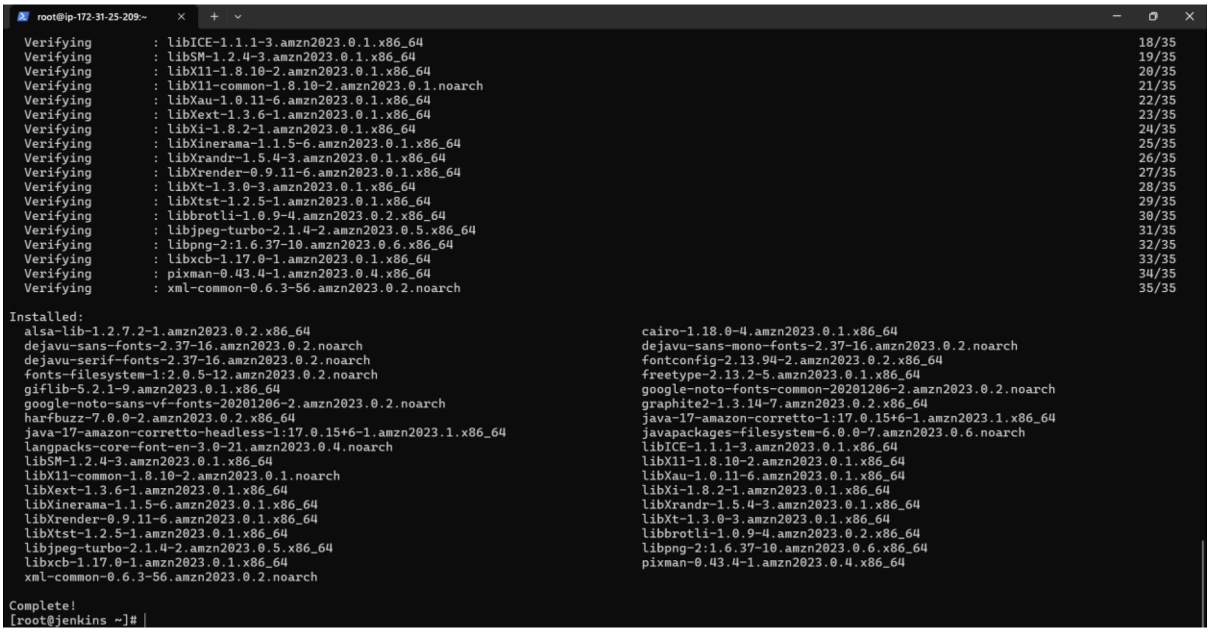
Successfully add to the project repository.



Jenkins server is connected and hostname is changed.

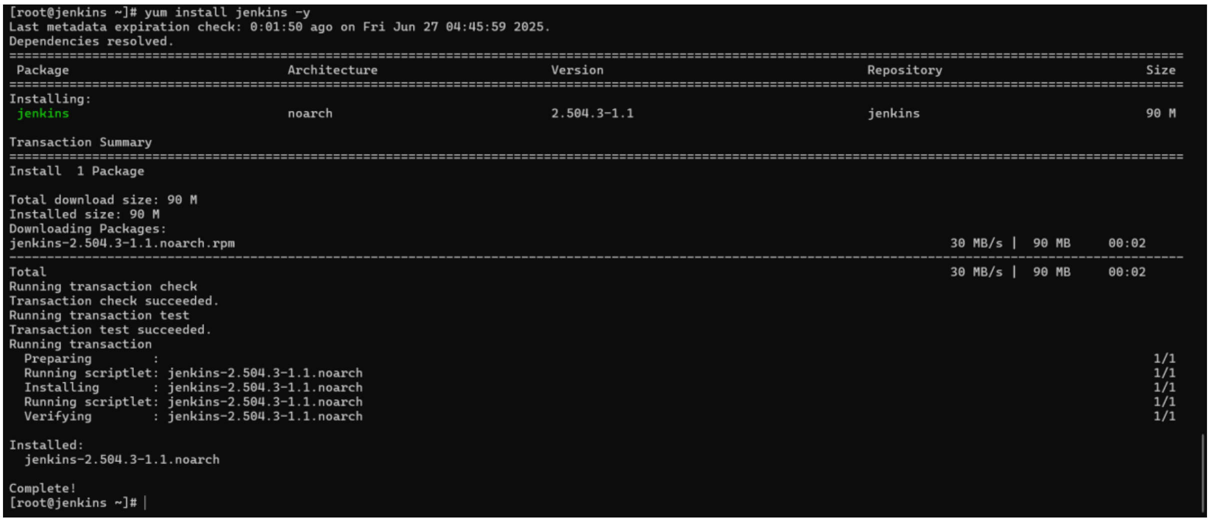
```
root@ip-172-31-25-209:~
[root@jenkins ~]# wget -O /etc/yum.repos.d/jenkins.repo \
     https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2025-06-27 04:45:34--  https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.34.133, 2a04:4e42:77::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.34.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 85
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo          100%[===========================================================================>]      85  --.-KB/s    in 0s

2025-06-27 04:45:34 (3.44 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [85/85]

[root@jenkins ~]# rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
[root@jenkins ~]#  yum upgrade
Jenkins-stable                                                                                                676 kB/s |  31 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@jenkins ~]# |
```

Jenkins packages installed.



```
root@ip-172-31-25-209:~
  Verifying        : libICE-1.1.1-3.amzn2023.0.1.x86_64                                                                          18/35
  Verifying        : libSM-1.2.4-3.amzn2023.0.1.x86_64                                                                           19/35
  Verifying        : libX11-1.8.10-2.amzn2023.0.1.x86_64                                                                         20/35
  Verifying        : libX11-common-1.8.10-2.amzn2023.0.1.noarch                                                                  21/35
  Verifying        : libXau-1.0.11-6.amzn2023.0.1.x86_64                                                                         22/35
  Verifying        : libXext-1.3.6-1.amzn2023.0.1.x86_64                                                                         23/35
  Verifying        : libXi-1.8.2-1.amzn2023.0.1.x86_64                                                                           24/35
  Verifying        : libXinerama-1.1.5-6.amzn2023.0.1.x86_64                                                                     25/35
  Verifying        : libXrandr-1.5.4-3.amzn2023.0.1.x86_64                                                                       26/35
  Verifying        : libXrender-0.9.11-6.amzn2023.0.1.x86_64                                                                     27/35
  Verifying        : libXt-1.3.0-3.amzn2023.0.1.x86_64                                                                           28/35
  Verifying        : libXtst-1.2.5-1.amzn2023.0.1.x86_64                                                                         29/35
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                                       30/35
  Verifying        : libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64                                                                   31/35
  Verifying        : libpng-2:1.6.37-10.amzn2023.0.6.x86_64                                                                      32/35
  Verifying        : libxcb-1.17.0-1.amzn2023.0.1.x86_64                                                                         33/35
  Verifying        : pixman-0.43.4-1.amzn2023.0.4.x86_64                                                                         34/35
  Verifying        : xml-common-0.6.3-56.amzn2023.0.2.noarch                                                                     35/35

Installed:
  alsa-lib-1.2.7.2-1.amzn2023.0.2.x86_64                                   cairo-1.18.0-4.amzn2023.0.1.x86_64
  dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch                            dejavu-sans-mono-fonts-2.37-16.amzn2023.0.2.noarch
  dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch                           fontconfig-2.13.94-2.amzn2023.0.2.x86_64
  fonts-filesystem-1:2.0.5-12.amzn2023.0.2.noarch                          freetype-2.13.2-5.amzn2023.0.1.x86_64
  giflib-5.2.1-9.amzn2023.0.1.x86_64                                       google-noto-fonts-common-20201206-2.amzn2023.0.2.noarch
  google-noto-sans-vf-fonts-20201206-2.amzn2023.0.2.noarch                 graphite2-1.3.14-7.amzn2023.0.2.x86_64
  harfbuzz-7.0.0-2.amzn2023.0.2.x86_64                                     java-17-amazon-corretto-1:17.0.15+6-1.amzn2023.1.x86_64
  java-17-amazon-corretto-headless-1:17.0.15+6-1.amzn2023.1.x86_64         javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
  langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch                        libICE-1.1.1-3.amzn2023.0.1.x86_64
  libSM-1.2.4-3.amzn2023.0.1.x86_64                                        libX11-1.8.10-2.amzn2023.0.1.x86_64
  libX11-common-1.8.10-2.amzn2023.0.1.noarch                               libXau-1.0.11-6.amzn2023.0.1.x86_64
  libXext-1.3.6-1.amzn2023.0.1.x86_64                                      libXi-1.8.2-1.amzn2023.0.1.x86_64
  libXinerama-1.1.5-6.amzn2023.0.1.x86_64                                  libXrandr-1.5.4-3.amzn2023.0.1.x86_64
  libXrender-0.9.11-6.amzn2023.0.1.x86_64                                  libXt-1.3.0-3.amzn2023.0.1.x86_64
  libXtst-1.2.5-1.amzn2023.0.1.x86_64                                      libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64                                libpng-2:1.6.37-10.amzn2023.0.6.x86_64
  libxcb-1.17.0-1.amzn2023.0.1.x86_64                                      pixman-0.43.4-1.amzn2023.0.4.x86_64
  xml-common-0.6.3-56.amzn2023.0.2.noarch

Complete!
[root@jenkins ~]# |
```

Installed successfully.



```
[root@jenkins ~]# yum install jenkins -y
Last metadata expiration check: 0:01:50 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
================================================================================================================================================
 Package                             Architecture                       Version                                   Repository                   Size
================================================================================================================================================
Installing:
 jenkins                             noarch                             2.504.3-1.1                               jenkins                      90 M

Transaction Summary
================================================================================================================================================
Install  1 Package

Total download size: 90 M
Installed size: 90 M
Downloading Packages:
jenkins-2.504.3-1.1.noarch.rpm                                                                                 30 MB/s |  90 MB     00:02
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                          30 MB/s |  90 MB     00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                    1/1
  Running scriptlet: jenkins-2.504.3-1.1.noarch                                                                                         1/1
  Installing       : jenkins-2.504.3-1.1.noarch                                                                                         1/1
  Running scriptlet: jenkins-2.504.3-1.1.noarch                                                                                         1/1
  Verifying        : jenkins-2.504.3-1.1.noarch                                                                                         1/1

Installed:
  jenkins-2.504.3-1.1.noarch

Complete!
[root@jenkins ~]# |
```

Jenkins is installed successfully.

```
[root@jenkins ~]# systemctl enable Jenkins
Failed to enable unit: Unit file Jenkins.service does not exist.
[root@jenkins ~]# systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@jenkins ~]# systemctl start jenkins
[root@jenkins ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
     Active: active (running) since Fri 2025-06-27 04:49:44 UTC; 7s ago
   Main PID: 26444 (java)
      Tasks: 48 (limit: 4656)
     Memory: 714.9M
        CPU: 17.075s
     CGroup: /system.slice/jenkins.service
             └─26444 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jun 27 04:49:40 jenkins.example.com jenkins[26444]: 63458ca172624c49811206630784a40f
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: *************************************************************
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: *************************************************************
Jun 27 04:49:40 jenkins.example.com jenkins[26444]: *************************************************************
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.409+0000 [id=31]        INFO        jenkins.InitReactorRunner$1#onAttained: Complet>
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.420+0000 [id=23]        INFO        hudson.lifecycle.Lifecycle#onReady: Jenkins is >
Jun 27 04:49:44 jenkins.example.com systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.456+0000 [id=49]        INFO        h.m.DownloadService$Downloadable#load: Obtained>
Jun 27 04:49:44 jenkins.example.com jenkins[26444]: 2025-06-27 04:49:44.457+0000 [id=49]        INFO        hudson.util.Retrier#start: Performed the action>
lines 1-20/20 (END)
```

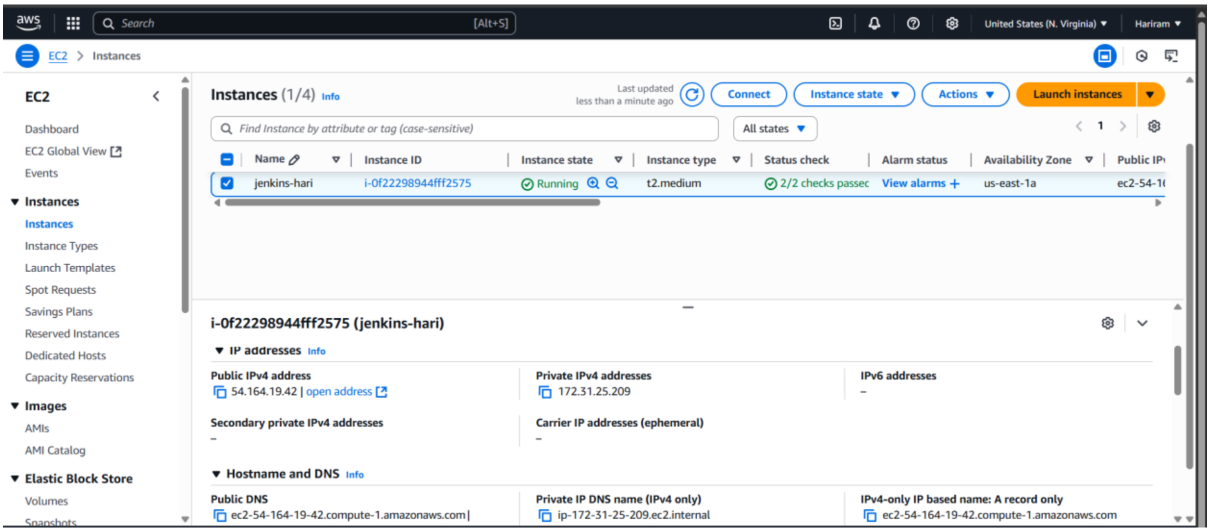
Jenkins is enabled.

```
[root@jenkins ~]# yum install git -y
Last metadata expiration check: 0:04:23 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
================================================================================================
 Package               Architecture     Version                        Repository          Size
================================================================================================
Installing:
 git                   x86_64           2.47.1-1.amzn2023.0.3          amazonlinux         52 k
Installing dependencies:
 git-core              x86_64           2.47.1-1.amzn2023.0.3          amazonlinux        4.5 M
 git-core-doc          noarch           2.47.1-1.amzn2023.0.3          amazonlinux        2.8 M
 perl-Error            noarch           1:0.17029-5.amzn2023.0.2       amazonlinux         41 k
 perl-File-Find        noarch           1.37-477.amzn2023.0.7          amazonlinux         25 k
 perl-Git              noarch           2.47.1-1.amzn2023.0.3          amazonlinux         40 k
 perl-TermReadKey      x86_64           2.38-9.amzn2023.0.2            amazonlinux         36 k
 perl-lib              x86_64           0.65-477.amzn2023.0.7          amazonlinux         15 k

Transaction Summary
================================================================================================
Install  8 Packages

Total download size: 7.5 M
Installed size: 37 M
Downloading Packages:
(1/8): git-2.47.1-1.amzn2023.0.3.x86_64.rpm                        1.1 MB/s |  52 kB     00:00
(2/8): perl-Error-0.17029-5.amzn2023.0.2.noarch.rpm                1.5 MB/s |  41 kB     00:00
(3/8): git-core-doc-2.47.1-1.amzn2023.0.3.noarch.rpm                25 MB/s | 2.8 MB     00:00
(4/8): perl-File-Find-1.37-477.amzn2023.0.7.noarch.rpm             672 kB/s |  25 kB     00:00
(5/8): perl-Git-2.47.1-1.amzn2023.0.3.noarch.rpm                   1.8 MB/s |  40 kB     00:00
(6/8): perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64.rpm             1.6 MB/s |  36 kB     00:00
(7/8): perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm                   628 kB/s |  15 kB     00:00
(8/8): git-core-2.47.1-1.amzn2023.0.3.x86_64.rpm                    25 MB/s | 4.5 MB     00:00
------------------------------------------------------------------------------------------------
Total                                                               35 MB/s | 7.5 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                        1/1
  Installing       : git-core-2.47.1-1.amzn2023.0.3.x86_64                                  1/8
```

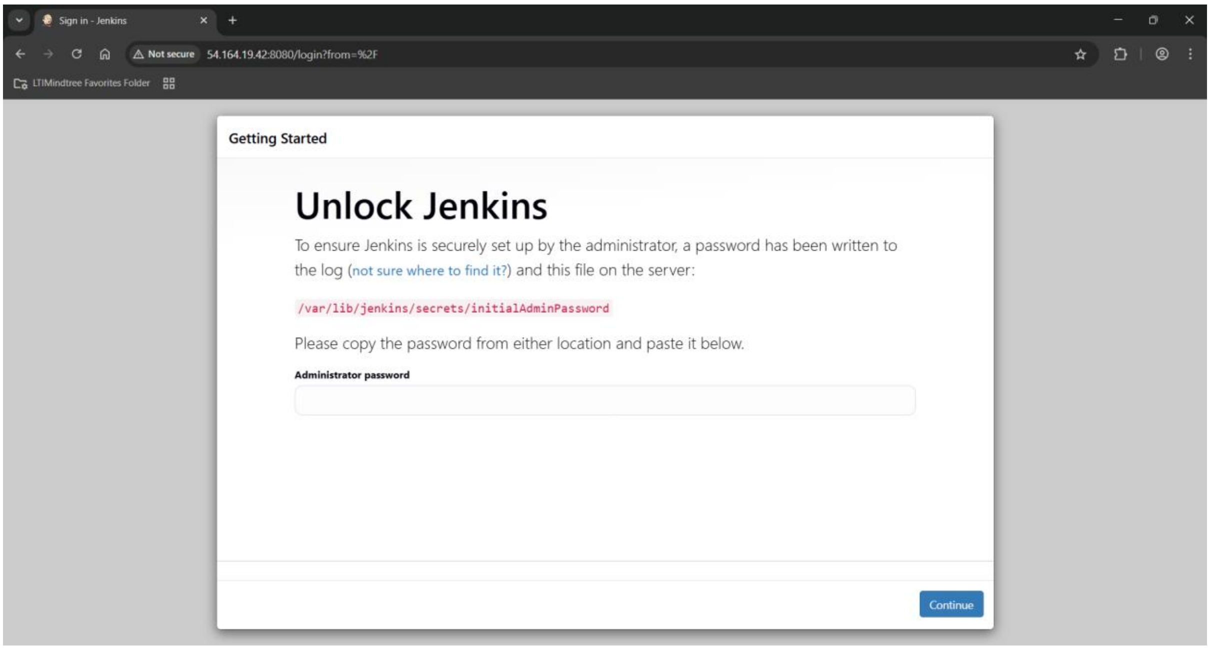
Git package is installed.

```
[root@jenkins ~]# yum install maven
Last metadata expiration check: 0:05:07 ago on Fri Jun 27 04:45:59 2025.
Dependencies resolved.
================================================================================================
 Package                                    Architecture   Version                     Repository     Size
================================================================================================
Installing:
 maven                                      noarch         1:3.8.4-3.amzn2023.0.5      amazonlinux    18 k
Installing dependencies:
 apache-commons-cli                         noarch         1.5.0-3.amzn2023.0.3        amazonlinux    76 k
 apache-commons-codec                       noarch         1.15-6.amzn2023.0.3         amazonlinux   303 k
 apache-commons-io                          noarch         1:2.8.0-7.amzn2023.0.5      amazonlinux   283 k
 apache-commons-lang3                       noarch         3.12.0-7.amzn2023.0.3       amazonlinux   559 k
 atinject                                   noarch         1.0.5-3.amzn2023.0.3        amazonlinux    23 k
 cdi-api                                    noarch         2.0.2-6.amzn2023.0.3        amazonlinux    54 k
 google-guice                               noarch         4.2.3-8.amzn2023.0.6        amazonlinux   473 k
 guava                                      noarch         31.0.1-3.amzn2023.0.6       amazonlinux   2.4 M
 httpcomponents-client                      noarch         4.5.13-4.amzn2023.0.4       amazonlinux   657 k
 httpcomponents-core                        noarch         4.4.13-6.amzn2023.0.3       amazonlinux   632 k
 jakarta-annotations                        noarch         1.3.5-13.amzn2023.0.3       amazonlinux    46 k
 jansi                                      x86_64         2.4.0-3.amzn2023.0.3        amazonlinux   113 k
 java-17-amazon-corretto-devel              x86_64         1:17.0.15+6-1.amzn2023.1    amazonlinux   142 k
 jcl-over-slf4j                             noarch         1.7.32-3.amzn2023.0.4       amazonlinux    25 k
 jsoup                                      noarch         1.16.1-4.amzn2023.0.2       amazonlinux   433 k
 jsr-305                                    noarch         3.0.2-5.amzn2023.0.4        amazonlinux    32 k
 maven-amazon-corretto17                    noarch         1:3.8.4-3.amzn2023.0.5      amazonlinux   9.4 k
 maven-lib                                  noarch         1:3.8.4-3.amzn2023.0.5      amazonlinux   1.5 M
 maven-resolver                             noarch         1:1.7.3-3.amzn2023.0.4      amazonlinux   557 k
 maven-shared-utils                         noarch         3.3.4-4.amzn2023.0.3        amazonlinux   152 k
 maven-wagon                                noarch         3.4.2-6.amzn2023.0.4        amazonlinux   113 k
 plexus-cipher                              noarch         1.8-3.amzn2023.0.3          amazonlinux    27 k
 plexus-classworlds                         noarch         2.6.0-10.amzn2023.0.4       amazonlinux    61 k
 plexus-containers-component-annotations    noarch         2.1.0-9.amzn2023.0.4        amazonlinux    19 k
 plexus-interpolation                       noarch         1.26-10.amzn2023.0.4        amazonlinux    80 k
 plexus-sec-dispatcher                      noarch         2.0-3.amzn2023.0.3          amazonlinux    34 k
 plexus-utils                               noarch         3.3.0-9.amzn2023.0.4        amazonlinux   254 k
 publicsuffix-list                          noarch         20240212-61.amzn2023        amazonlinux    89 k
 sisu                                       noarch         1:0.3.4-9.amzn2023.0.4      amazonlinux   510 k
 slf4j                                      noarch         1.7.32-3.amzn2023.0.4       amazonlinux    70 k

Transaction Summary
```

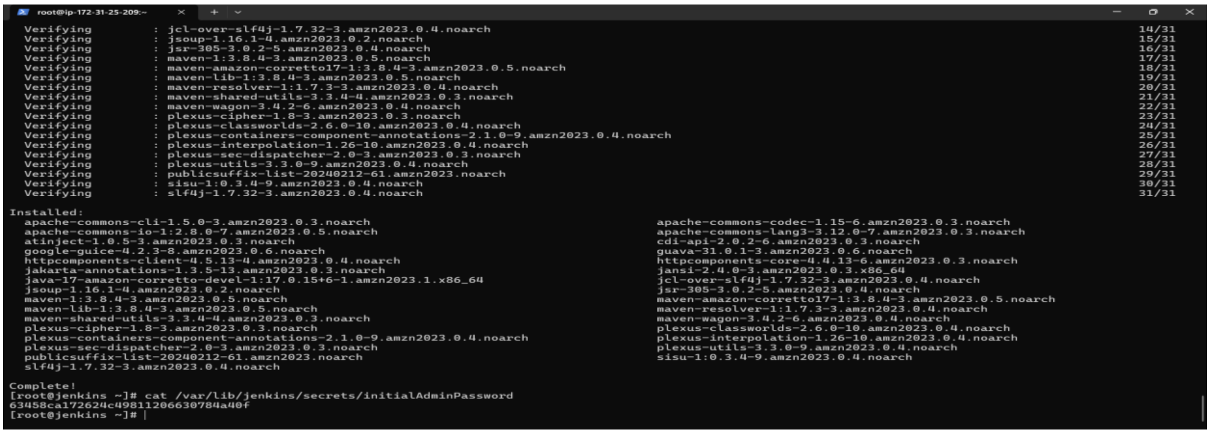
Maven package is installed.



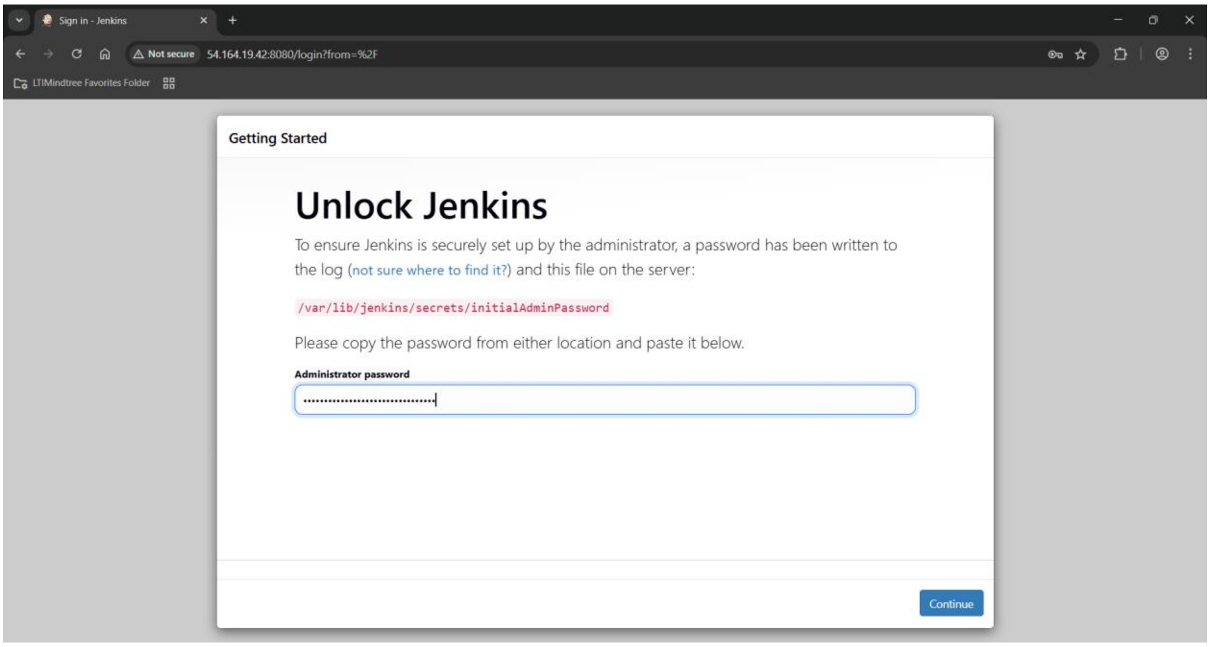
Jenkins public IP is copied.



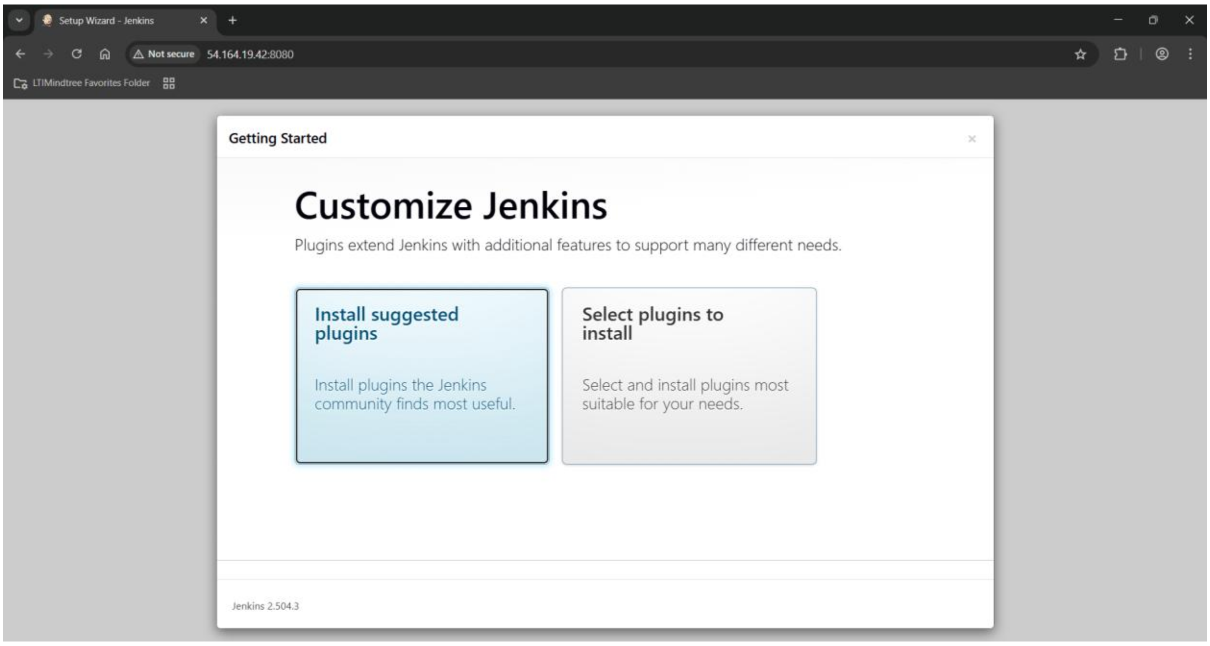
Through Jenkins public IP and port 8080, Jenkins dashboard is opened.



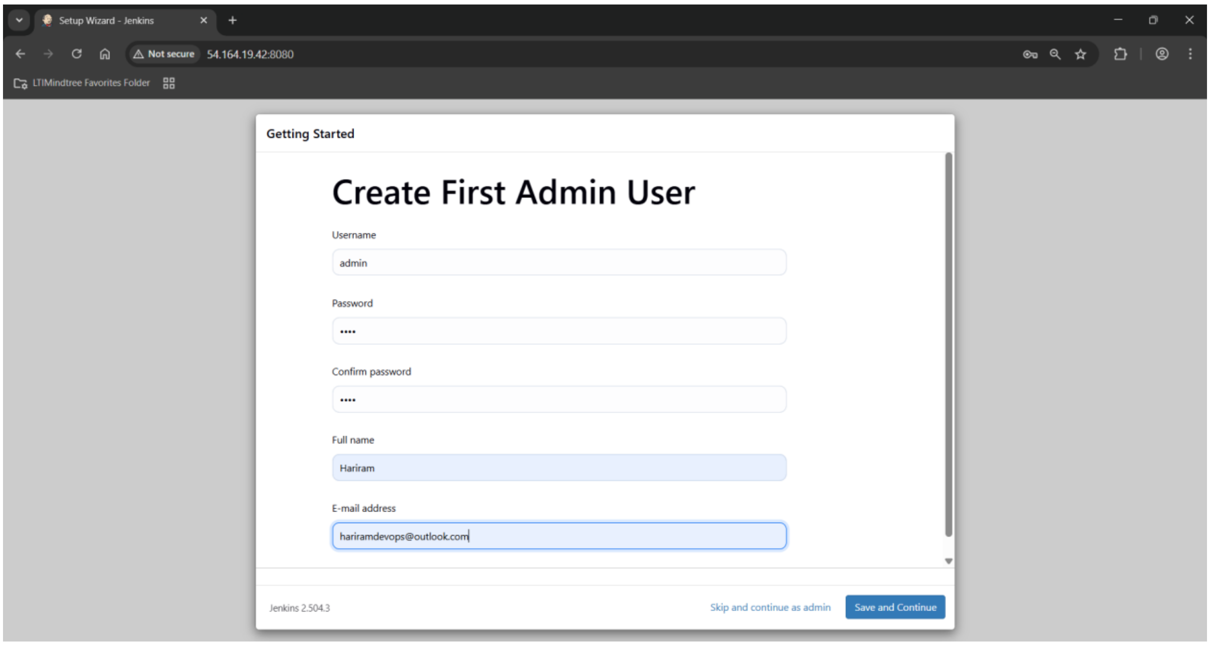
Administrator password is taken from Jenkins machine.



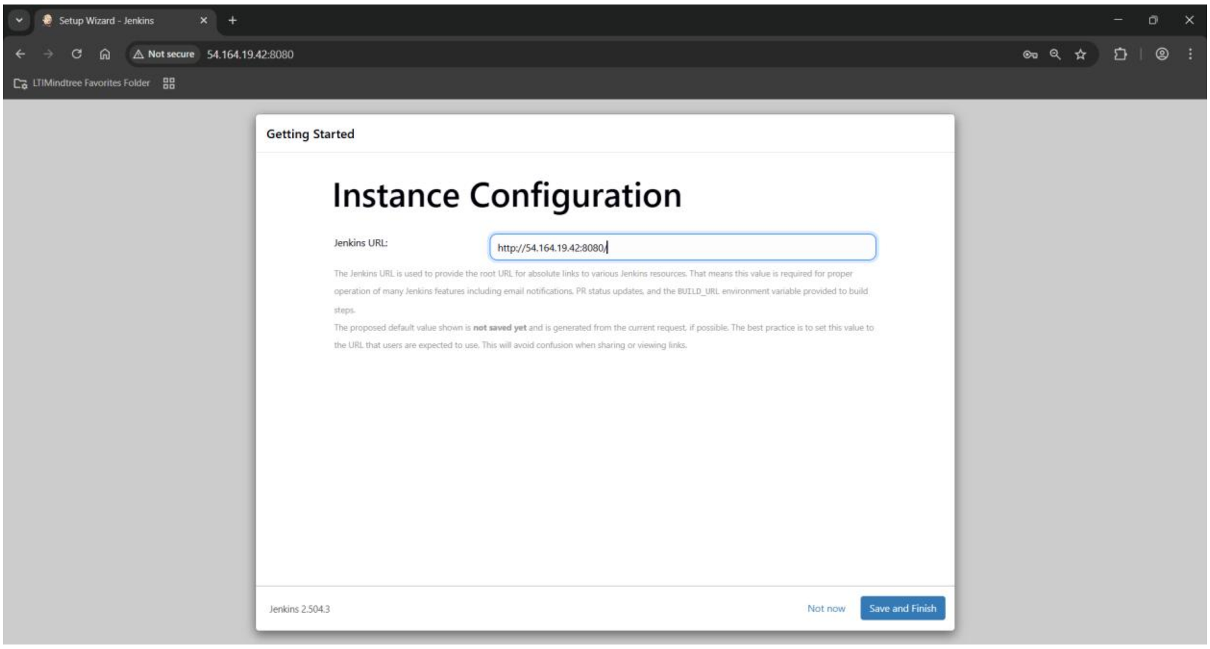
Pasted in Jenkins dashboard to access it.



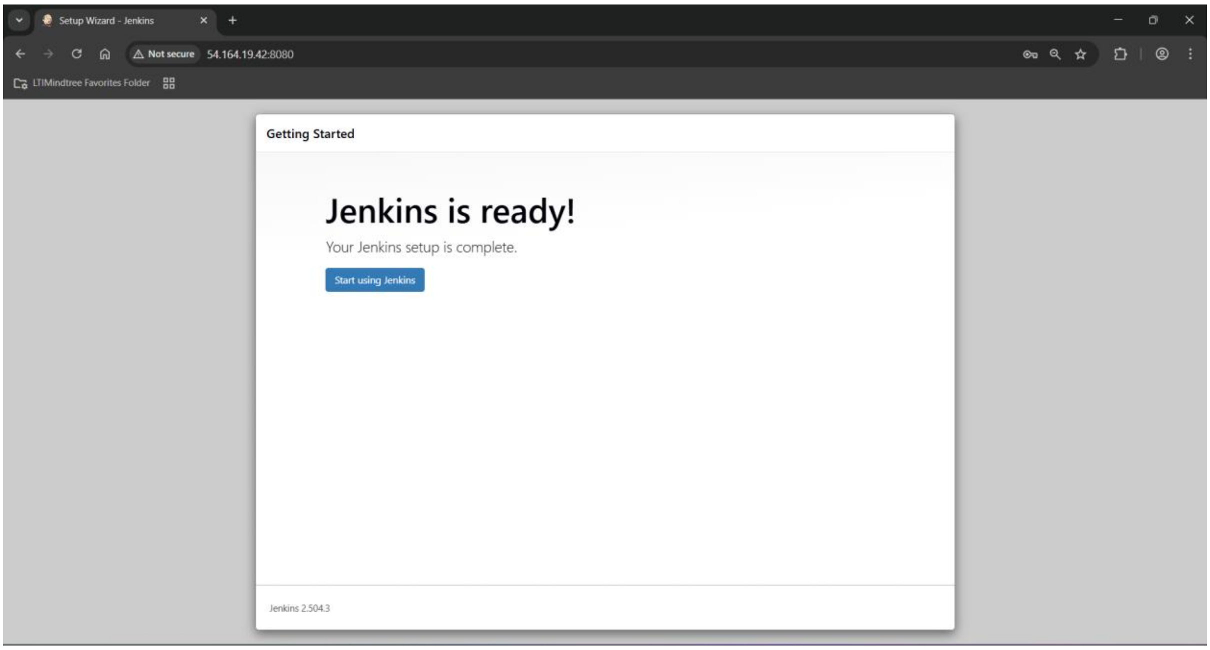
Install suggested plugins.



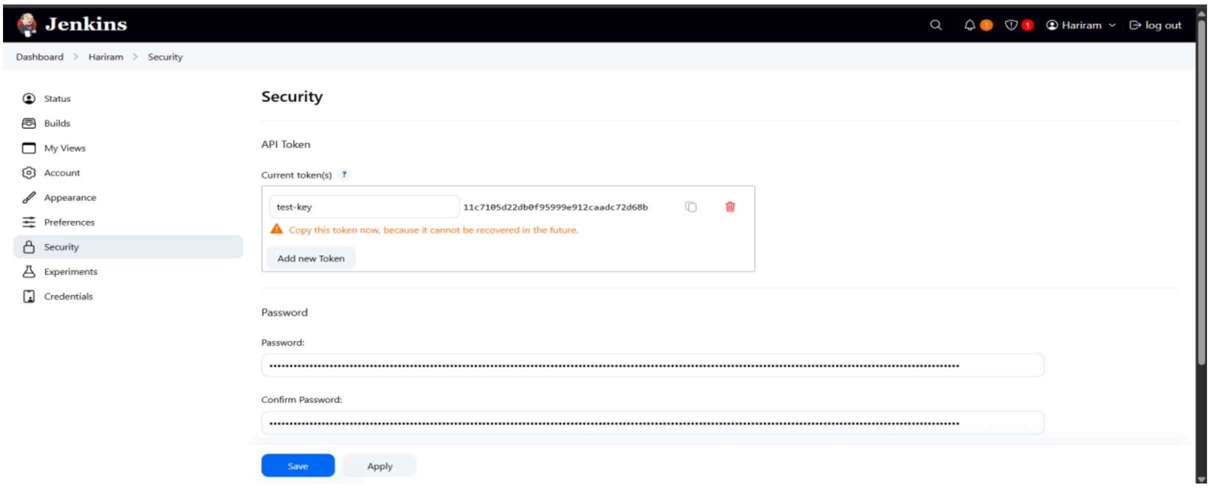
Created admin user with username: admin and password: admin.



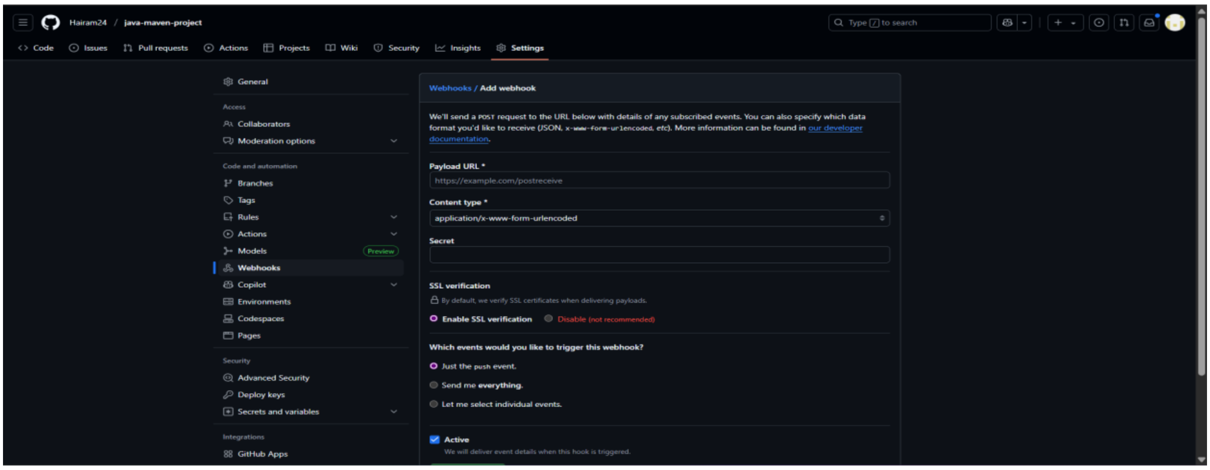
This is the Jenkins url.



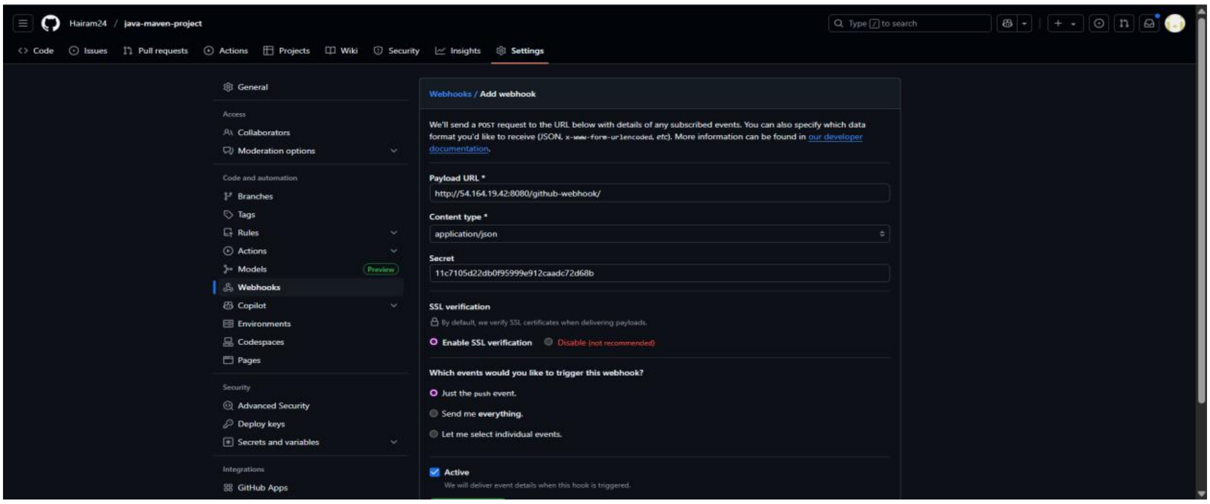
Jenkins is ready to use.



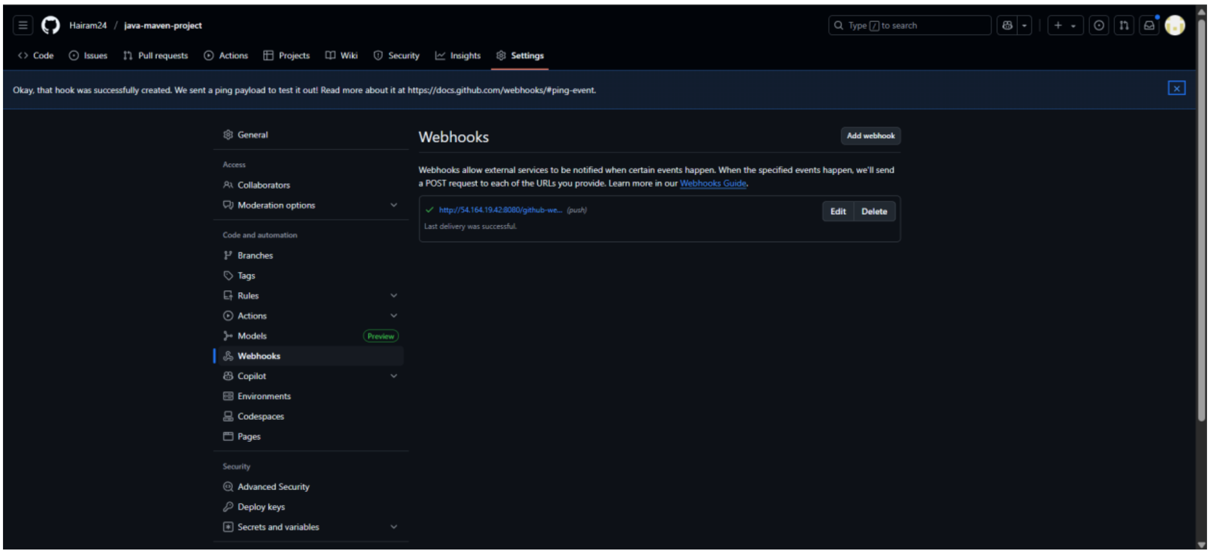
In dashboard > profile > security , create a API token.



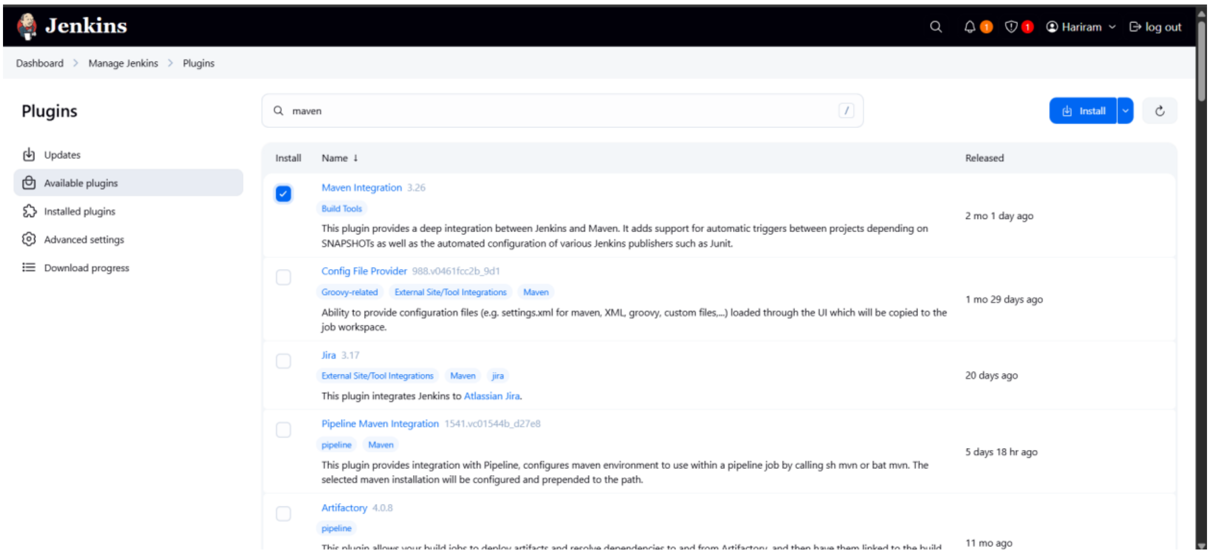
API key pasted in repo setting > webhook.



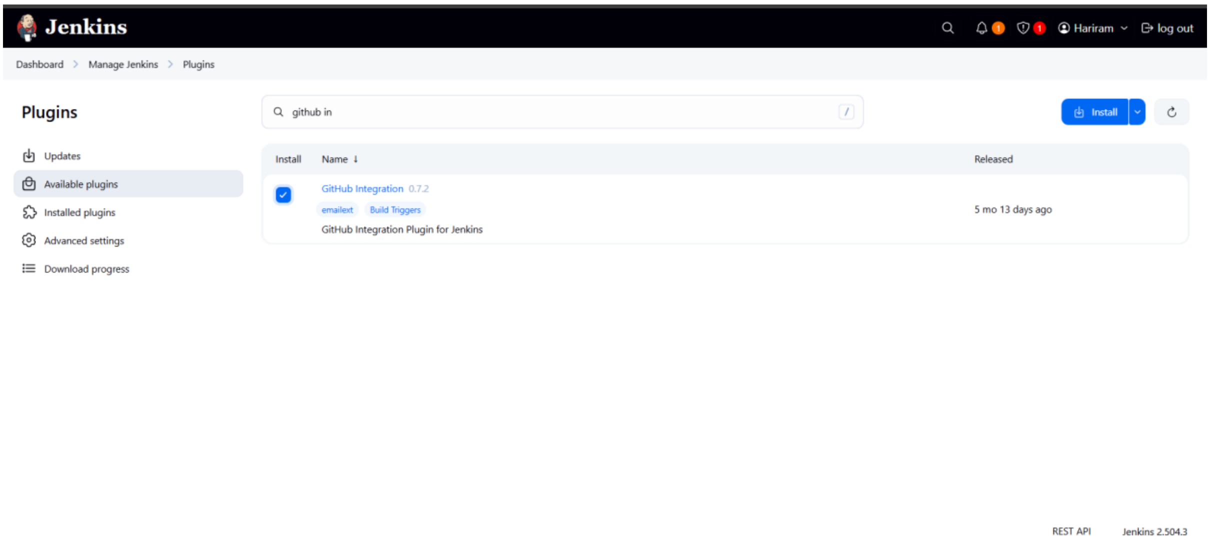
Connection type : application/json is selected.



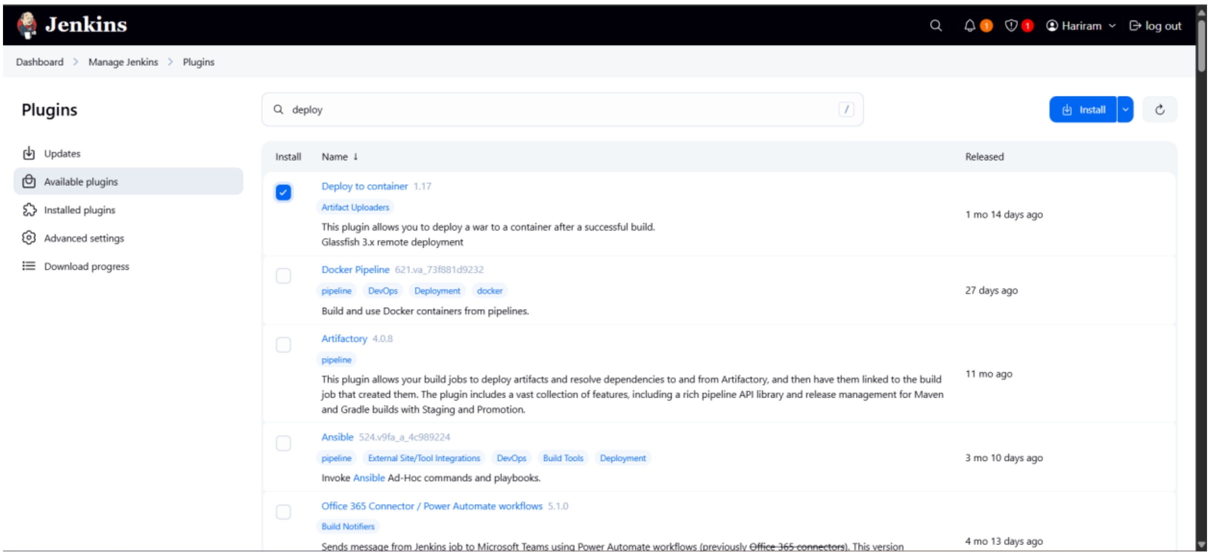
Webhook is created successfully.



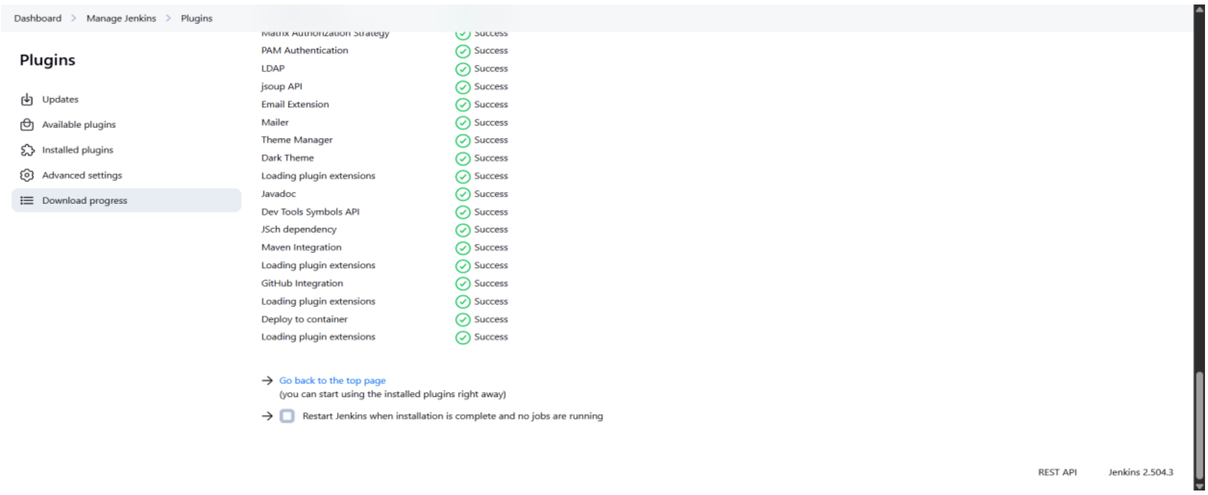
Maven plugin is installed.



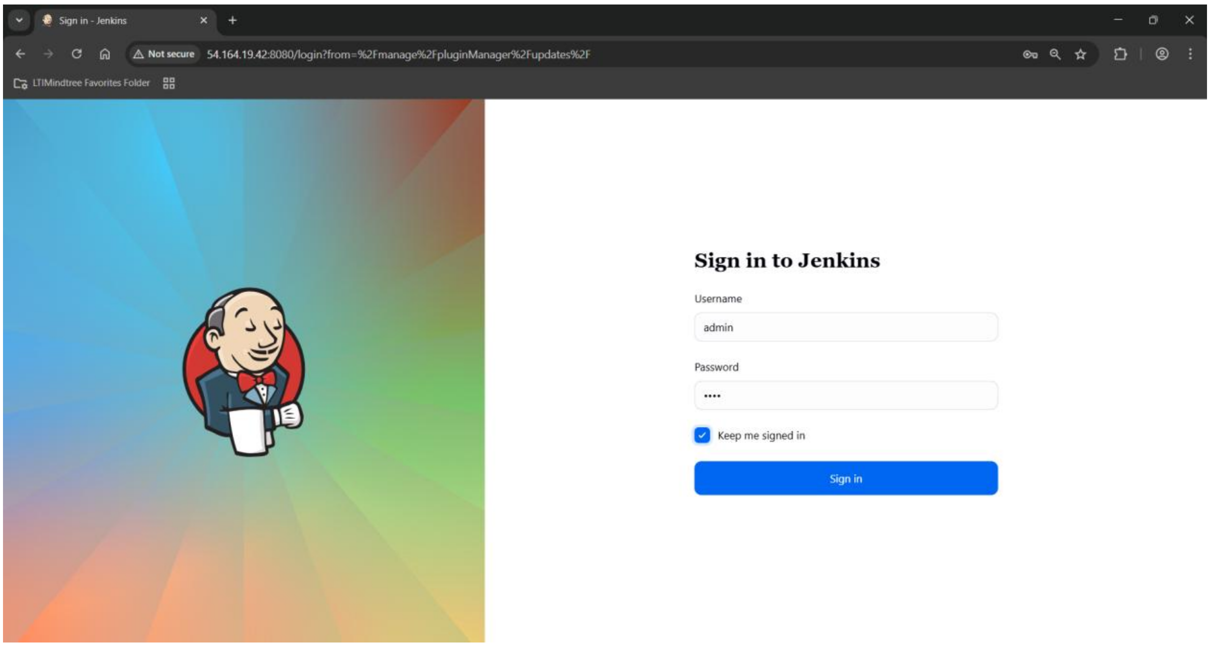
GitHub integration plugin is installed.



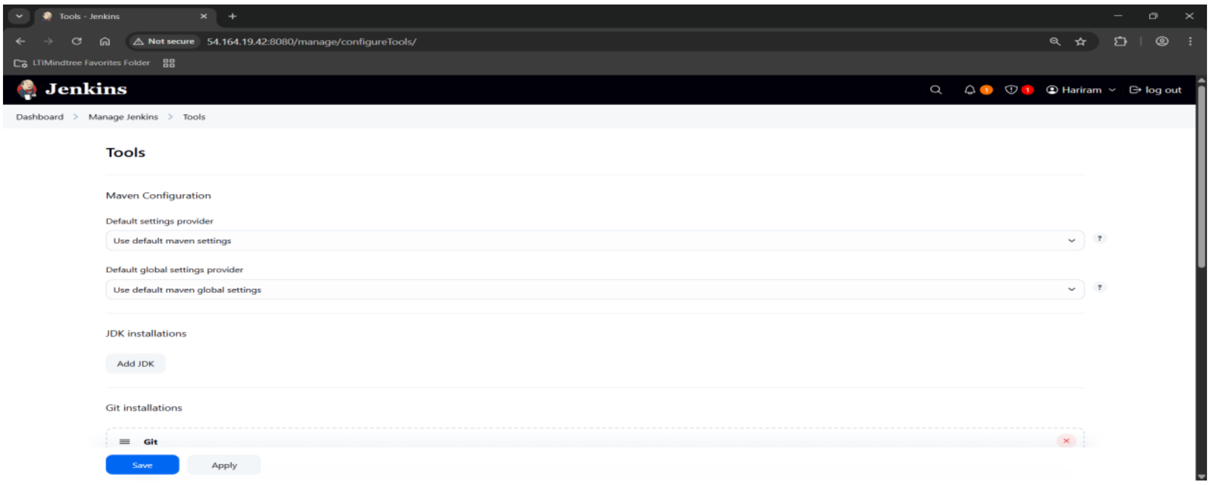
Deploy to container plugin is installed.



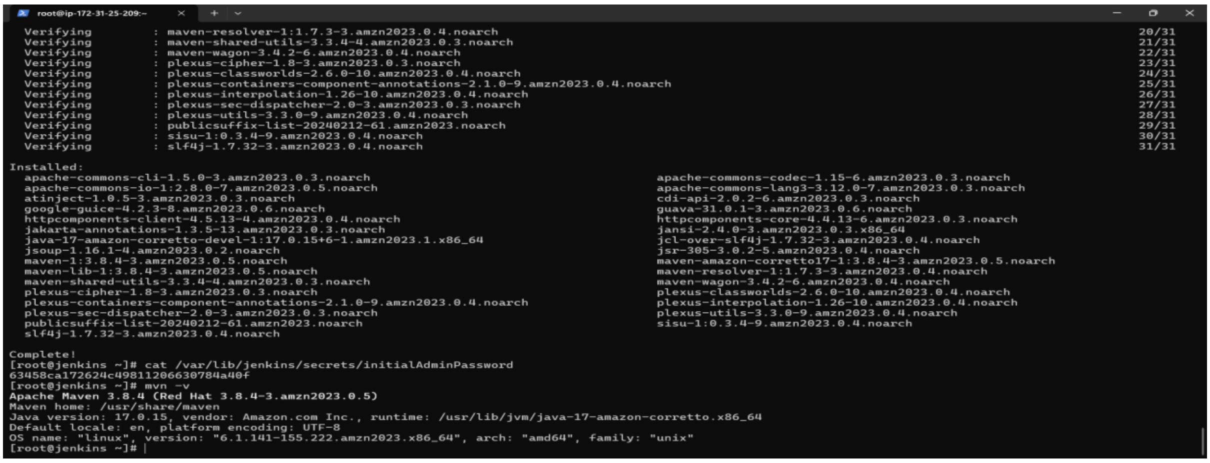
Restarted the Jenkins.



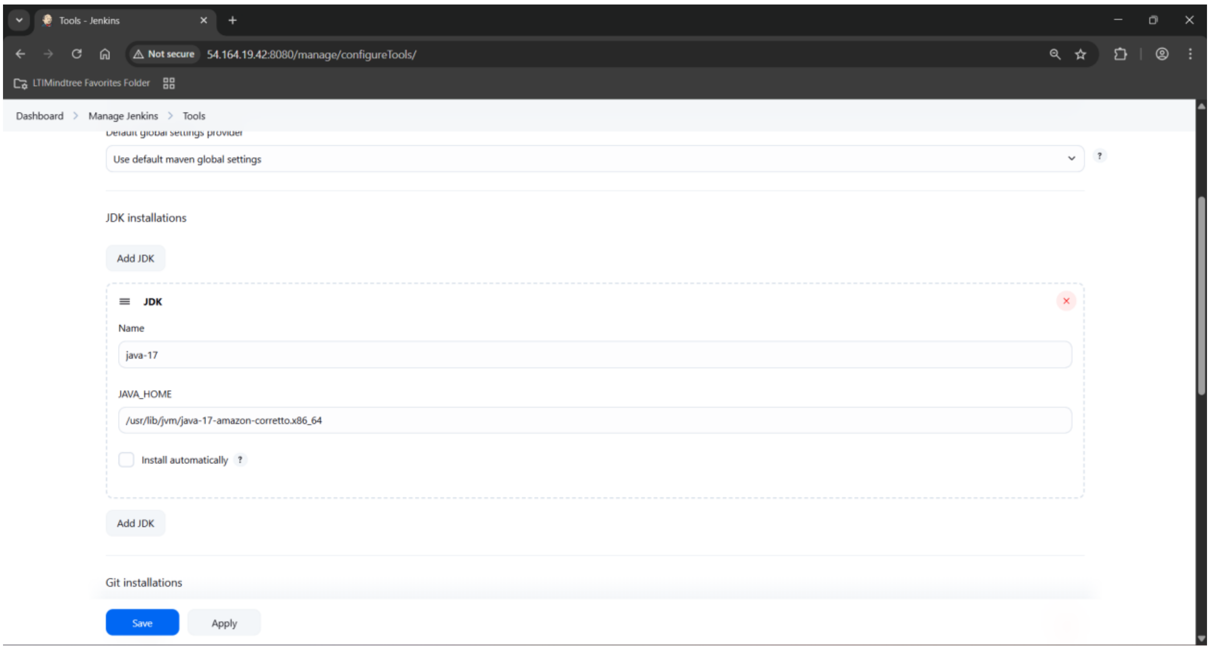
Logged in to the Jenkins.



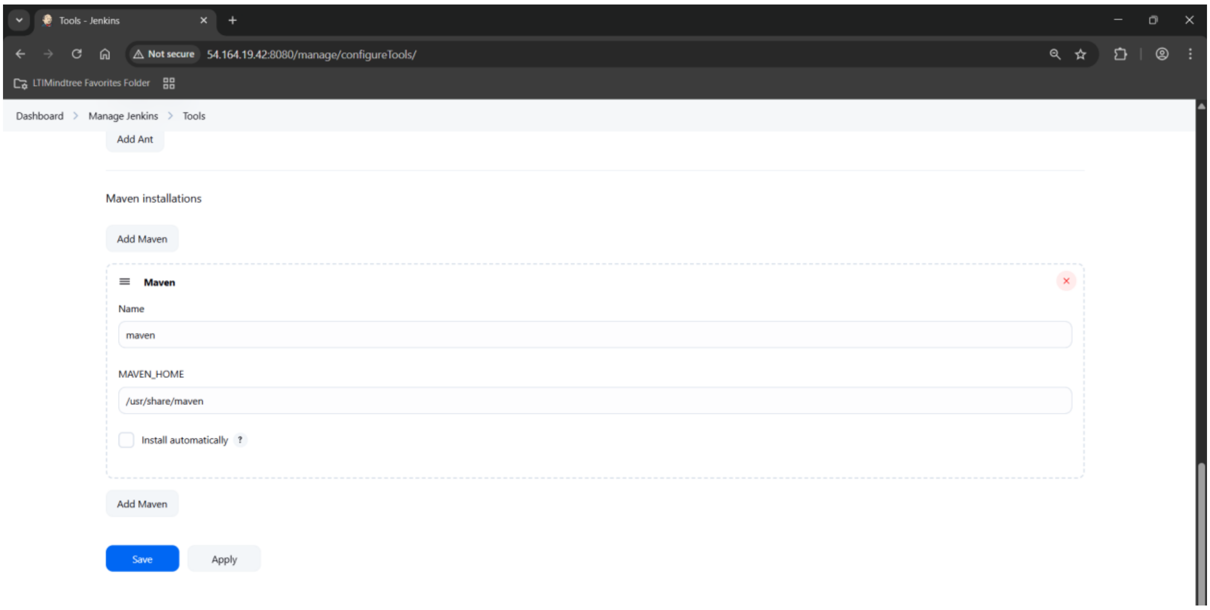
In manage Jenkins > tools add the JDK and Maven path.



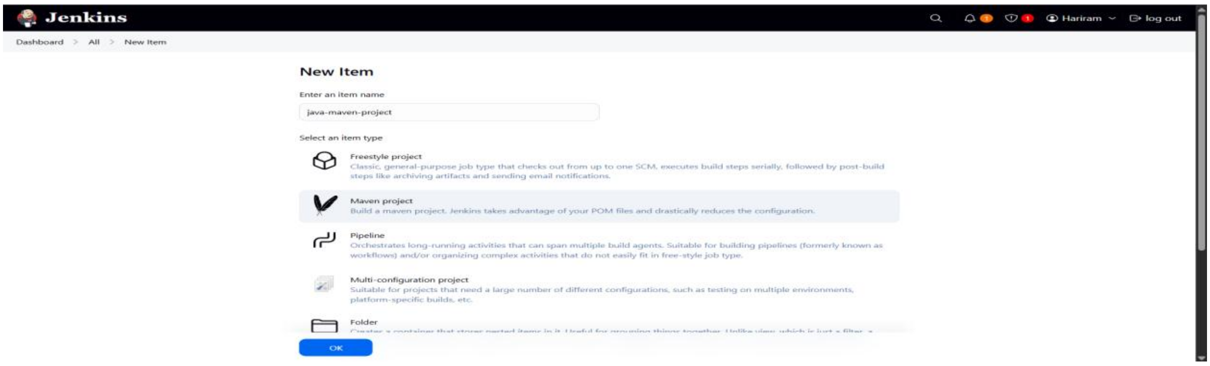
Using mvn -v java and maven path is copied.



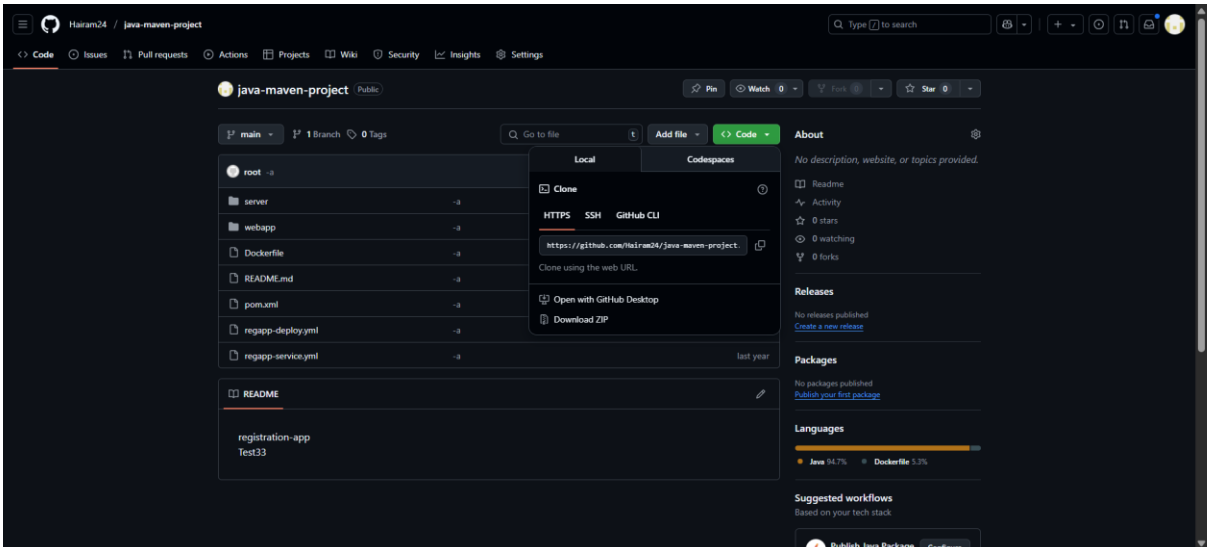
JDK path is added.



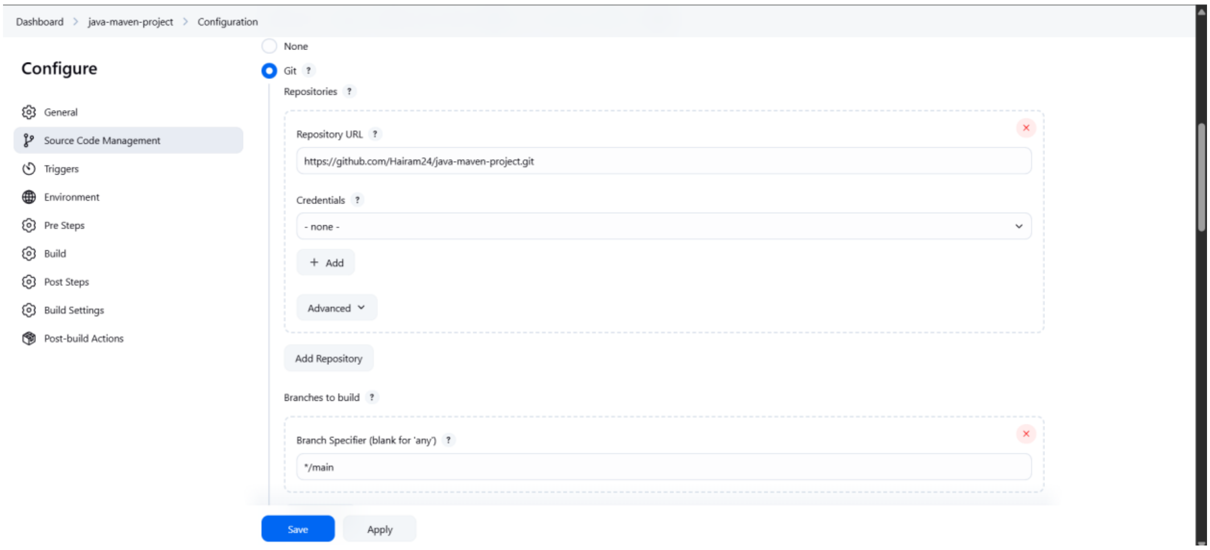
Maven path is added.



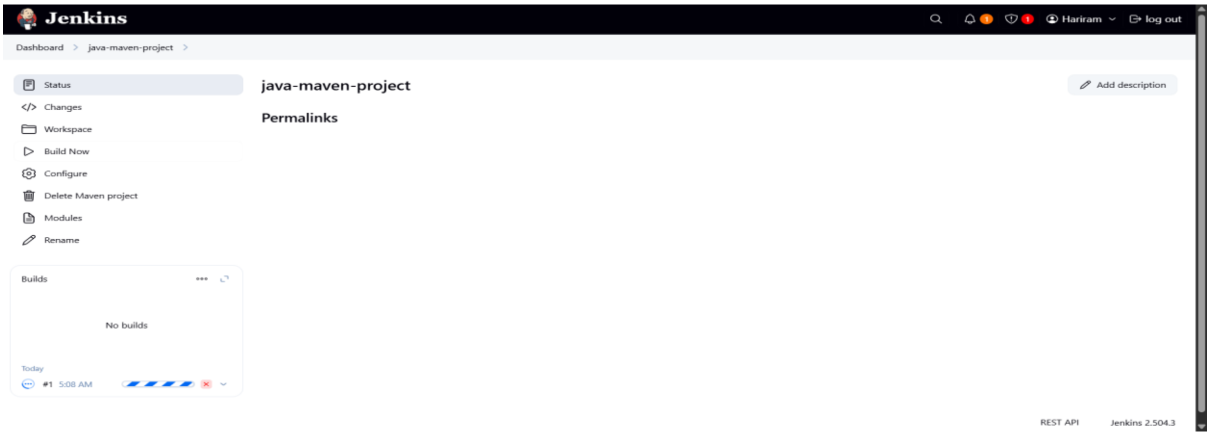
Create a new maven project.



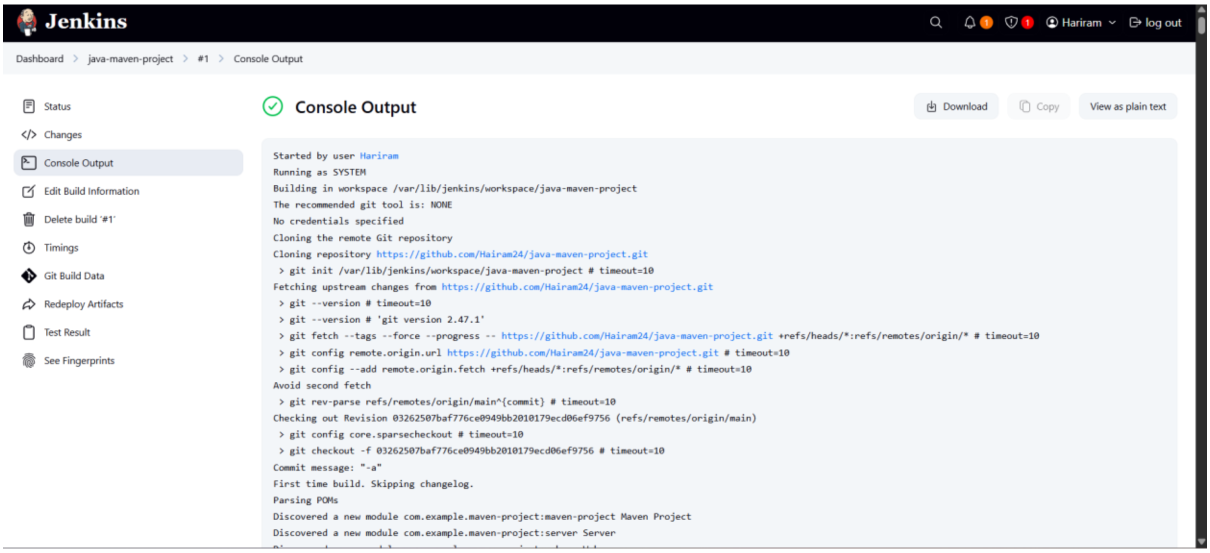
Copy the https repo link.



Add the git repository.

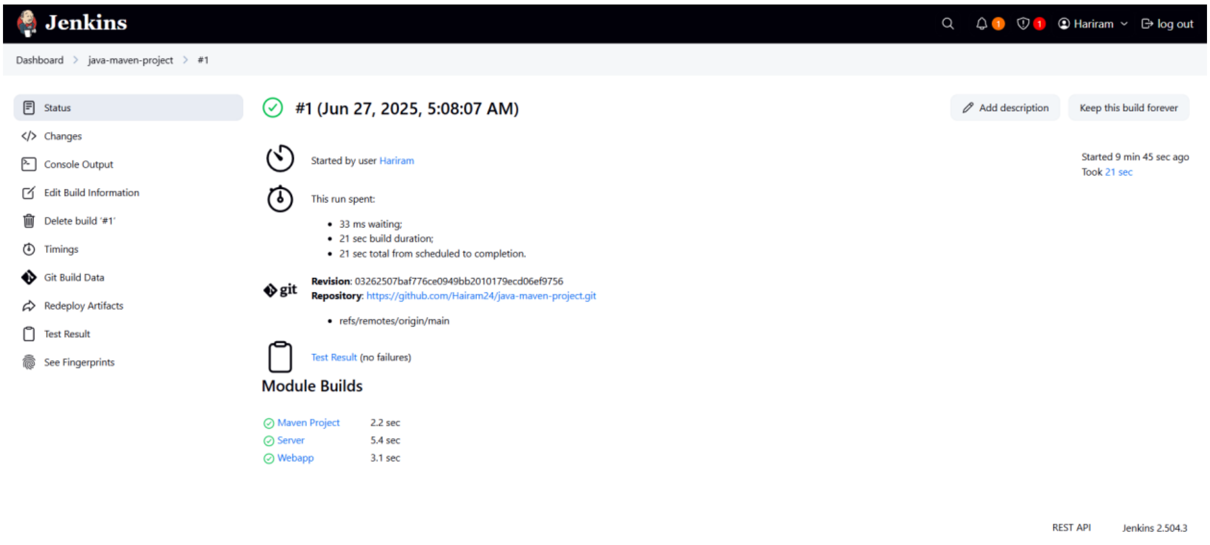


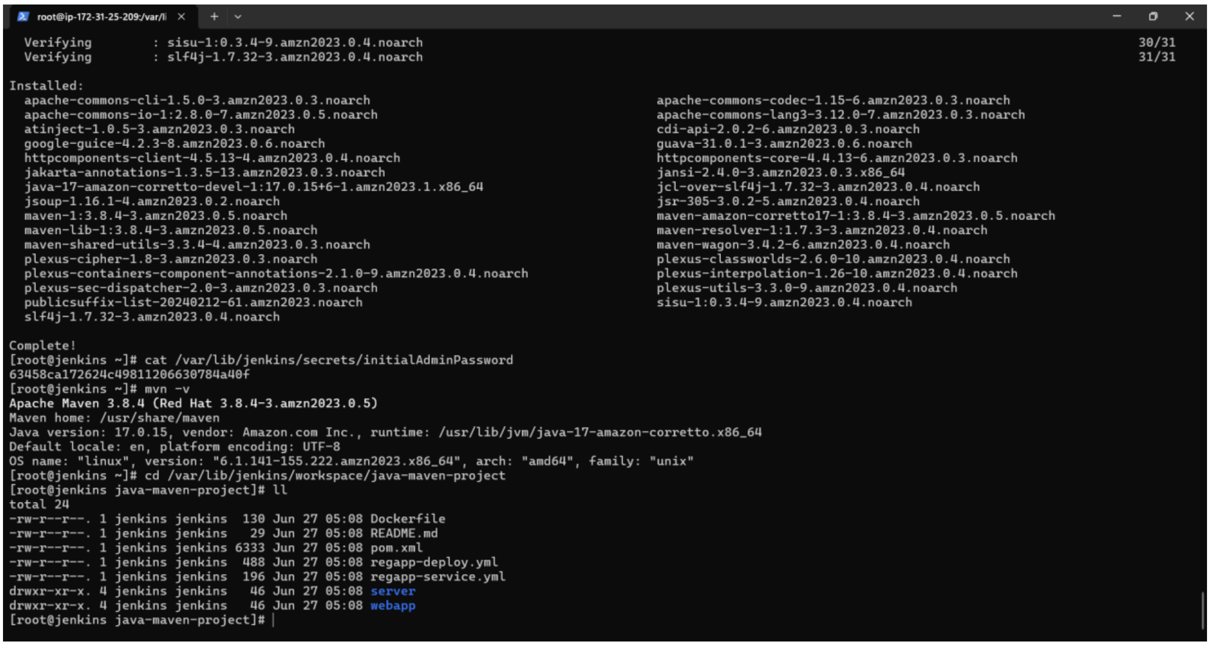
Build now to check.



Build successfully.

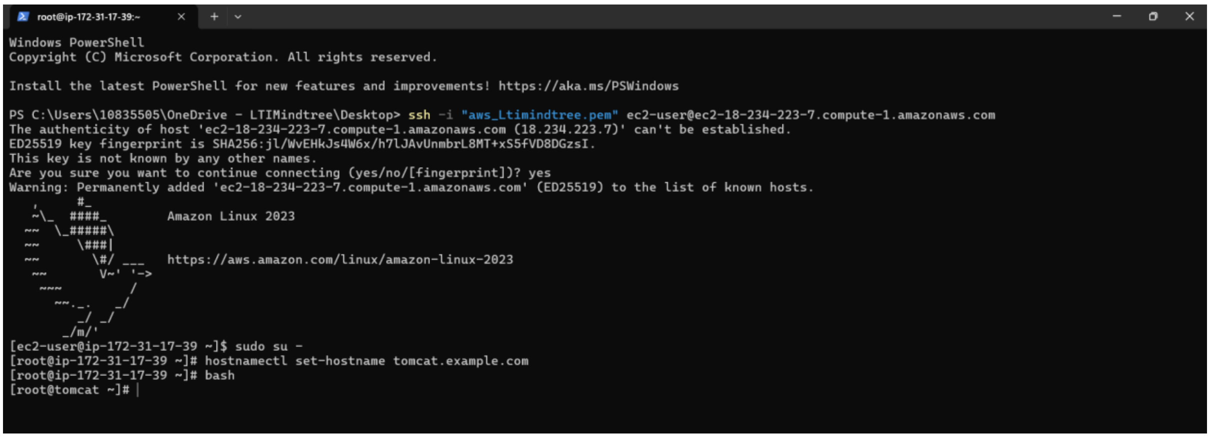
Build finished success.





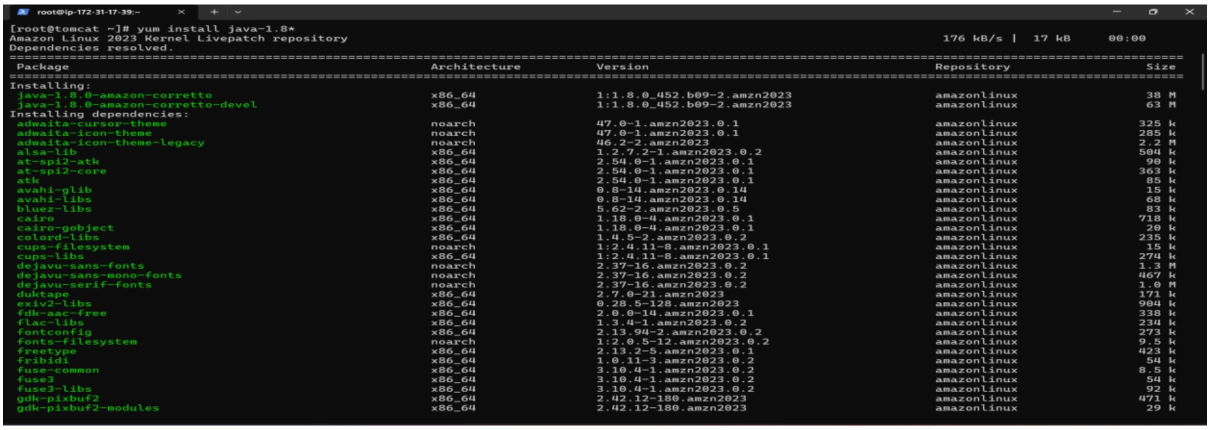
```
root@ip-172-31-25-209:/var/li
  Verifying        : sisu-1:0.3.4-9.amzn2023.0.4.noarch                                    30/31
  Verifying        : slf4j-1.7.32-3.amzn2023.0.4.noarch                                    31/31

Installed:
  apache-commons-cli-1.5.0-3.amzn2023.0.3.noarch                    apache-commons-codec-1.15-6.amzn2023.0.3.noarch
  apache-commons-io-1:2.8.0-7.amzn2023.0.5.noarch                   apache-commons-lang3-3.12.0-7.amzn2023.0.3.noarch
  atinject-1.0.5-3.amzn2023.0.3.noarch                              cdi-api-2.0.2-6.amzn2023.0.3.noarch
  google-guice-4.2.3-8.amzn2023.0.6.noarch                          guava-31.0.1-3.amzn2023.0.6.noarch
  httpcomponents-client-4.5.13-4.amzn2023.0.4.noarch                httpcomponents-core-4.4.13-6.amzn2023.0.3.noarch
  jakarta-annotations-1.3.5-13.amzn2023.0.3.noarch                  jansi-2.4.0-3.amzn2023.0.3.x86_64
  java-17-amazon-corretto-devel-1:17.0.15+6-1.amzn2023.1.x86_64     jcl-over-slf4j-1.7.32-3.amzn2023.0.4.noarch
  jsoup-1.16.1-4.amzn2023.0.2.noarch                                jsr-305-3.0.2-5.amzn2023.0.4.noarch
  maven-1:3.8.4-3.amzn2023.0.5.noarch                               maven-amazon-corretto17-1:3.8.4-3.amzn2023.0.5.noarch
  maven-lib-1:3.8.4-3.amzn2023.0.5.noarch                           maven-resolver-1:1.7.3-3.amzn2023.0.4.noarch
  maven-shared-utils-3.3.4-4.amzn2023.0.3.noarch                    maven-wagon-3.4.2-6.amzn2023.0.4.noarch
  plexus-cipher-1.8-3.amzn2023.0.3.noarch                           plexus-classworlds-2.6.0-10.amzn2023.0.4.noarch
  plexus-containers-component-annotations-2.1.0-9.amzn2023.0.4.noarch  plexus-interpolation-1.26-10.amzn2023.0.4.noarch
  plexus-sec-dispatcher-2.0-3.amzn2023.0.3.noarch                   plexus-utils-3.3.0-9.amzn2023.0.4.noarch
  publicsuffix-list-20240212-61.amzn2023.noarch                     sisu-1:0.3.4-9.amzn2023.0.4.noarch
  slf4j-1.7.32-3.amzn2023.0.4.noarch

Complete!
[root@jenkins ~]# cat /var/lib/jenkins/secrets/initialAdminPassword
63458ca172624c4981120663078 4a40f
[root@jenkins ~]# mvn -v
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.15, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.141-155.222.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@jenkins ~]# cd /var/lib/jenkins/workspace/java-maven-project
[root@jenkins java-maven-project]# ll
total 24
-rw-r--r--. 1 jenkins jenkins  130 Jun 27 05:08 Dockerfile
-rw-r--r--. 1 jenkins jenkins   29 Jun 27 05:08 README.md
-rw-r--r--. 1 jenkins jenkins 6333 Jun 27 05:08 pom.xml
-rw-r--r--. 1 jenkins jenkins  488 Jun 27 05:08 regapp-deploy.yml
-rw-r--r--. 1 jenkins jenkins  196 Jun 27 05:08 regapp-service.yml
drwxr-xr-x. 4 jenkins jenkins   46 Jun 27 05:08 server
drwxr-xr-x. 4 jenkins jenkins   46 Jun 27 05:08 webapp
[root@jenkins java-maven-project]# |
```

WAR file created successfully.



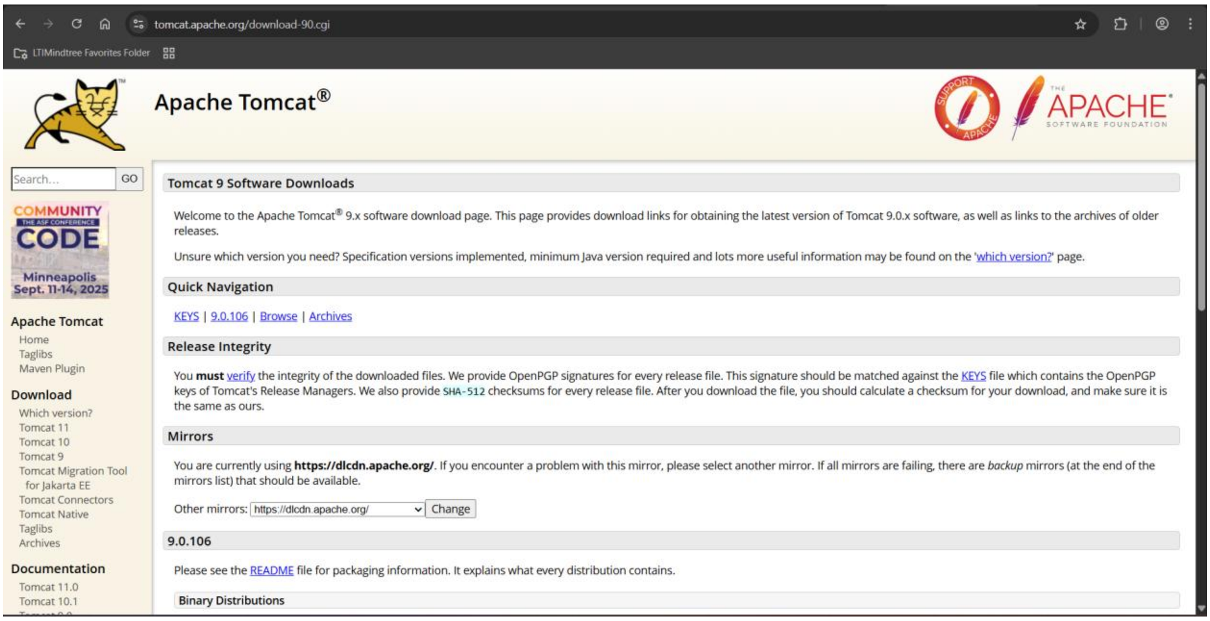
```
root@ip-172-31-17-39:~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\10835505\OneDrive - LTIMindtree\Desktop> ssh -i "aws_Ltimindtree.pem" ec2-user@ec2-18-234-223-7.compute-1.amazonaws.com
The authenticity of host 'ec2-18-234-223-7.compute-1.amazonaws.com (18.234.223.7)' can't be established.
ED25519 key fingerprint is SHA256:jl/WvEHkJs4W6x/h7lJAvUnmbrL8MT+xS5fVD8DGzsI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-18-234-223-7.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-17-39 ~]$ sudo su -
[root@ip-172-31-17-39 ~]# hostnamectl set-hostname tomcat.example.com
[root@ip-172-31-17-39 ~]# bash
[root@tomcat ~]# |
```

Tomcat machine is connected and hostname is changed.



```
root@ip-172-31-17-39:~
[root@tomcat ~]# yum install java-1.8*
Amazon Linux 2023 Kernel Livepatch repository                                  176 kB/s |  17 kB     00:00
Dependencies resolved.
==========================================================================================================
 Package                               Architecture   Version                          Repository     Size
==========================================================================================================
Installing:
 java-1.8.0-amazon-corretto            x86_64         1:1.8.0_452.b09-2.amzn2023       amazonlinux    38 M
 java-1.8.0-amazon-corretto-devel      x86_64         1:1.8.0_452.b09-2.amzn2023       amazonlinux    63 M
Installing dependencies:
 adwaita-cursor-theme                  noarch         47.0-1.amzn2023.0.1              amazonlinux   325 k
 adwaita-icon-theme                    noarch         47.0-1.amzn2023.0.1              amazonlinux   285 k
 adwaita-icon-theme-legacy             noarch         46.2-2.amzn2023                  amazonlinux   2.2 M
 alsa-lib                              x86_64         1.2.7.2-1.amzn2023.0.2           amazonlinux   504 k
 at-spi2-atk                           x86_64         2.54.0-1.amzn2023.0.1            amazonlinux    90 k
 at-spi2-core                          x86_64         2.54.0-1.amzn2023.0.1            amazonlinux   363 k
 atk                                   x86_64         2.54.0-1.amzn2023.0.1            amazonlinux    85 k
 avahi-glib                            x86_64         0.8-14.amzn2023.0.14             amazonlinux    15 k
 avahi-libs                            x86_64         0.8-14.amzn2023.0.14             amazonlinux    68 k
 bluez-libs                            x86_64         5.62-2.amzn2023.0.5              amazonlinux    83 k
 cairo                                 x86_64         1.18.0-4.amzn2023.0.1            amazonlinux   718 k
 cairo-gobject                         x86_64         1.18.0-4.amzn2023.0.1            amazonlinux    20 k
 colord-libs                           x86_64         1.4.5-2.amzn2023.0.2             amazonlinux   235 k
 cups-filesystem                       noarch         1:2.4.11-8.amzn2023.0.1          amazonlinux    15 k
 cups-libs                             x86_64         1:2.4.11-8.amzn2023.0.1          amazonlinux   274 k
 dejavu-sans-fonts                     noarch         2.37-16.amzn2023.0.2             amazonlinux   1.3 M
 dejavu-sans-mono-fonts                noarch         2.37-16.amzn2023.0.2             amazonlinux   467 k
 dejavu-serif-fonts                    noarch         2.37-16.amzn2023.0.2             amazonlinux   1.0 M
 duktape                               x86_64         2.7.0-21.amzn2023                amazonlinux   171 k
 exiv2-libs                            x86_64         0.28.5-128.amzn2023              amazonlinux   904 k
 fdk-aac-free                          x86_64         2.0.0-14.amzn2023.0.1            amazonlinux   338 k
 flac-libs                             x86_64         1.3.4-1.amzn2023.0.2             amazonlinux   234 k
 fontconfig                            x86_64         2.13.94-2.amzn2023.0.2           amazonlinux   273 k
 fonts-filesystem                      noarch         1:2.0.5-12.amzn2023.0.2          amazonlinux   9.5 k
 freetype                              x86_64         2.13.2-5.amzn2023.0.1            amazonlinux   423 k
 fribidi                               x86_64         1.0.11-3.amzn2023.0.2            amazonlinux    54 k
 fuse-common                           x86_64         3.10.4-1.amzn2023.0.2            amazonlinux   8.5 k
 fuse3                                 x86_64         3.10.4-1.amzn2023.0.2            amazonlinux    54 k
 fuse3-libs                            x86_64         3.10.4-1.amzn2023.0.2            amazonlinux    92 k
 gdk-pixbuf2                           x86_64         2.42.12-180.amzn2023             amazonlinux   471 k
 gdk-pixbuf2-modules                   x86_64         2.42.12-180.amzn2023             amazonlinux    29 k
```

Java package is installed.



Copy the tomcat 9 tar file link.

```
[root@tomcat ~]# wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
--2025-06-27 05:26:35--  https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.106/bin/apache-tomcat-9.0.106.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13007554 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.106.tar.gz'

apache-tomcat-9.0.106.tar.gz          100%[==========================================================================>]  12.40M  --.-KB/s    in 0.09s

2025-06-27 05:26:35 (131 MB/s) - 'apache-tomcat-9.0.106.tar.gz' saved [13007554/13007554]

[root@tomcat ~]#
```

Download the tar file.

```
root@ip-172-31-17-39:~

[root@tomcat ~]# tar -zvxf apache-tomcat-9.0.106.tar.gz
apache-tomcat-9.0.106/conf/
apache-tomcat-9.0.106/conf/catalina.policy
apache-tomcat-9.0.106/conf/catalina.properties
apache-tomcat-9.0.106/conf/context.xml
apache-tomcat-9.0.106/conf/jaspic-providers.xml
apache-tomcat-9.0.106/conf/jaspic-providers.xsd
apache-tomcat-9.0.106/conf/logging.properties
apache-tomcat-9.0.106/conf/server.xml
apache-tomcat-9.0.106/conf/tomcat-users.xml
apache-tomcat-9.0.106/conf/tomcat-users.xsd
apache-tomcat-9.0.106/conf/web.xml
apache-tomcat-9.0.106/bin/
apache-tomcat-9.0.106/lib/
apache-tomcat-9.0.106/logs/
apache-tomcat-9.0.106/temp/
apache-tomcat-9.0.106/webapps/
apache-tomcat-9.0.106/webapps/ROOT/
apache-tomcat-9.0.106/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/
apache-tomcat-9.0.106/webapps/docs/META-INF/
apache-tomcat-9.0.106/webapps/docs/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.106/webapps/docs/annotationapi/
apache-tomcat-9.0.106/webapps/docs/api/
apache-tomcat-9.0.106/webapps/docs/appdev/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/src/mypackage/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/WEB-INF/
apache-tomcat-9.0.106/webapps/docs/appdev/sample/web/images/
apache-tomcat-9.0.106/webapps/docs/architecture/
```

Unzip the tar file.

```
[root@tomcat ~]# cd apache-tomcat-9.0.106
[root@tomcat apache-tomcat-9.0.106]# cd bin
[root@tomcat bin]# chmod +x startup.sh
[root@tomcat bin]# chmod +x shutdown.sh
[root@tomcat bin]# ln -s /opt/apache-tomcat-9.0.106/bin/startup.sh /usr/local/bin/tomcatup
[root@tomcat bin]# ln -s /opt/apache-tomcat-9.0.106/bin/shutdown.sh /usr/local/bin/tomcatdown
```

Tomcat installation is started.

```
[root@tomcat ~]# cd apache-tomcat-9.0.106
[root@tomcat apache-tomcat-9.0.106]# cd bin
[root@tomcat bin]# ./startup.sh
Using CATALINA_BASE:   /root/apache-tomcat-9.0.106
Using CATALINA_HOME:   /root/apache-tomcat-9.0.106
Using CATALINA_TMPDIR: /root/apache-tomcat-9.0.106/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /root/apache-tomcat-9.0.106/bin/bootstrap.jar:/root/apache-tomcat-9.0.106/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@tomcat bin]#
```

Tomcat is started.

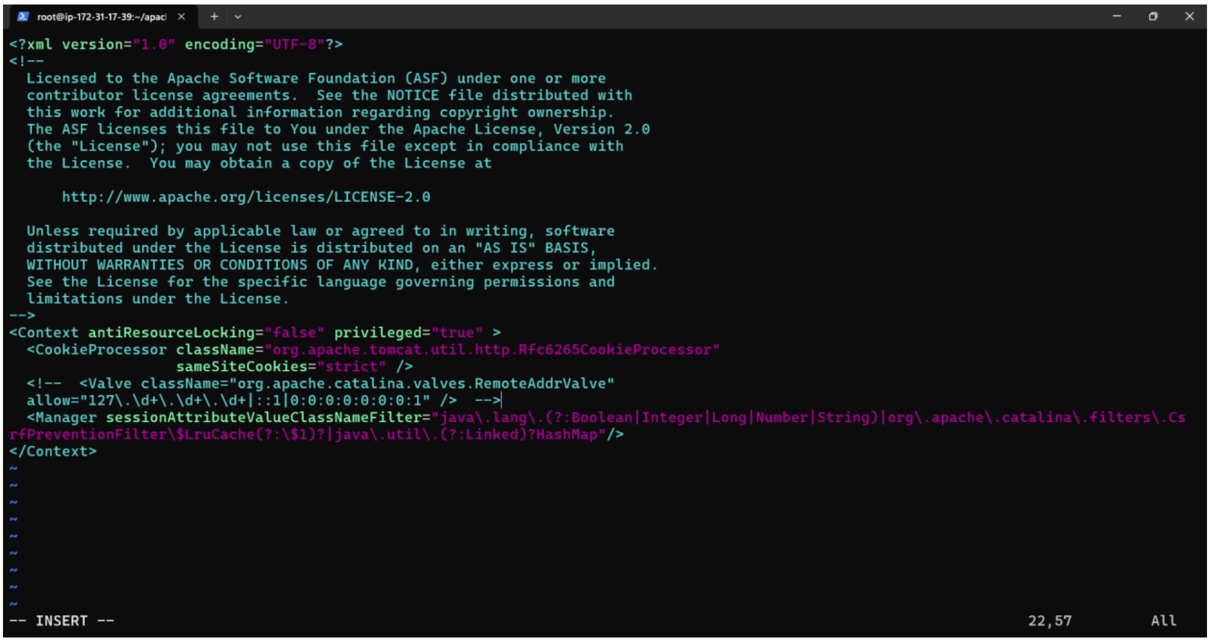
```
root@ip-172-31-17-39:~/apac
[root@tomcat apache-tomcat-9.0.106]# vi ./webapps/examples/META-INF/context.xml
```

```
root@ip-172-31-17-39:~/apac
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<Context>
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
                   sameSiteCookies="strict" />
  <!--  <Valve className="org.apache.catalina.valves.RemoteAddrValve"
  allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />  -->
</Context>
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --                                                       22,57         All
```

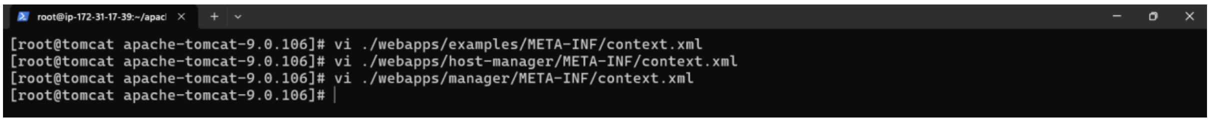
Commented the IP part.

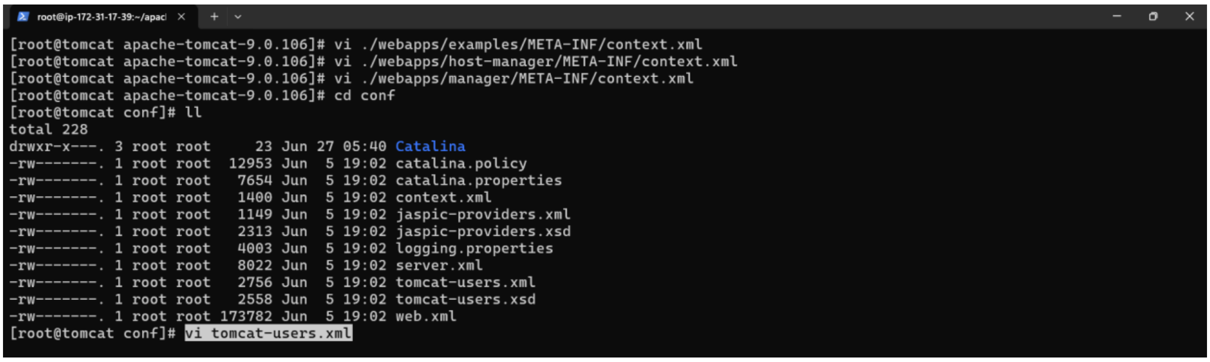
```
root@ip-172-31-17-39:~/apac
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Licensed to the Apache Software Foundation (ASF) under one or more
  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
  the License.  You may obtain a copy of the License at

      http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" >
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
                   sameSiteCookies="strict" />
  <!-- <Valve className="org.apache.catalina.valves.RemoteAddrValve"
  allow="127\.\d+\.\d+\.\d+|::1|0:0:0:0:0:0:0:1" />  -->
  <Manager sessionAttributeValueClassNameFilter="java\.lang\.(?:Boolean|Integer|Long|Number|String)|org\.apache\.catalina\.filters\.Cs
rfPreventionFilter\$LruCache(?:\$1)?|java\.util\.(?:Linked)?HashMap"/>
</Context>
~
~
~
~
~
~
~
~
~
-- INSERT --                                                       22,57         All
```

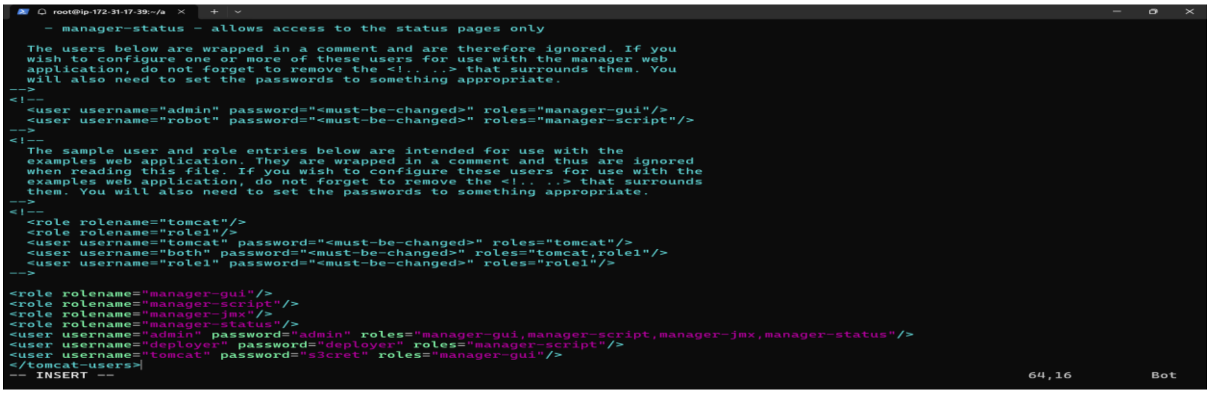
Commented the IP part.



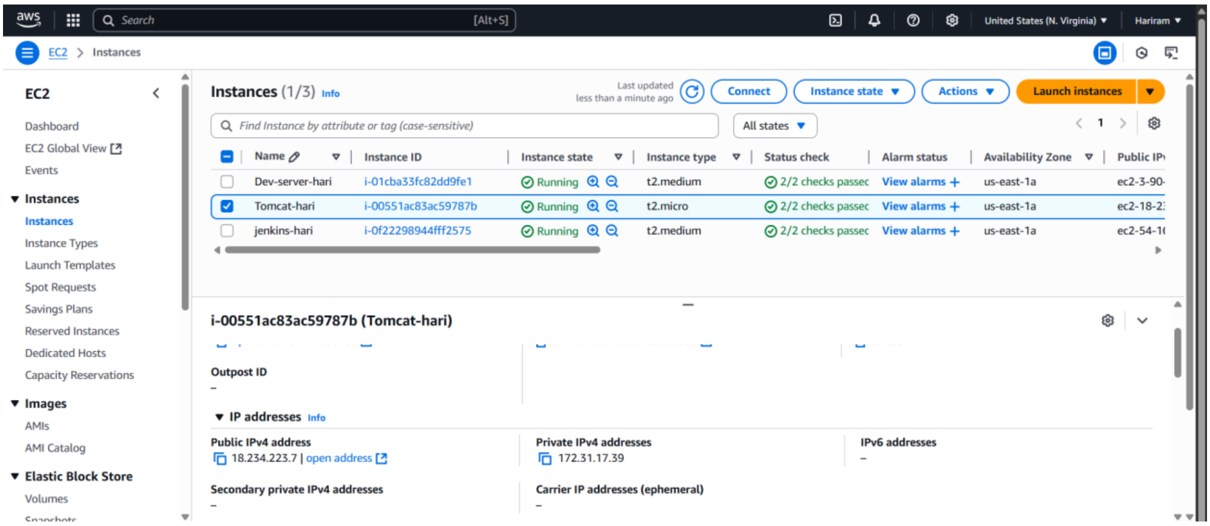
Commented the IP part.



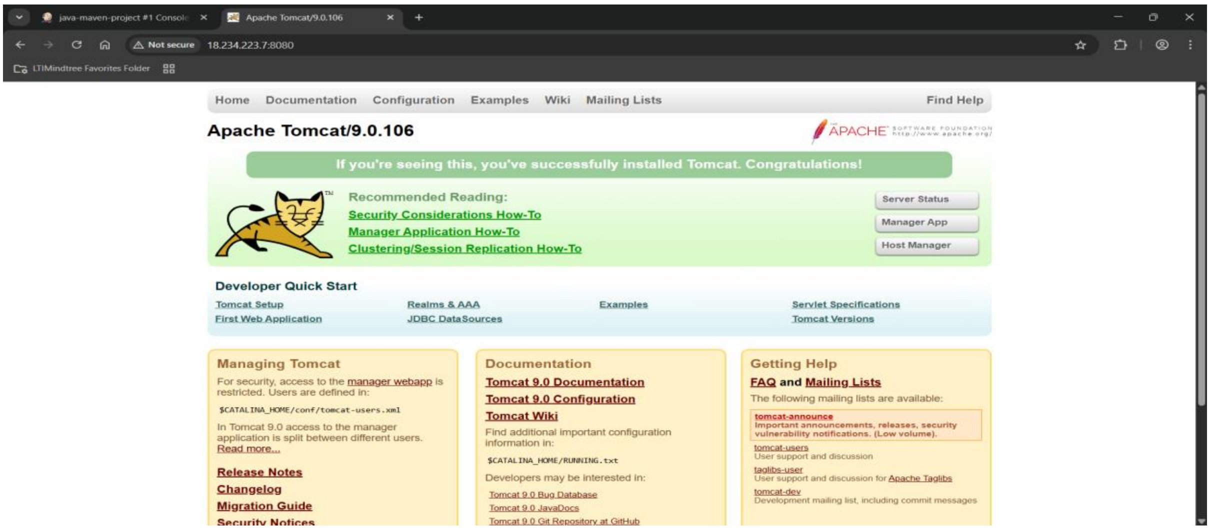




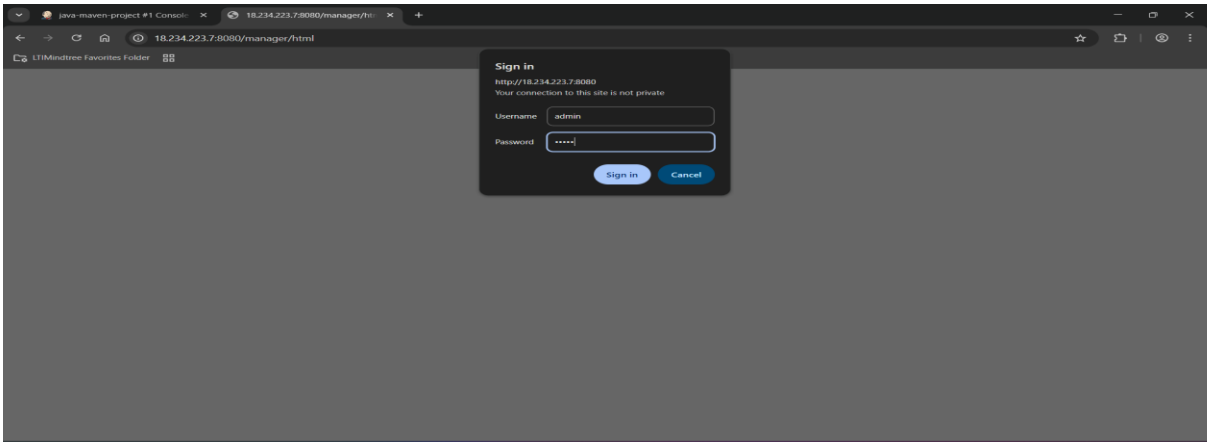
Roles are added.



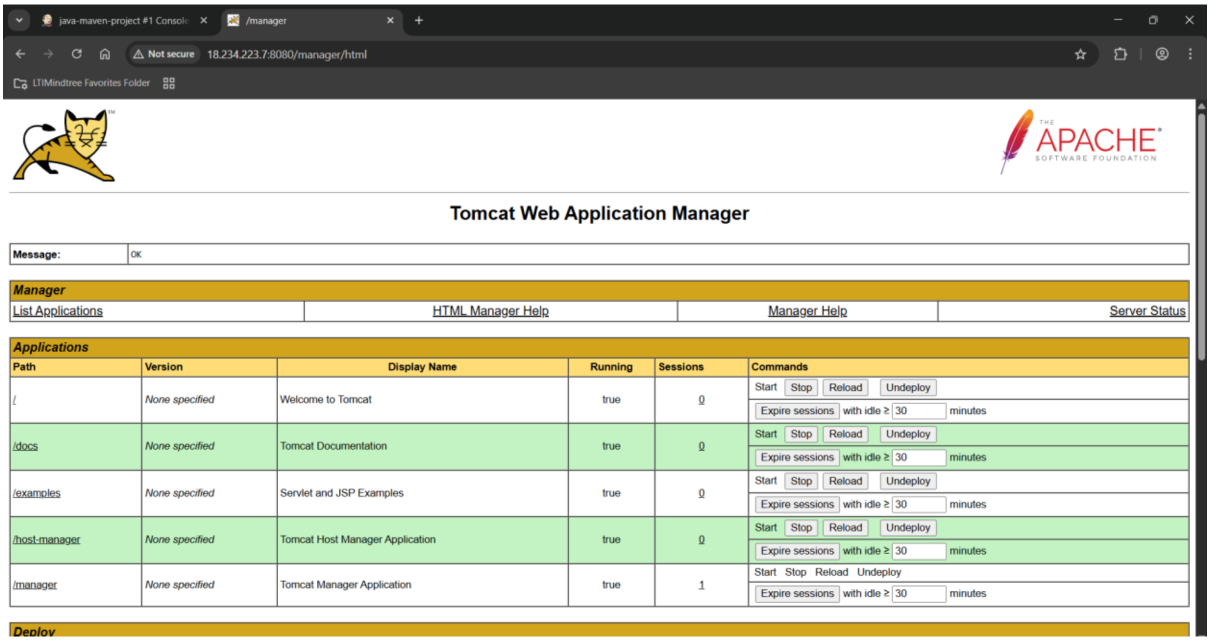
Tomcat mchine public IP is copied.



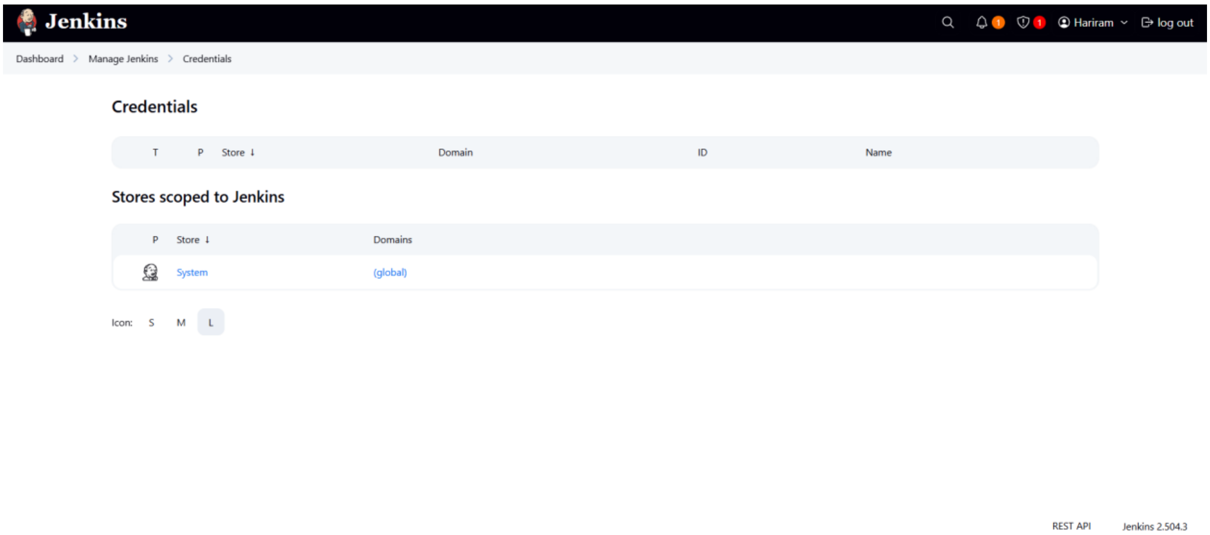
Tomcat dashboard is accessible.



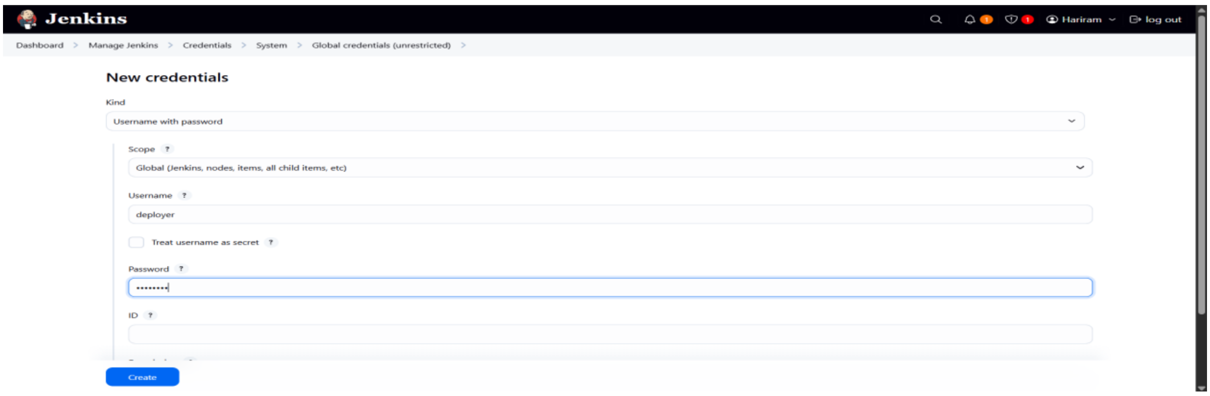
Login through role credentials.



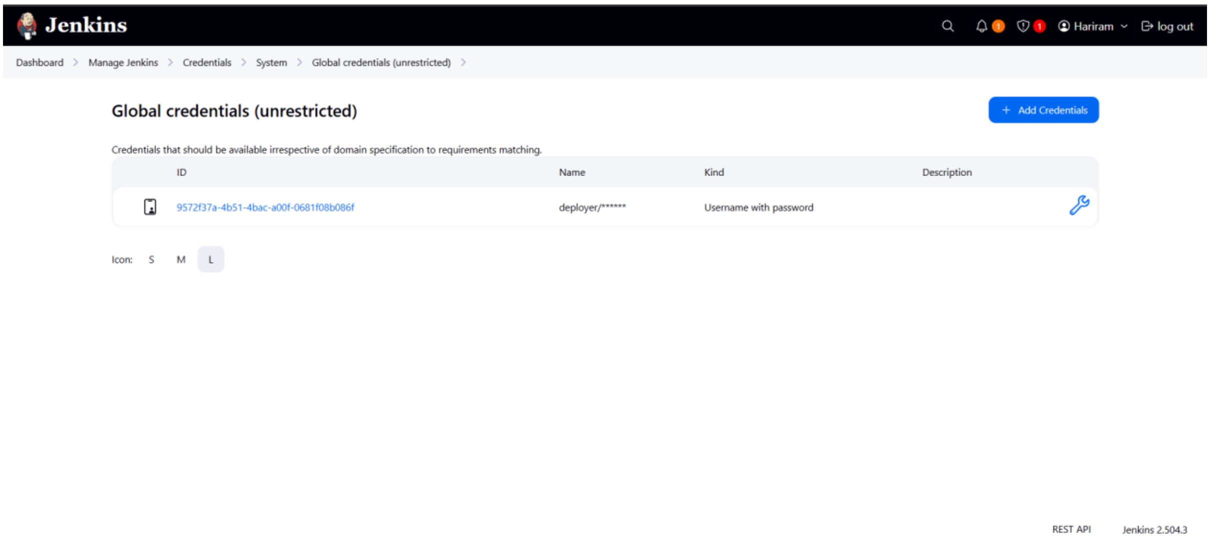
Tomcat setup is done.



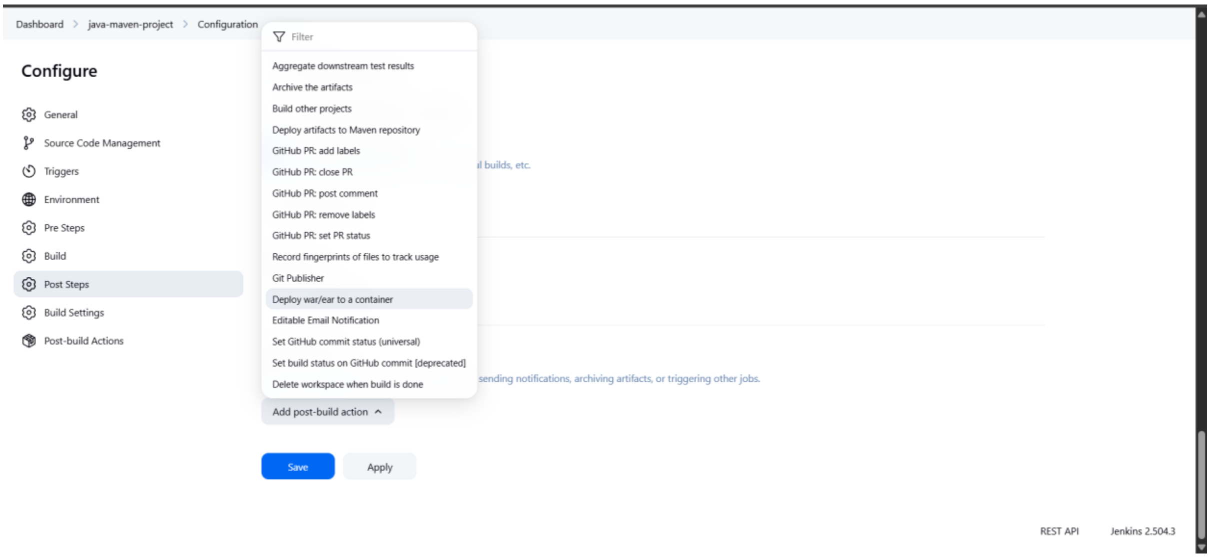
In Jenkins add global credentials.



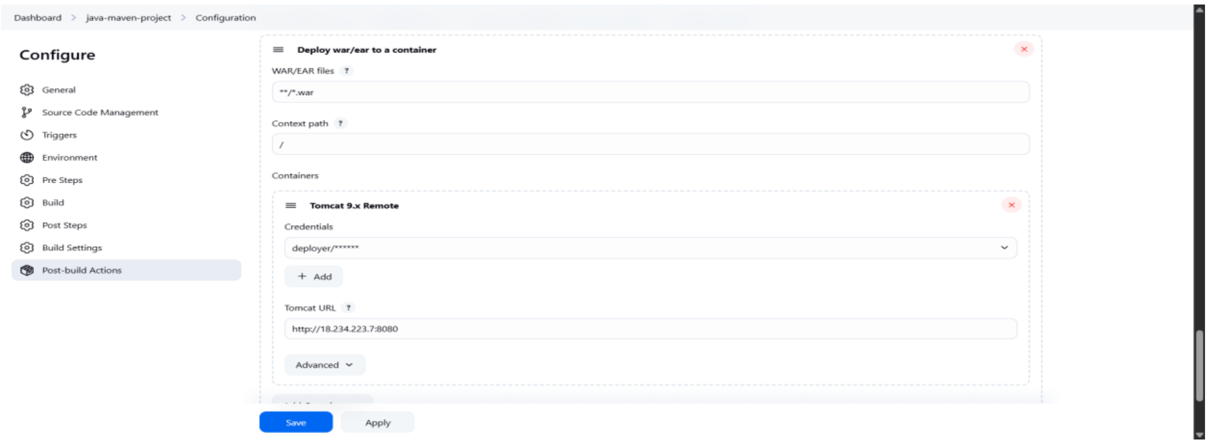
Username: deployer , Password: deployer.



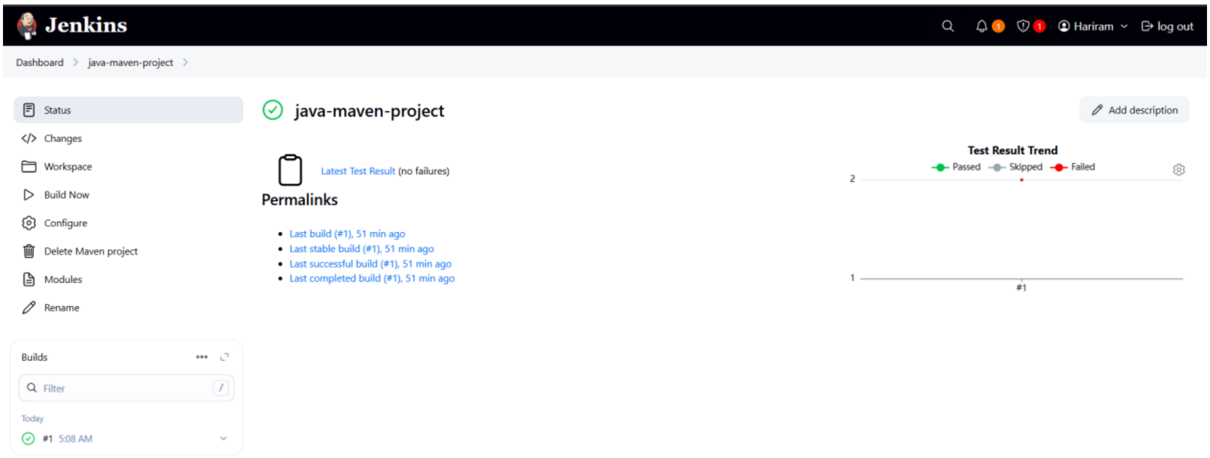
Credential is added successfully.



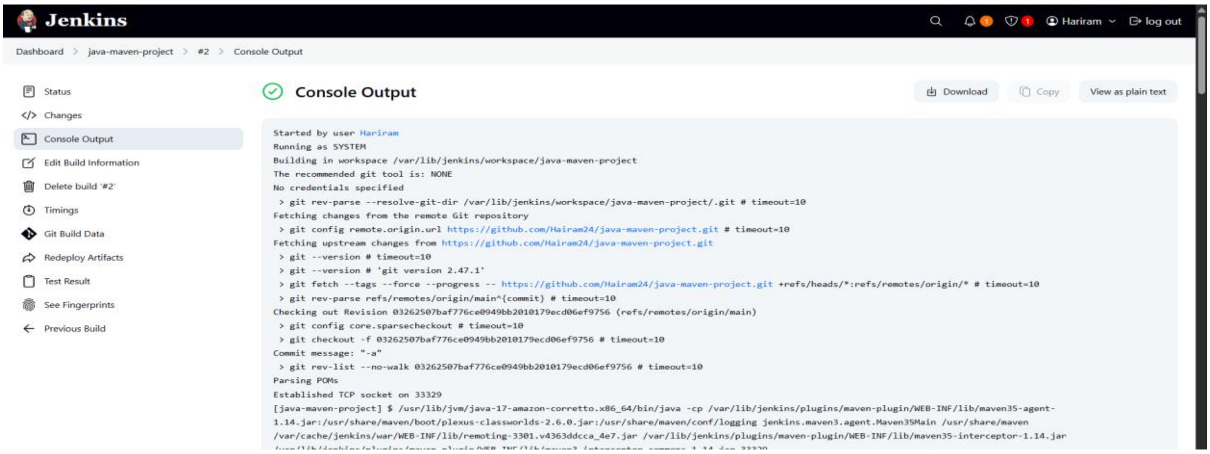
Add post build action. Use deploy war/ear to a container.



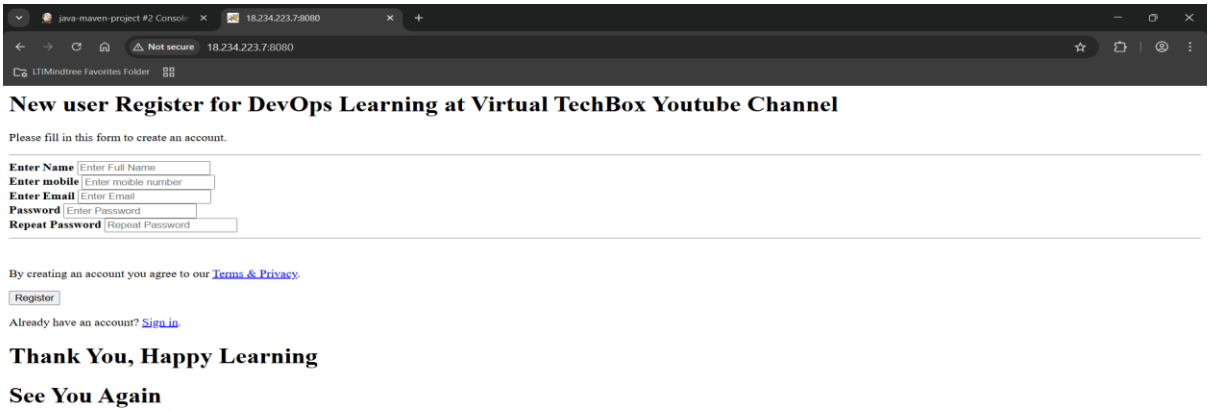
Entered the required details.



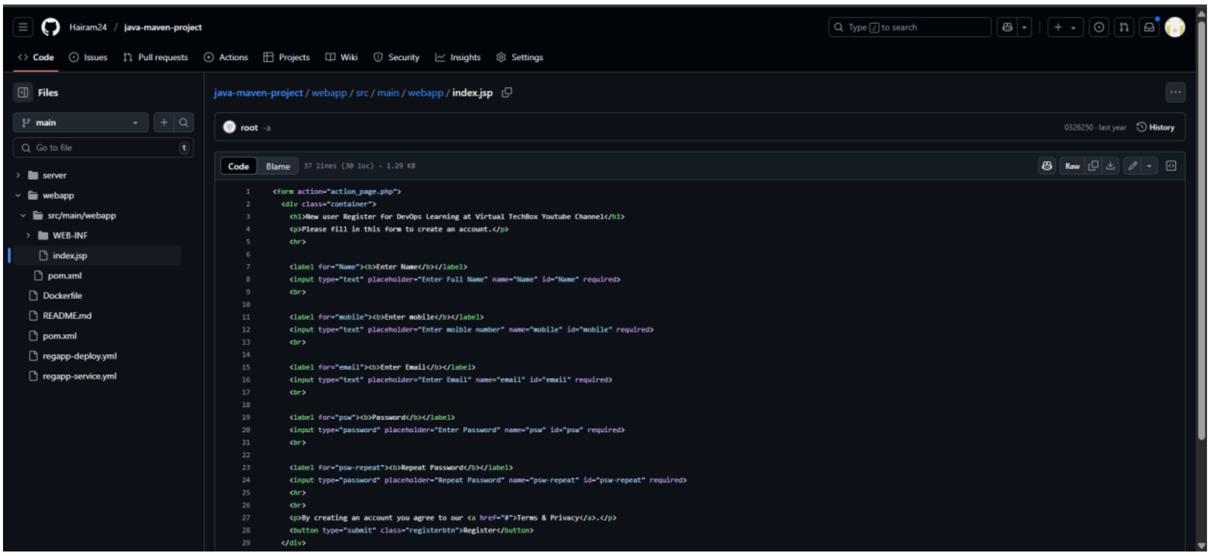
Build now.

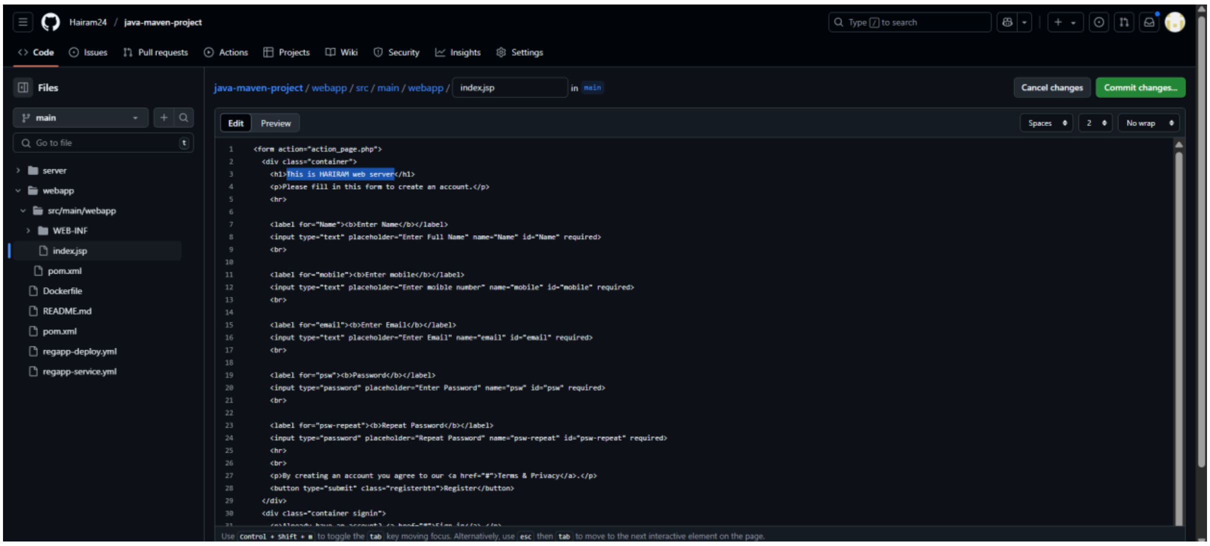


Build successfull.

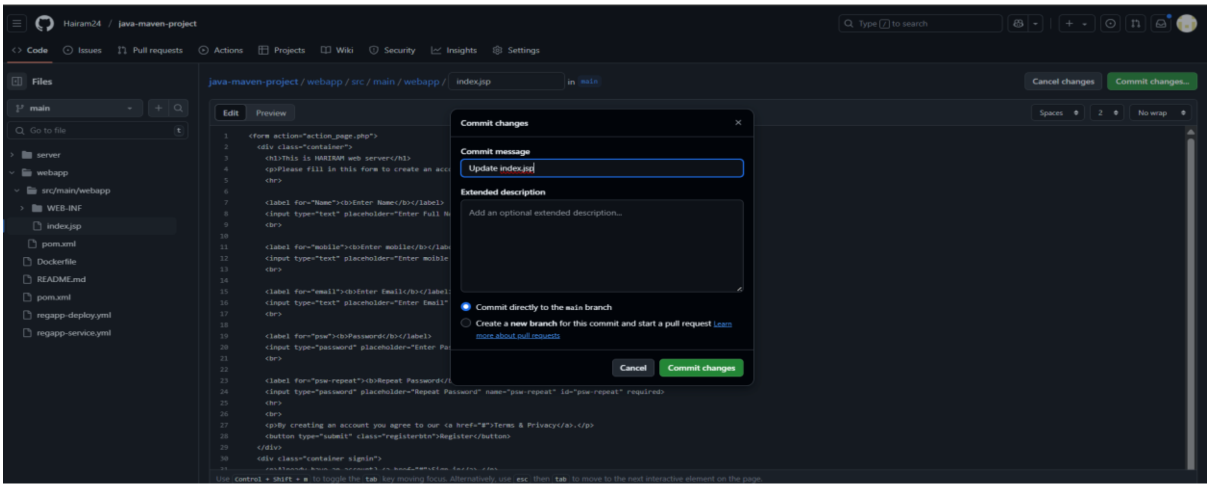


Web page is visible in tomcat.

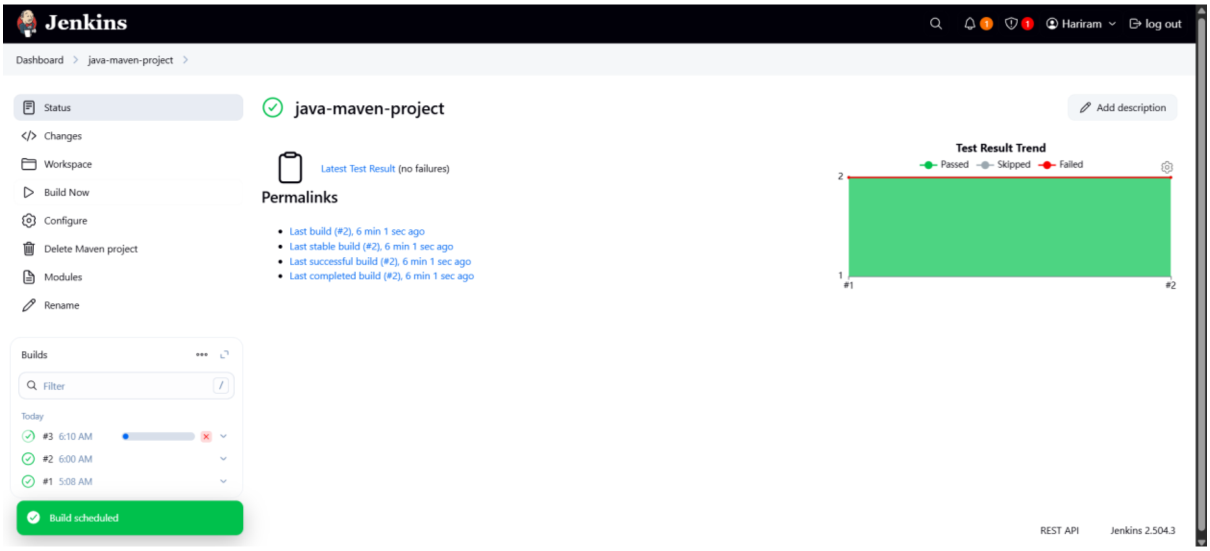




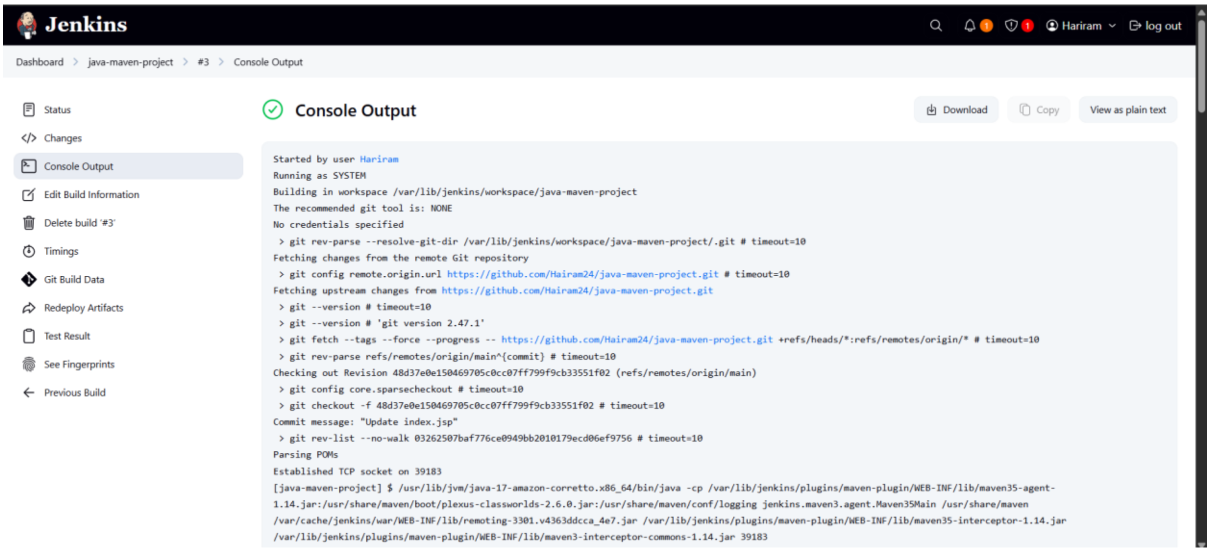
Made change in project index code.



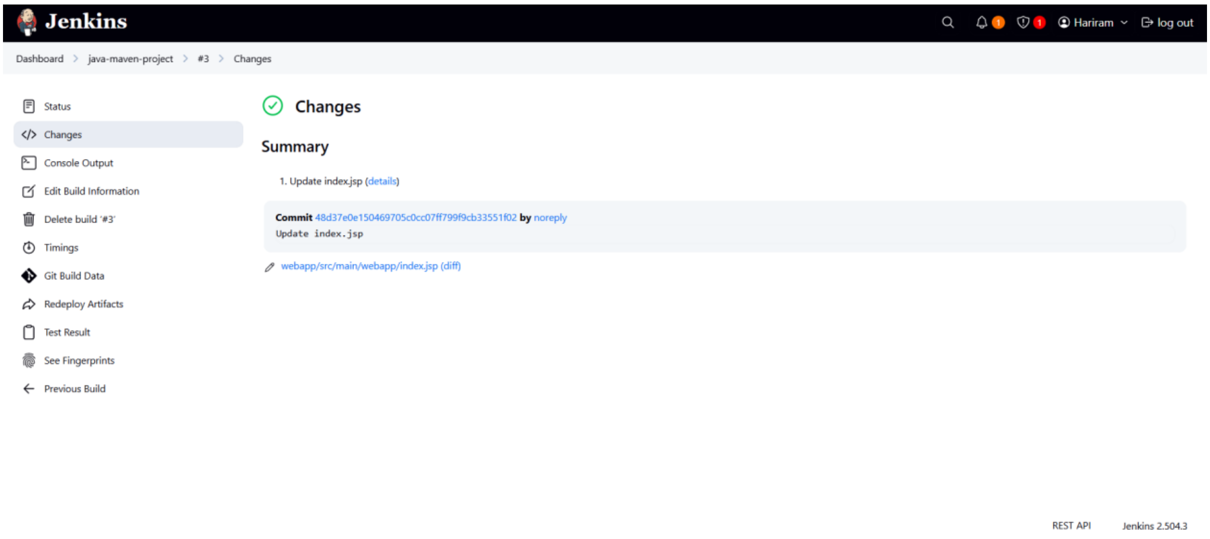
Commit the changes.



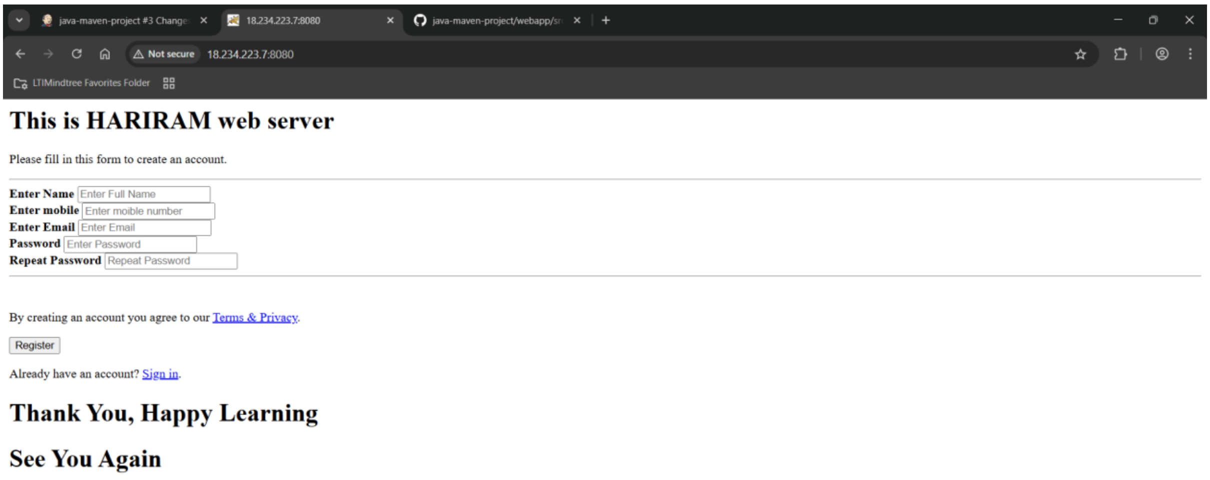
Manual Build.



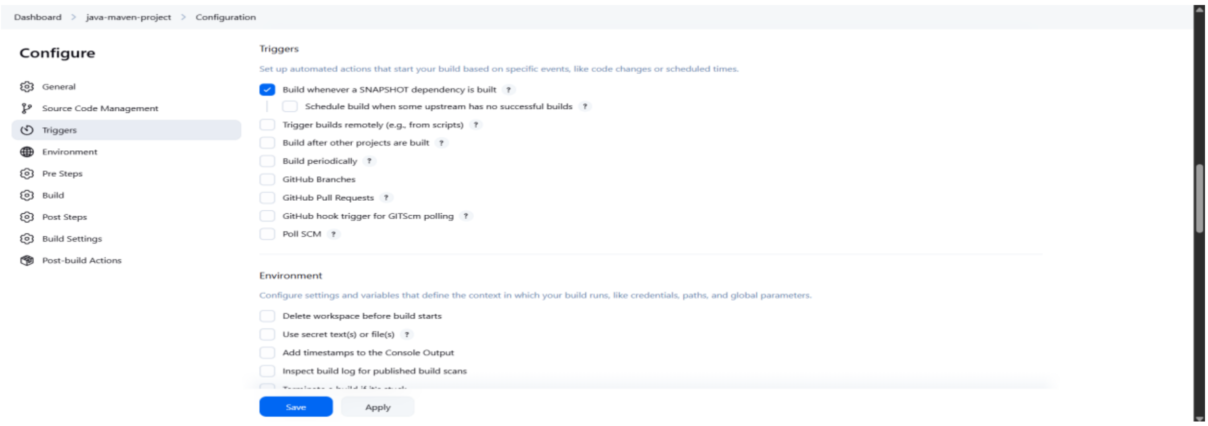
Build successfully.

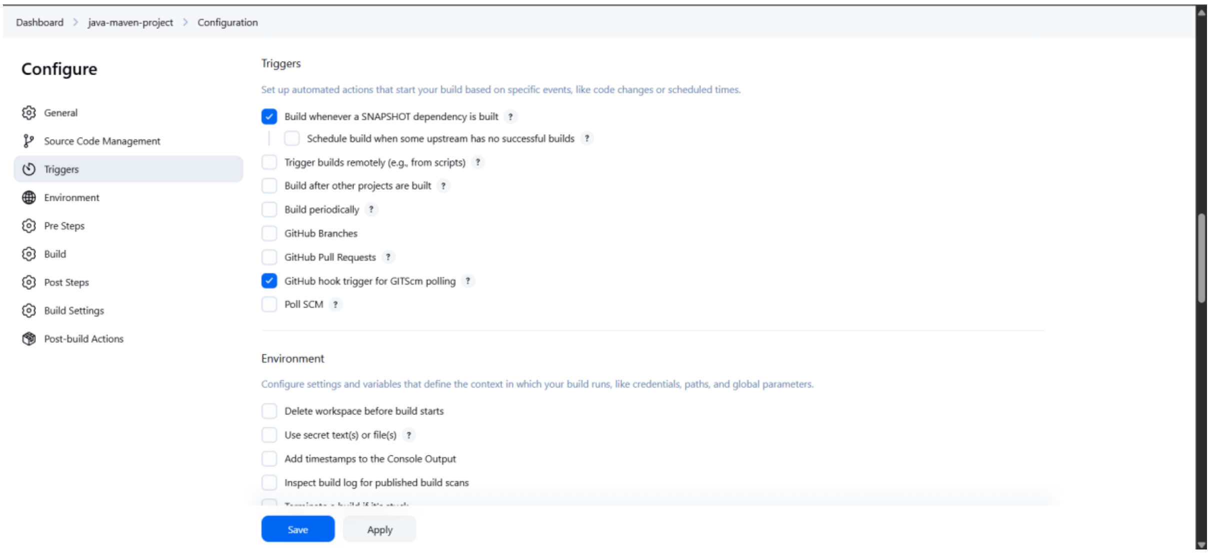


Changes made was updated.

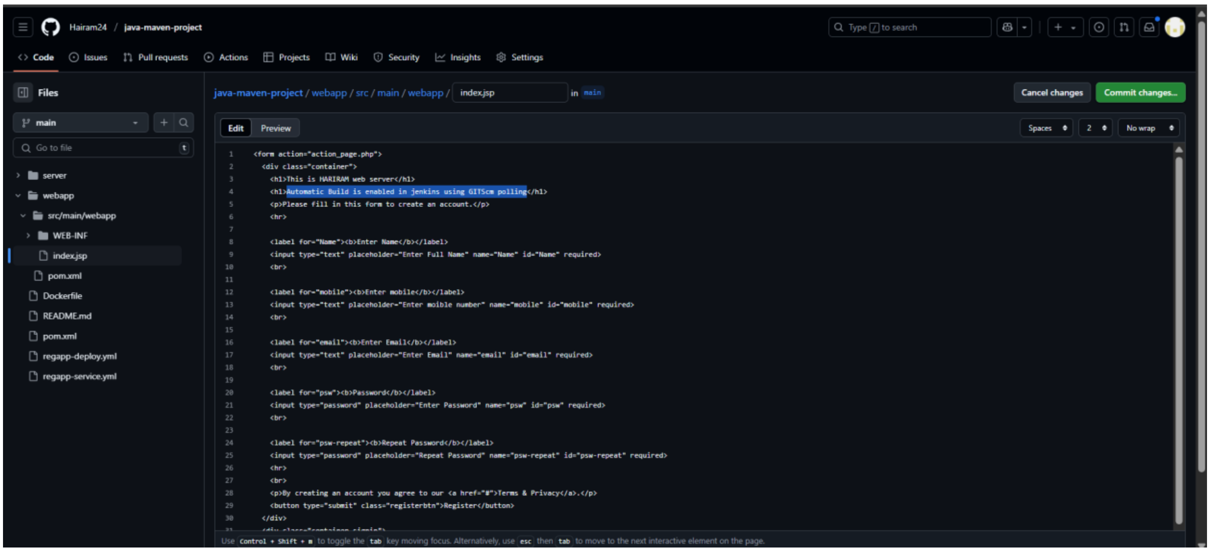


Reflected in the web page.

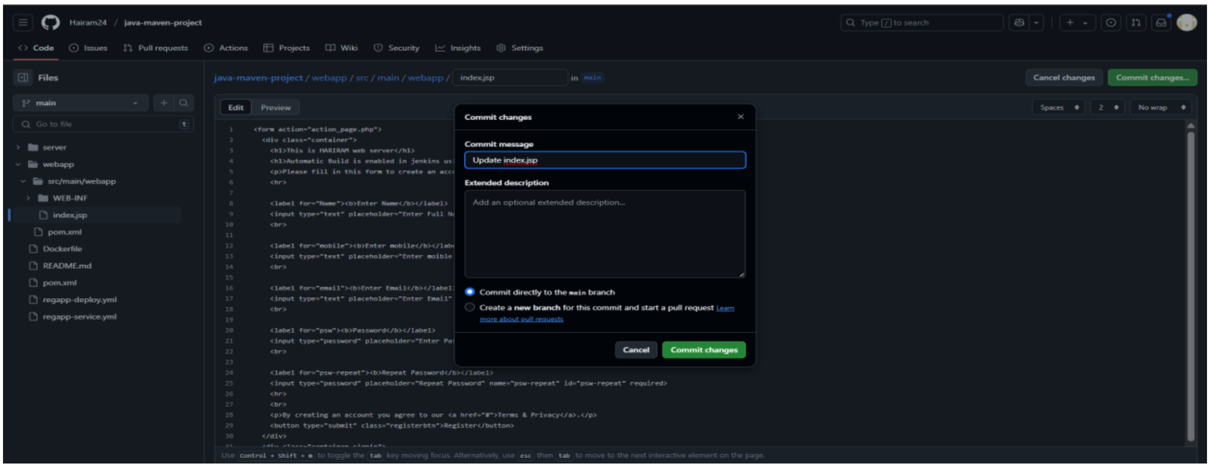




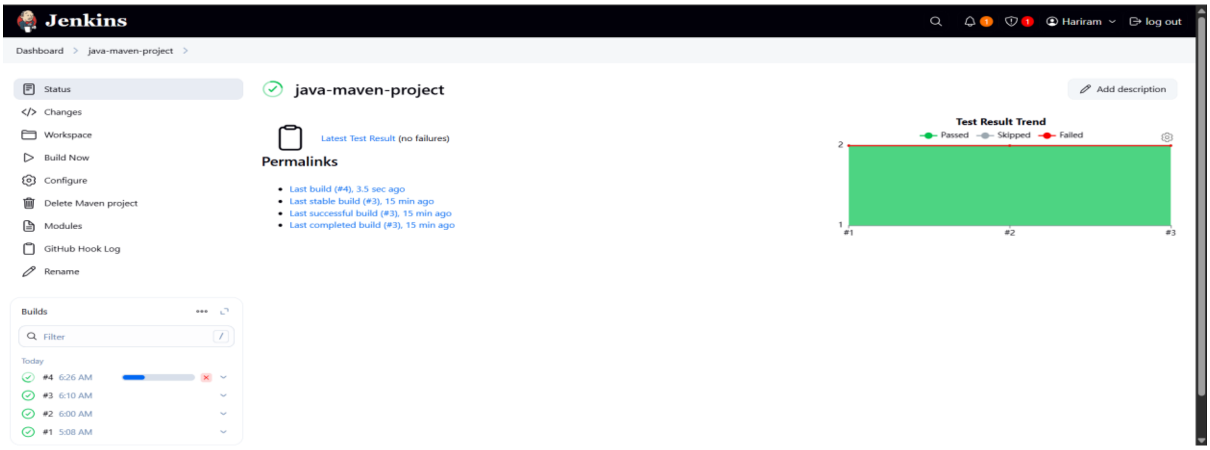
To automate build when code is changed, using GITscm polling.



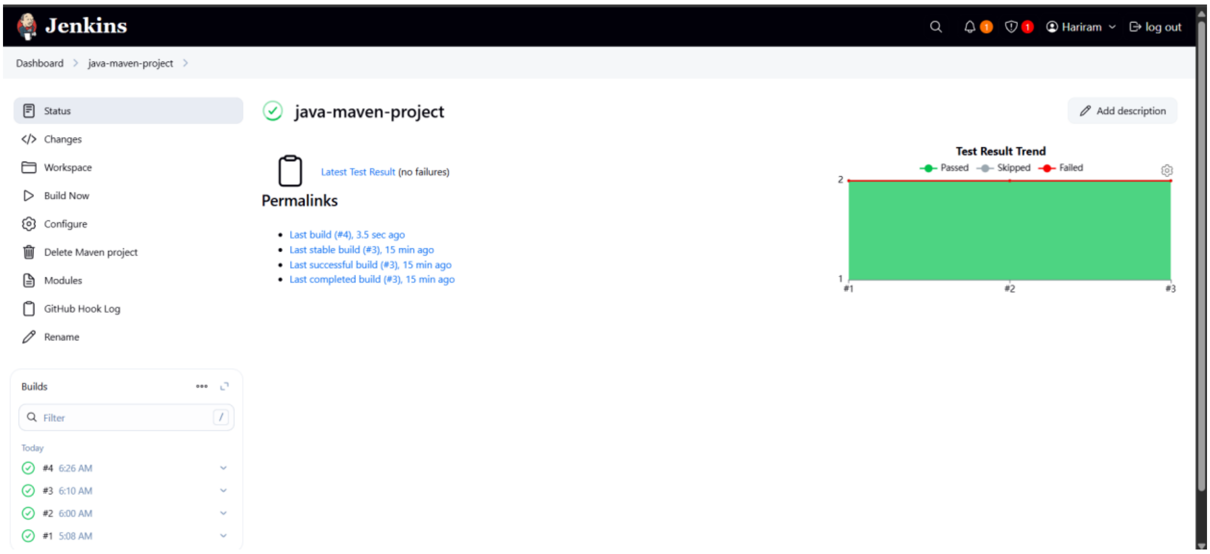
Made change in index code.



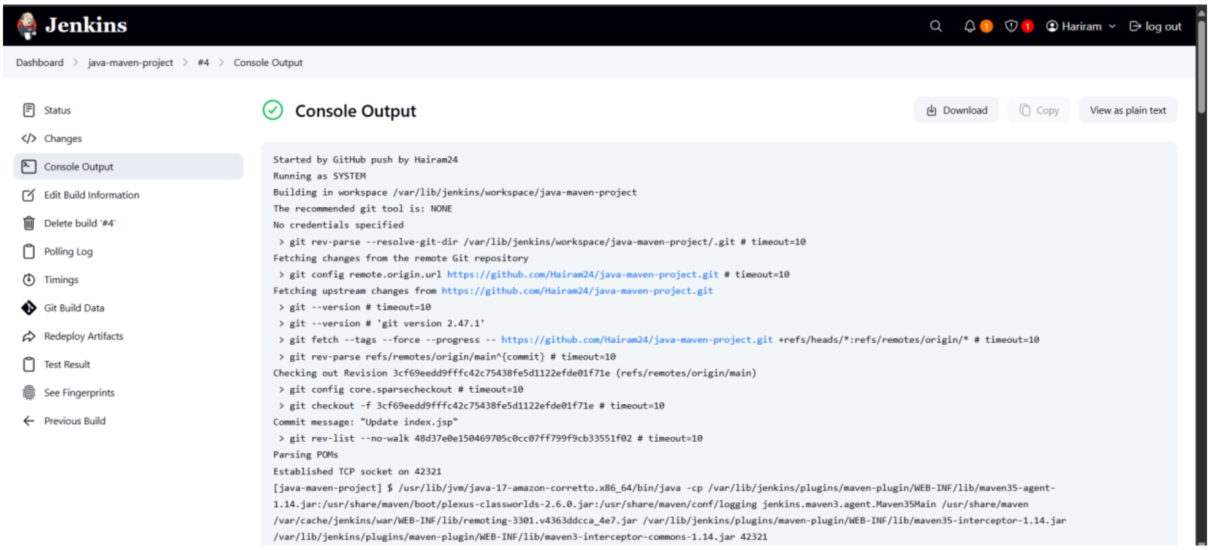
Commit the changes.



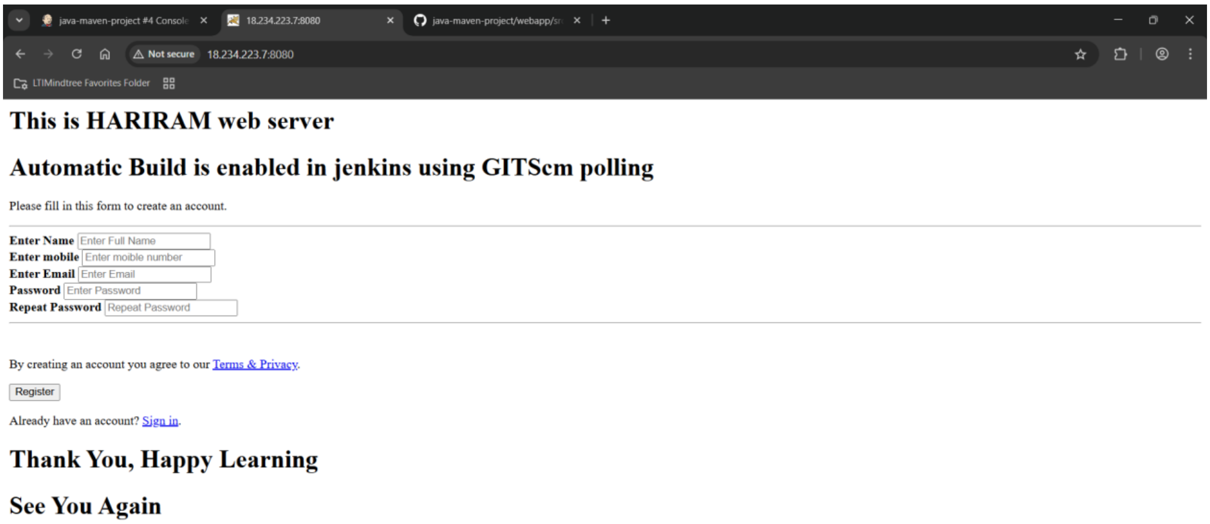
Automatically build is done.



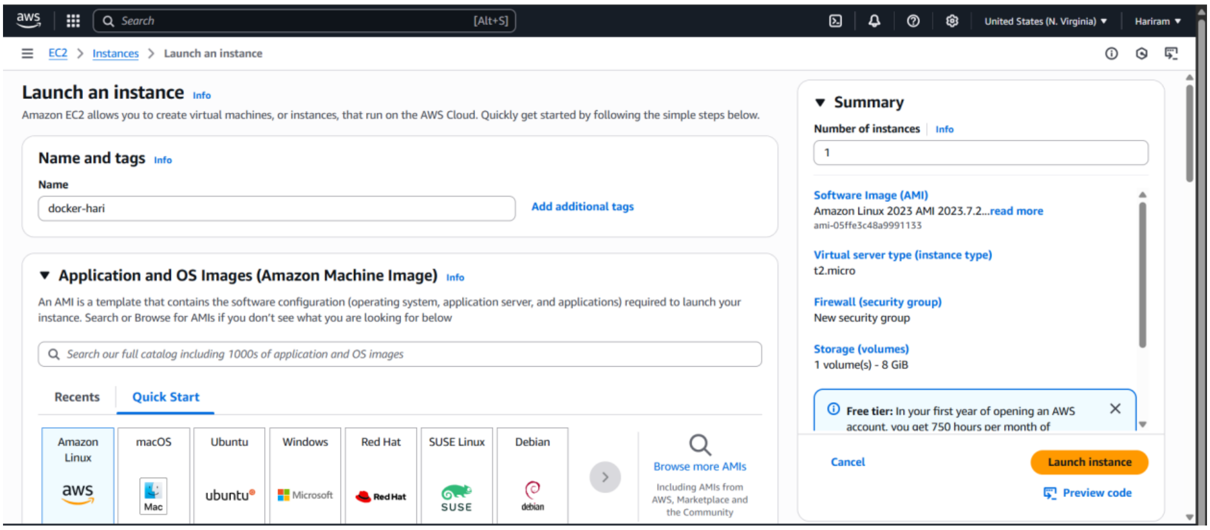
Build successfully.



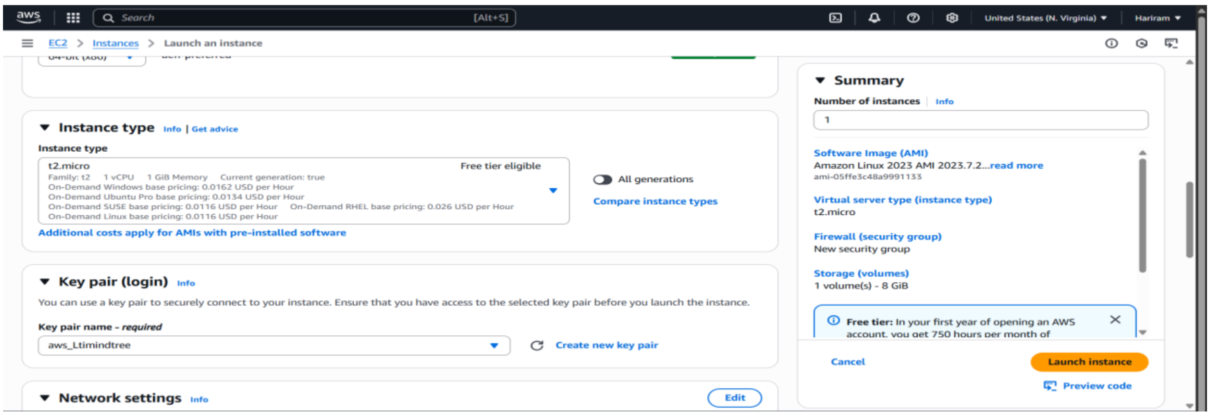
Console output.



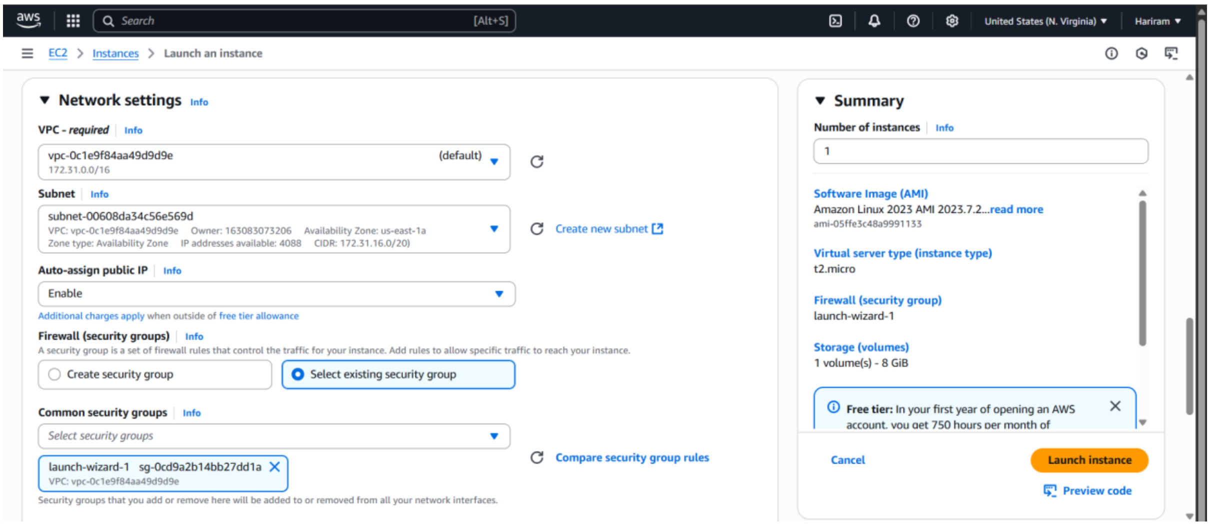
Reflected in we page.



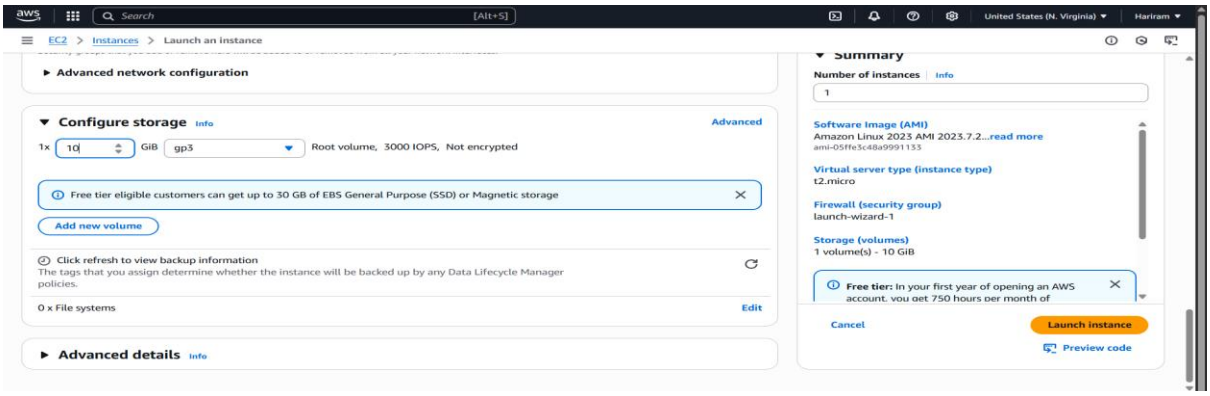
Docker machine is created with amazon linux.



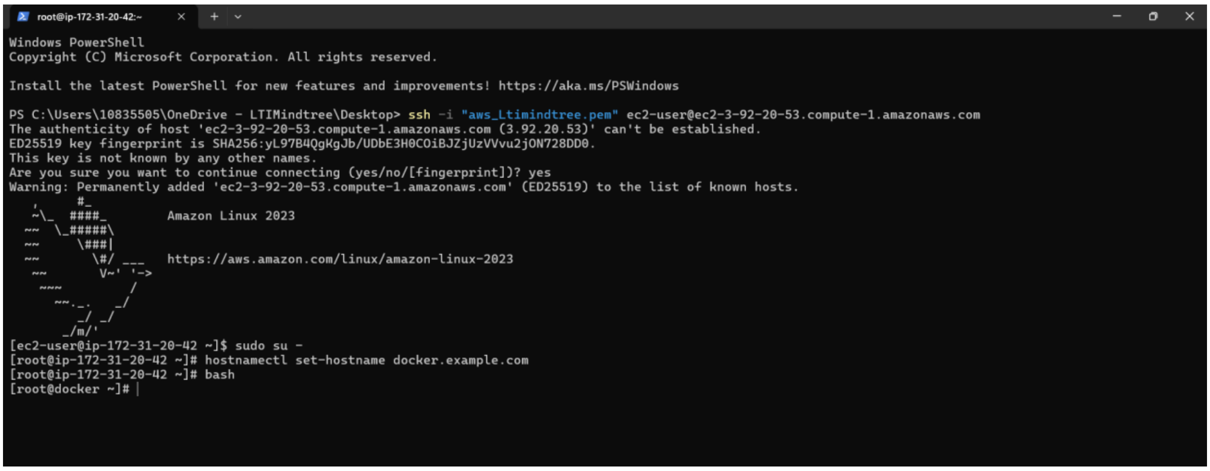
T2.micro is selected. Existing key pair is selected.



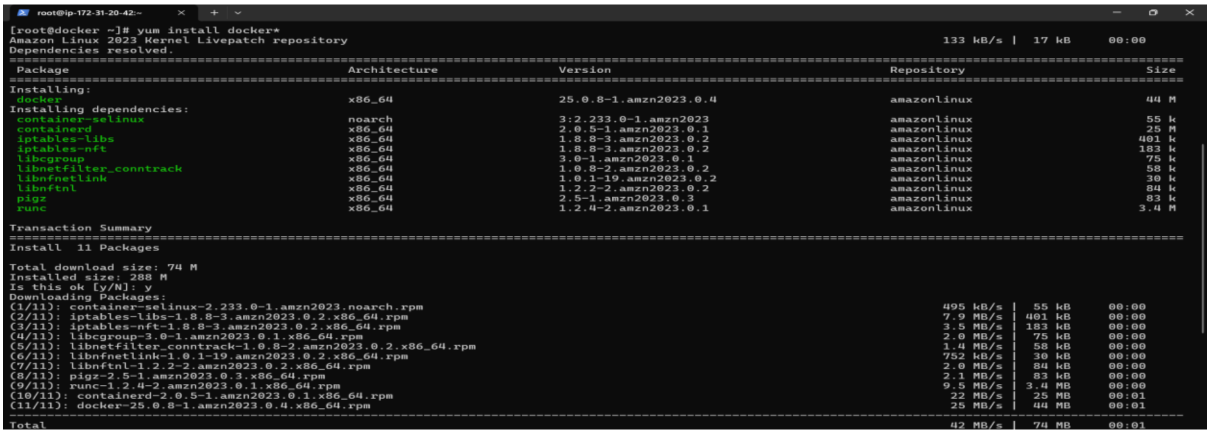
Existing security group is selected.



10 GB storage is selected.



Docker server connected and hostname is changed.

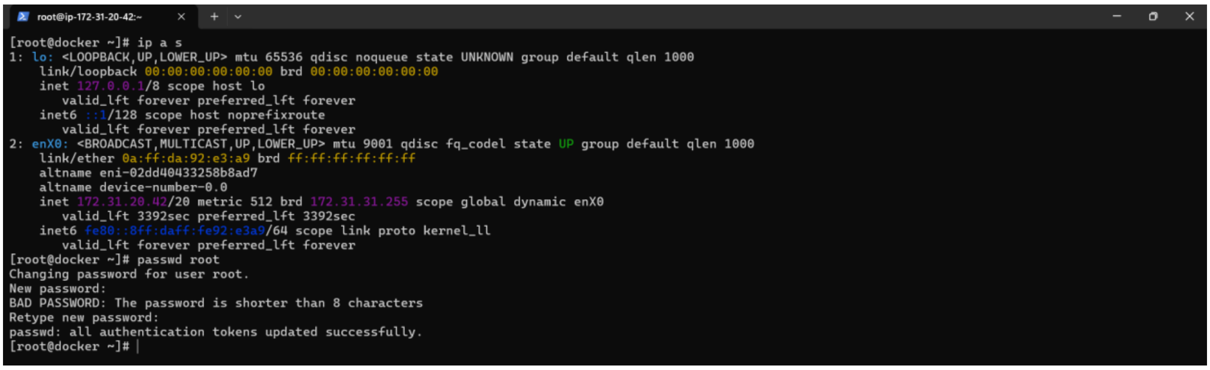


```
[root@docker ~]# yum install docker*
Amazon Linux 2023 Kernel Livepatch repository                       133 kB/s |  17 kB     00:00
Dependencies resolved.
==========================================================================================================
 Package                      Architecture      Version                        Repository            Size
==========================================================================================================
Installing:
 docker                       x86_64            25.0.8-1.amzn2023.0.4          amazonlinux           44 M
Installing dependencies:
 container-selinux            noarch            3:2.233.0-1.amzn2023           amazonlinux           55 k
 containerd                   x86_64            2.0.5-1.amzn2023.0.1           amazonlinux           25 M
 iptables-libs                x86_64            1.8.8-3.amzn2023.0.2           amazonlinux          401 k
 iptables-nft                 x86_64            1.8.8-3.amzn2023.0.2           amazonlinux          183 k
 libcgroup                    x86_64            3.0-1.amzn2023.0.1             amazonlinux           75 k
 libnetfilter_conntrack       x86_64            1.0.8-2.amzn2023.0.2           amazonlinux           58 k
 libnfnetlink                 x86_64            1.0.1-19.amzn2023.0.2          amazonlinux           30 k
 libnftnl                     x86_64            1.2.2-2.amzn2023.0.2           amazonlinux           84 k
 pigz                         x86_64            2.5-1.amzn2023.0.3             amazonlinux           83 k
 runc                         x86_64            1.2.4-2.amzn2023.0.1           amazonlinux          3.4 M

Transaction Summary
==========================================================================================================
Install  11 Packages

Total download size: 74 M
Installed size: 288 M
Is this ok [y/N]: y
Downloading Packages:
(1/11): container-selinux-2.233.0-1.amzn2023.noarch.rpm             495 kB/s |  55 kB     00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm               7.9 MB/s | 401 kB     00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                3.5 MB/s | 183 kB     00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                     2.0 MB/s |  75 kB     00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm      1.4 MB/s |  58 kB     00:00
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm               752 kB/s |  30 kB     00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                    2.0 MB/s |  84 kB     00:00
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm                          2.1 MB/s |  83 kB     00:00
(9/11): runc-1.2.4-2.amzn2023.0.1.x86_64.rpm                        9.5 MB/s | 3.4 MB     00:00
(10/11): containerd-2.0.5-1.amzn2023.0.1.x86_64.rpm                  22 MB/s |  25 MB     00:01
(11/11): docker-25.0.8-1.amzn2023.0.4.x86_64.rpm                     25 MB/s |  44 MB     00:01
----------------------------------------------------------------------------------------------------------
Total                                                                 42 MB/s |  74 MB     00:01
```

Docker package is installed.

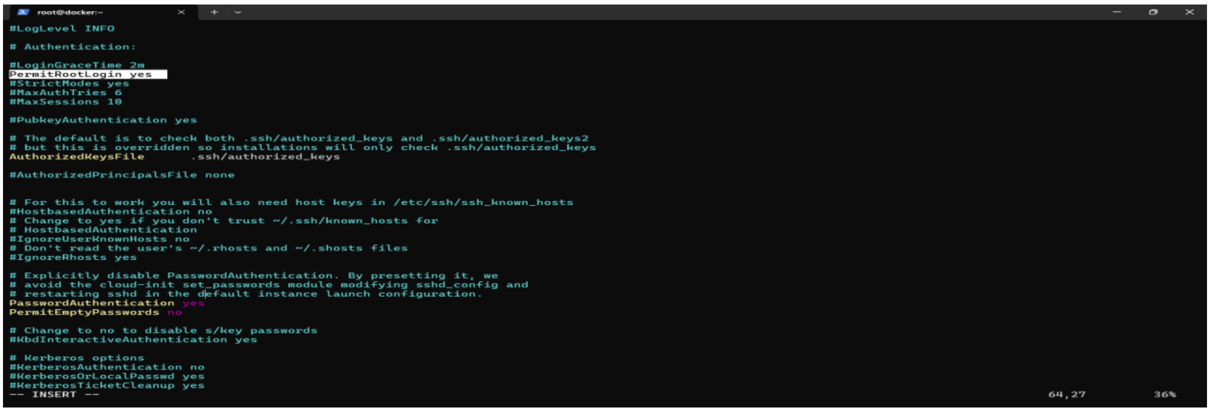
```
[root@jenkins ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:eqIEw/GnAcGb9VR+LbiXCqFIHpQpSkqOQqAb1B2mezI root@jenkins.example.com
The key's randomart image is:
+---[RSA 3072]----+
|oo+o.o...        |
|+=+o+... . .     |
|@.== o. o o .    |
|=Bo*.... o o     |
|o *E+.o S o      |
|   o++ o o       |
|    o o o        |
|   . . o         |
|                 |
+----[SHA256]-----+
[root@jenkins ~]#
```

SSH key is generated.

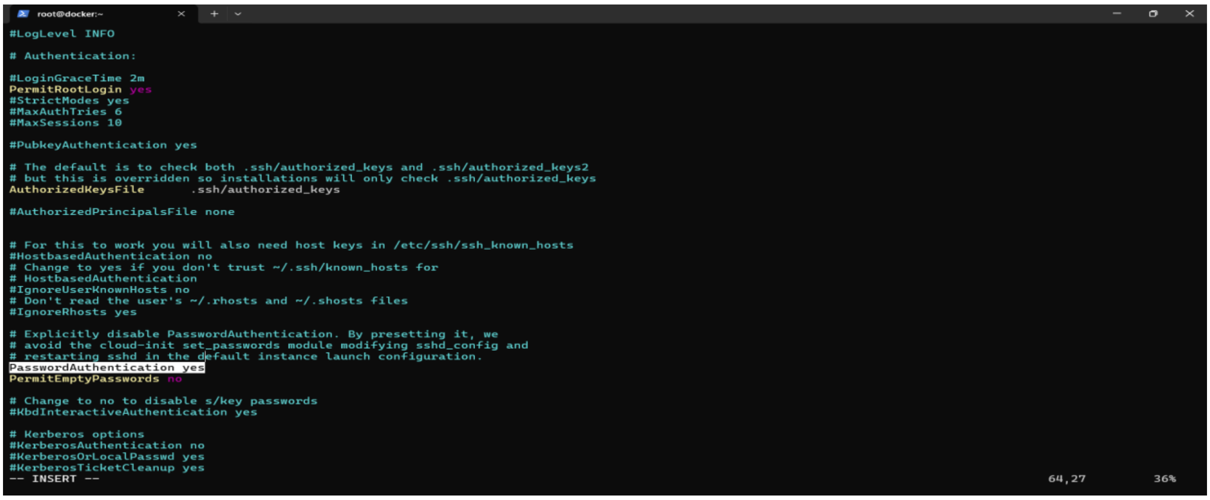


```
[root@docker ~]# ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 0a:ff:da:92:e3:a9 brd ff:ff:ff:ff:ff:ff
    altname eni-02dd40433258b8ad7
    altname device-number-0.0
    inet 172.31.20.42/20 metric 512 brd 172.31.31.255 scope global dynamic enX0
       valid_lft 3392sec preferred_lft 3392sec
    inet6 fe80::8ff:daff:fe92:e3a9/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
[root@docker ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@docker ~]#
```

Root password is set.



```
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
-- INSERT --                                                                        64,27         36%
```



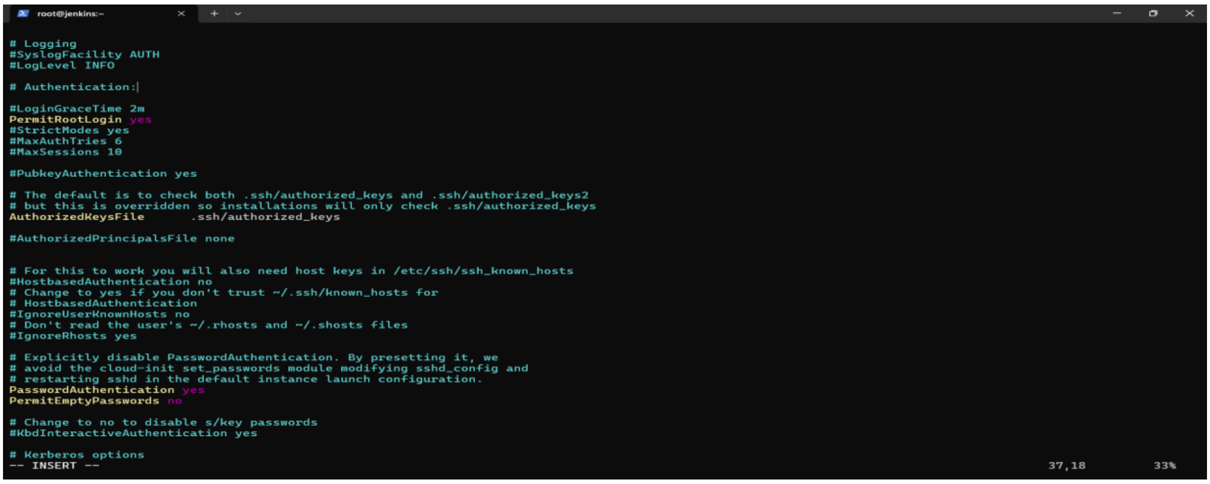
```
root@docker:~
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
-- INSERT --                                                    64,27        36%
```

```
root@docker:~
[root@docker ~]# vim /etc/ssh/sshd_config
[root@docker ~]# systemctl restart sshd
[root@docker ~]# systemctl enable sshd
[root@docker ~]# |
```

Sshd configuration is edited and sshd is restarted.

```
[root@jenkins ~]# ssh-copy-id root@172.31.20.42
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@172.31.20.42's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@172.31.20.42'"
and check to make sure that only the key(s) you wanted were added.

[root@jenkins ~]# |
```

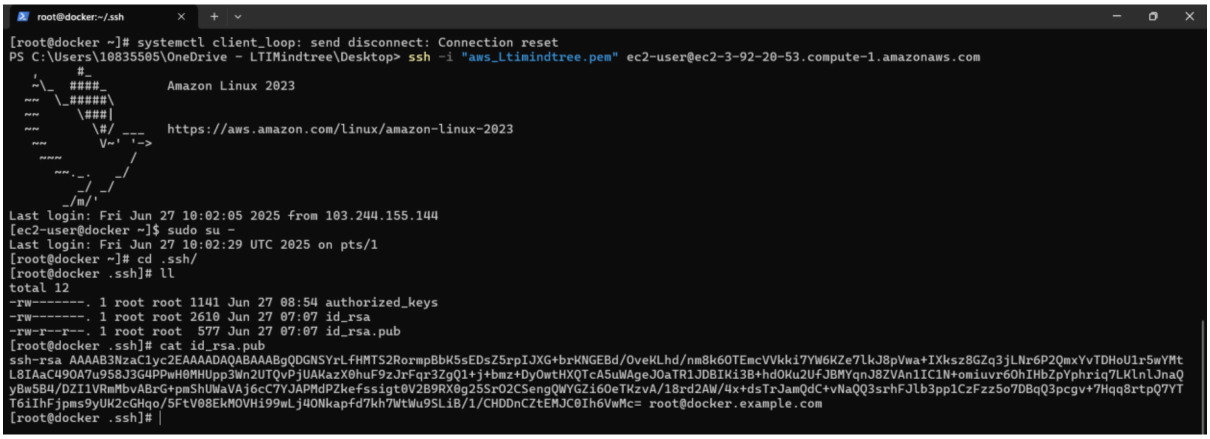
Ssh public key is added to Jenkins



```
root@jenkins:~
# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:|

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes

# Kerberos options
-- INSERT --                                                    37,18        33%
```

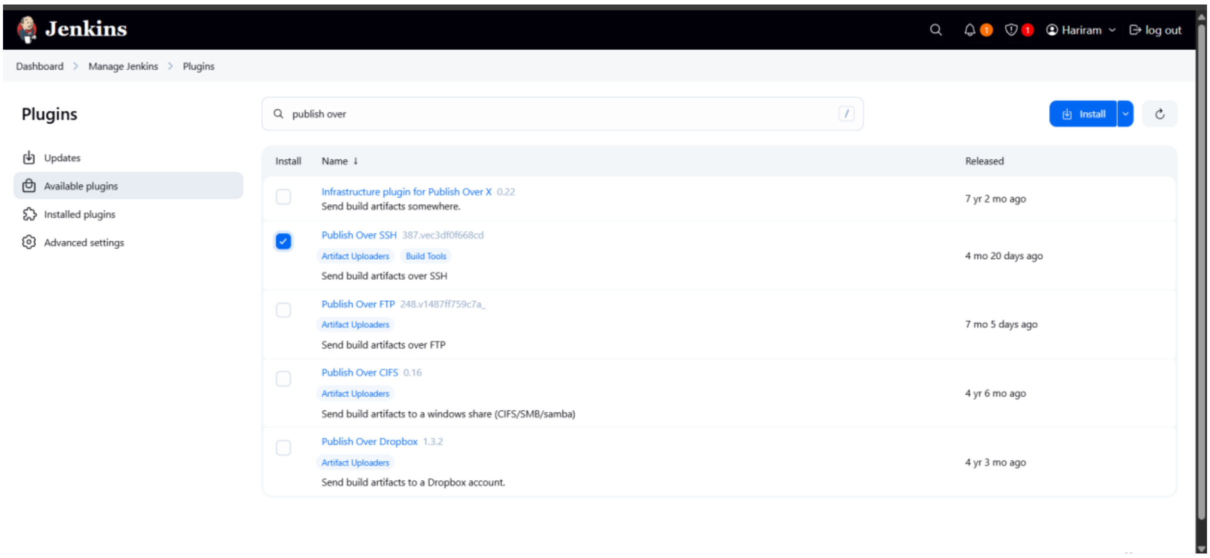
```
root@jenkins:~
[root@jenkins ~]# vim /etc/ssh/sshd_config
[root@jenkins ~]# systemctl start sshd
[root@jenkins ~]# systemctl enable sshd
[root@jenkins ~]# |
```

Jenkins sshd config is edited.

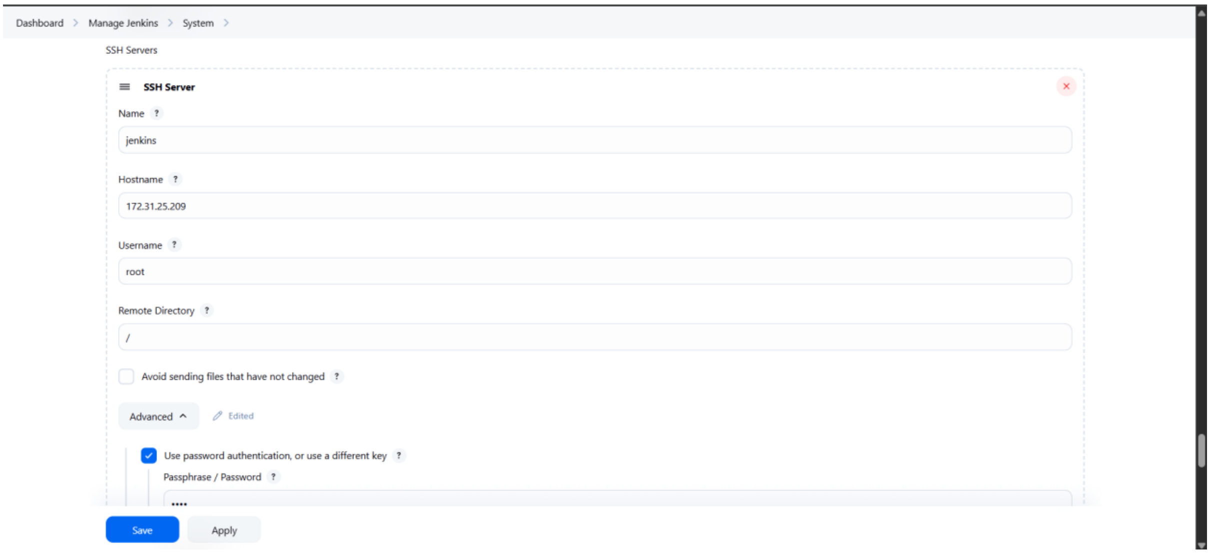


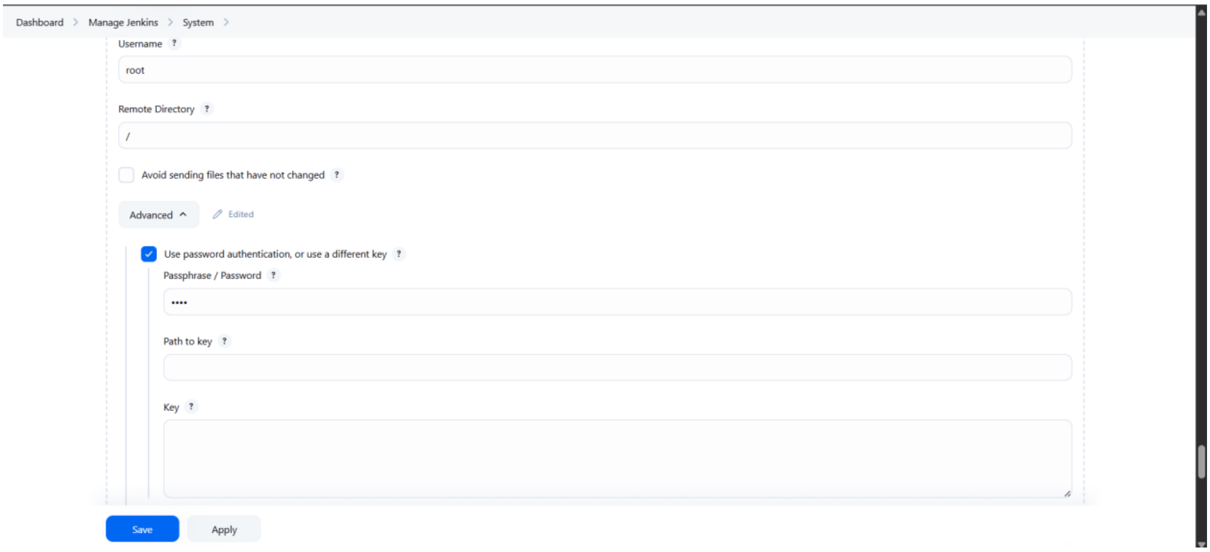
Public is copied.

Jenkins pub key is added to the docker authorized_keys

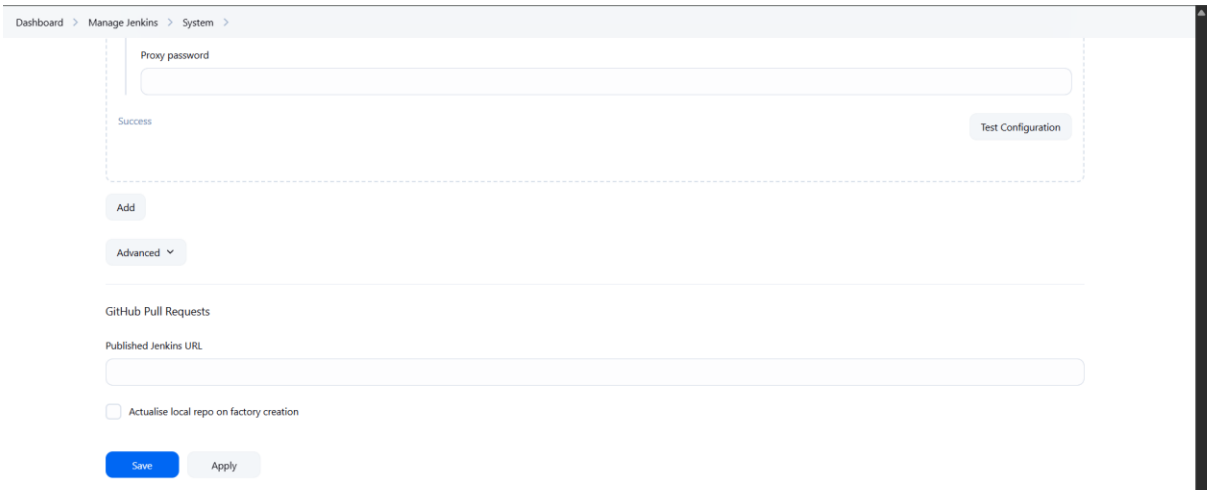


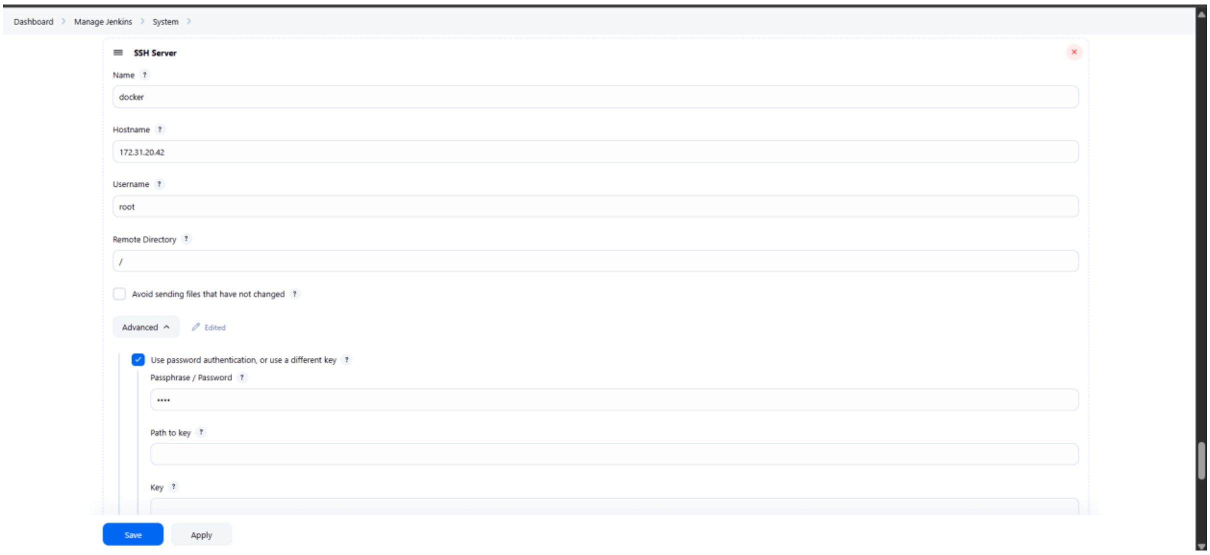
Publish over ssh plugin is installed.

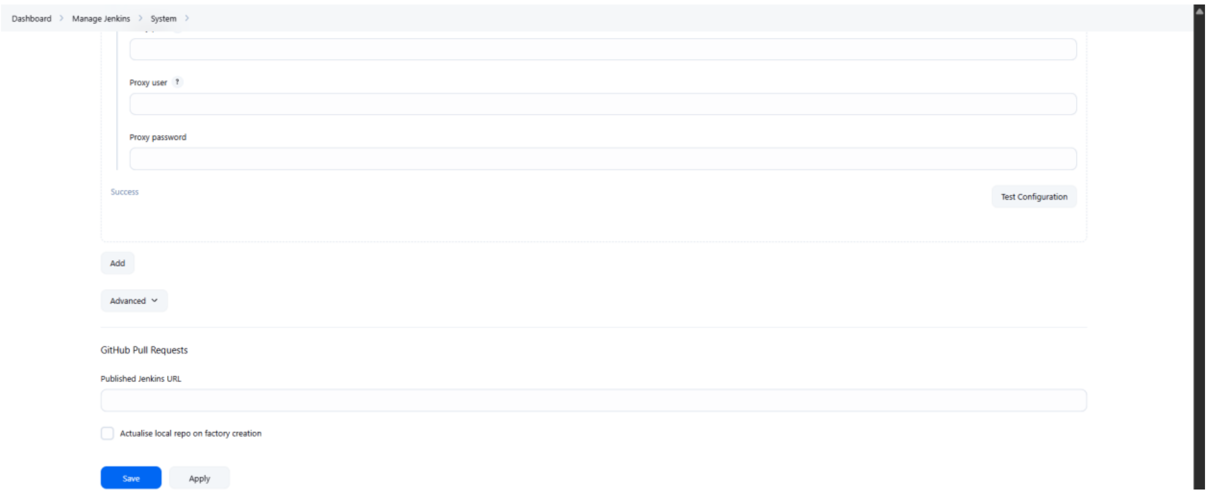




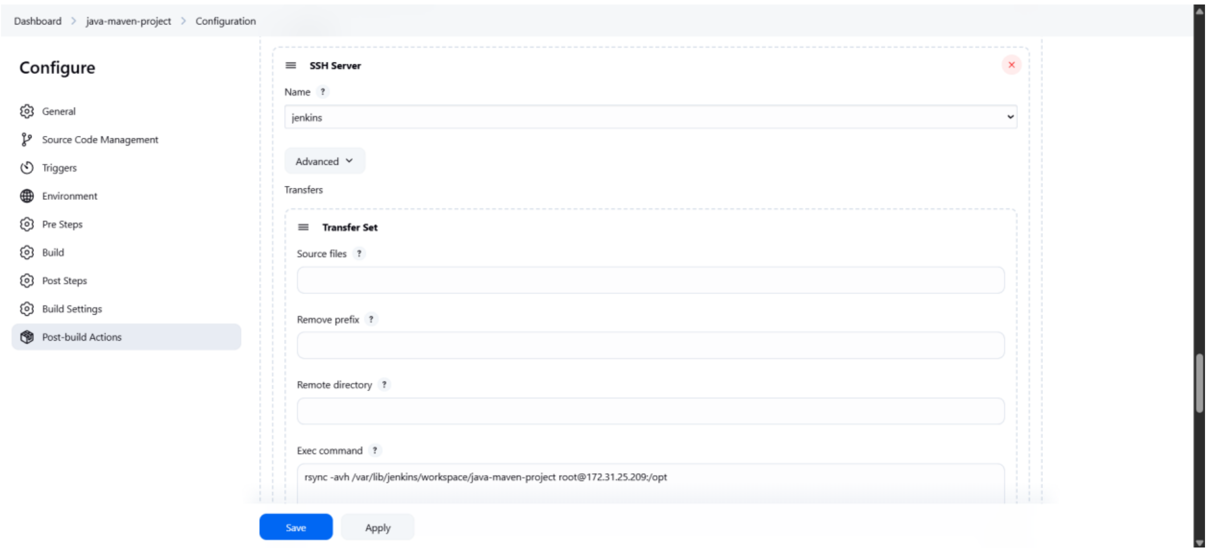
Jenkins ssh server is added.



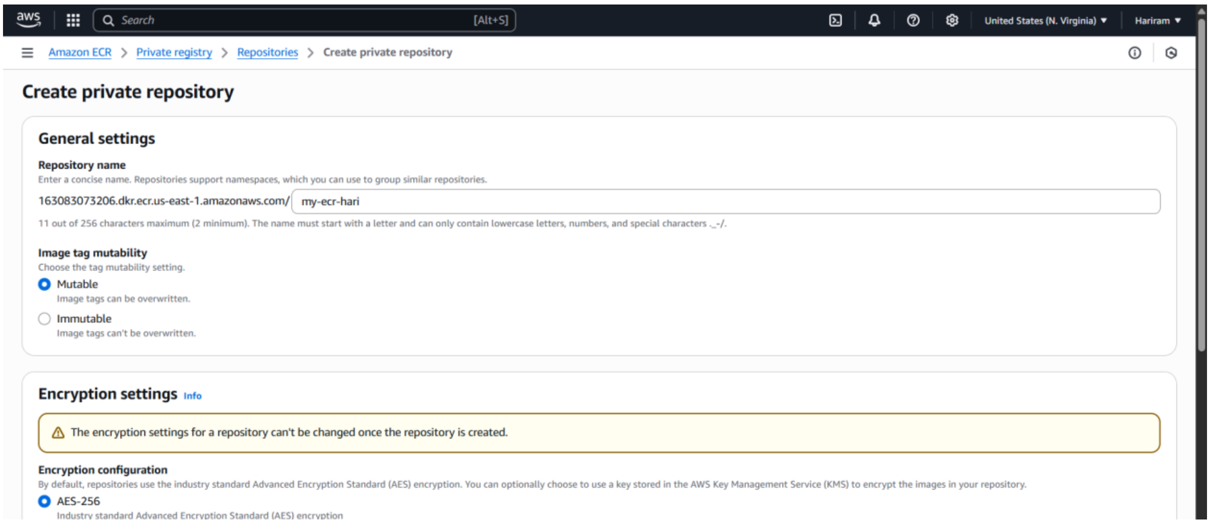




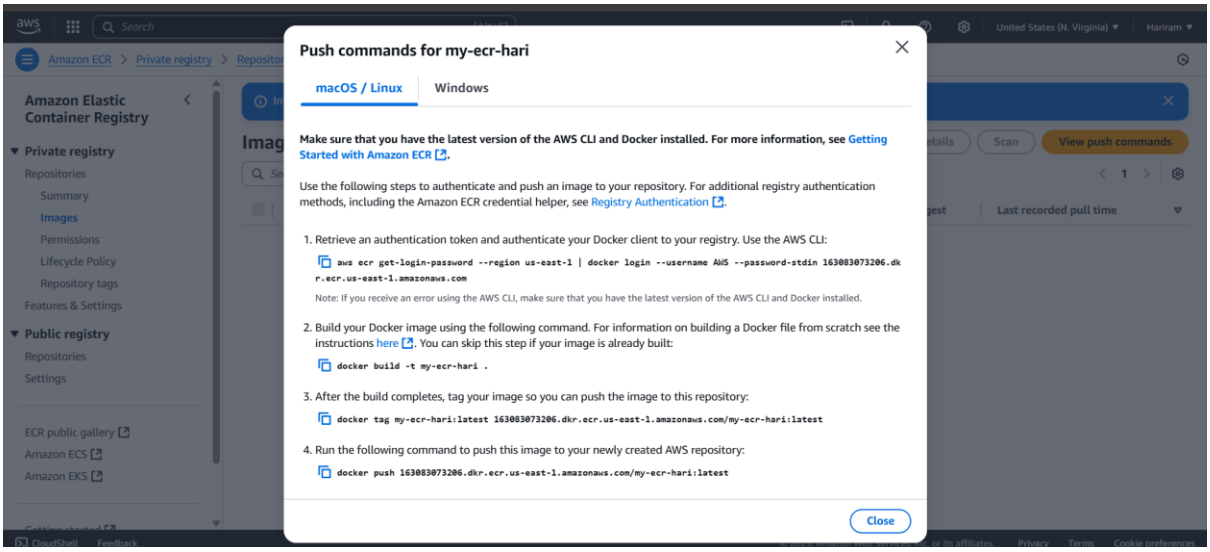
Docker ssh server is added.

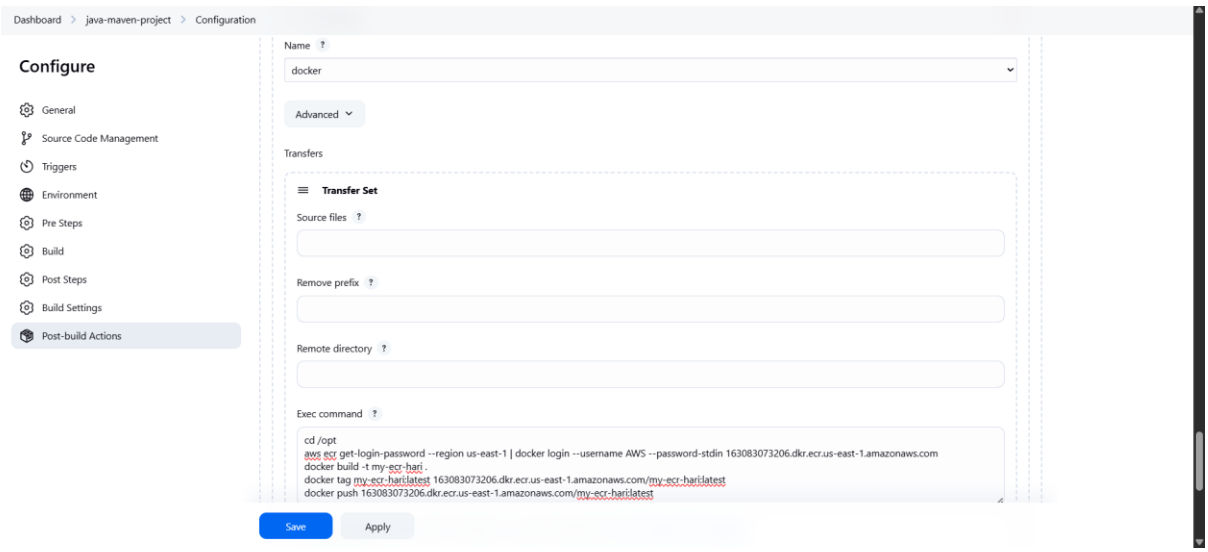


Jenkins post build is added with exec commands.

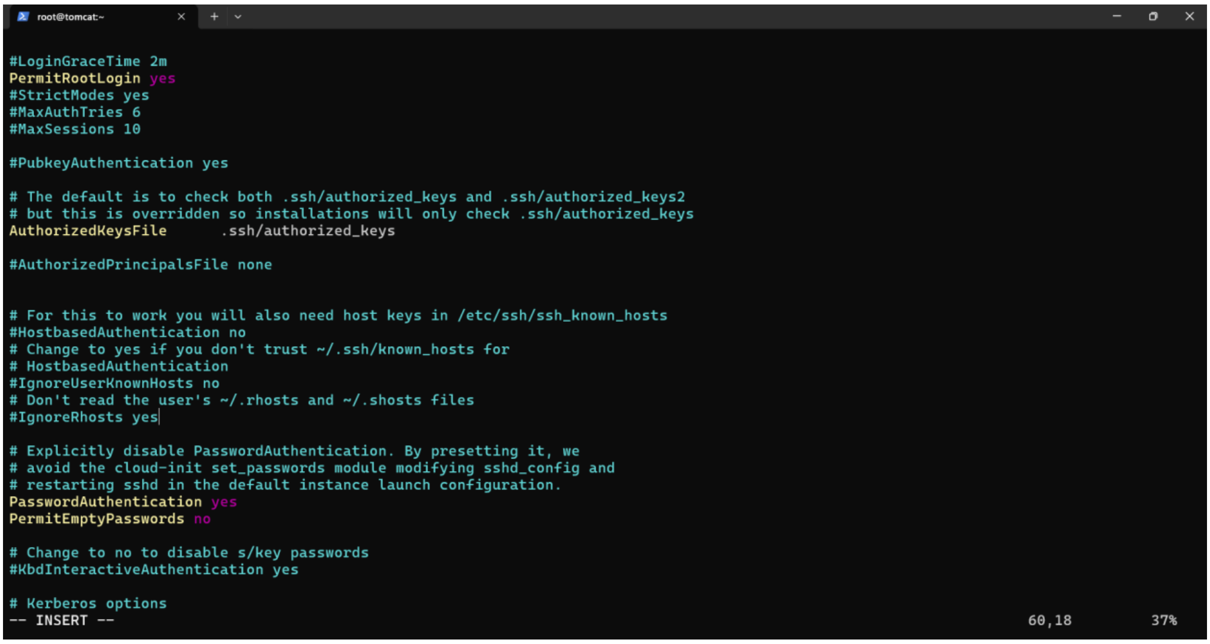

Amazon ECR > Private registry > Repositories > Create private repository
Create private repository
General settings
Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
163083073206.dkr.ecr.us-east-1.amazonaws.com/
my-ecr-hari
Image tag mutability
Choose the tag mutability setting.
Mutable
Image tags can be overwritten.
Immutable
Image tags can't be overwritten.
Encryption settings
The encryption settings for a repository can't be changed once the repository is created.
Encryption configuration
AES-256

Create a ECR.


Push commands for my-ecr-hari
macOS / Linux
Windows
Make sure that you have the latest version of the AWS CLI and Docker installed. For more information, see Getting Started with Amazon ECR.
Use the following steps to authenticate and push an image to your repository. For additional registry authentication methods, including the Amazon ECR credential helper, see Registry Authentication.
1. Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS CLI:
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 163083073206.dkr.ecr.us-east-1.amazonaws.com
Note: If you receive an error using the AWS CLI, make sure that you have the latest version of the AWS CLI and Docker installed.
2. Build your Docker image using the following command. For information on building a Docker file from scratch see the instructions here. You can skip this step if your image is already built:
docker build -t my-ecr-hari .
3. After the build completes, tag your image so you can push the image to this repository:
docker tag my-ecr-hari:latest 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest
4. Run the following command to push this image to your newly created AWS repository:
docker push 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest
Close

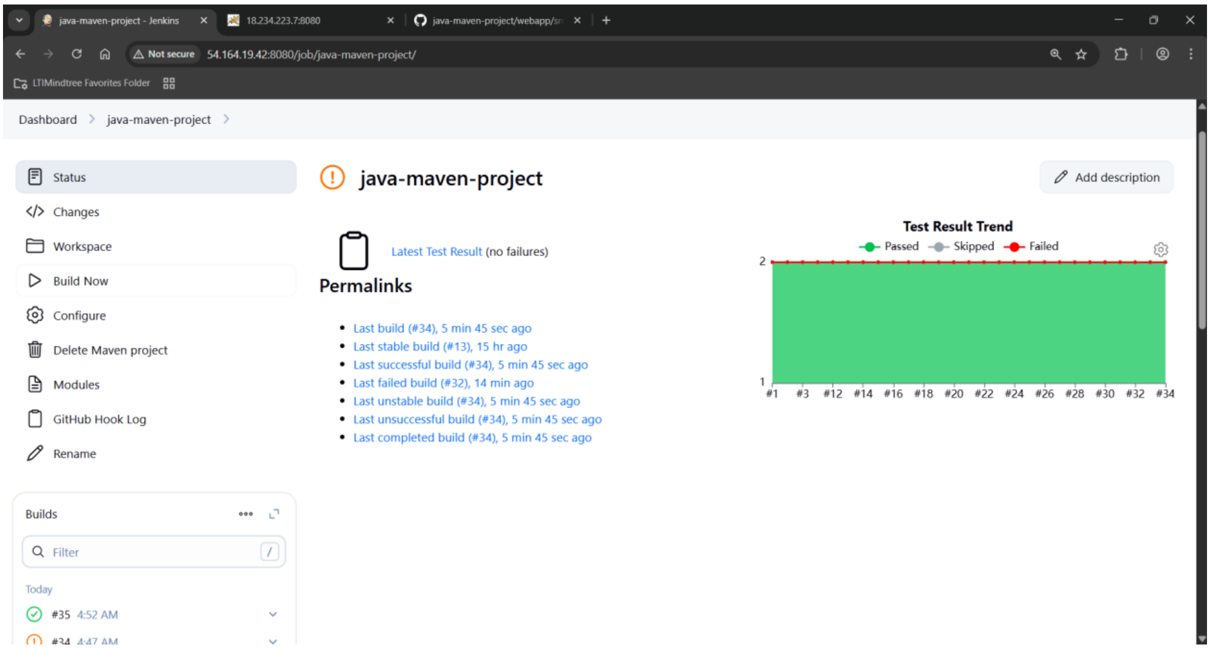

Dashboard > java-maven-project > Configuration
Configure
General
Source Code Management
Triggers
Environment
Pre Steps
Build
Post Steps
Build Settings
Post-build Actions
Name
docker
Advanced
Transfers
Transfer Set
Source files
Remove prefix
Remote directory
Exec command
cd /opt
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 163083073206.dkr.ecr.us-east-1.amazonaws.com
docker build -t my-ecr-hari .
docker tag my-ecr-hari:latest 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest
docker push 163083073206.dkr.ecr.us-east-1.amazonaws.com/my-ecr-hari:latest
Save
Apply

Copy and paste the push commands of ECR to docker ssh server.

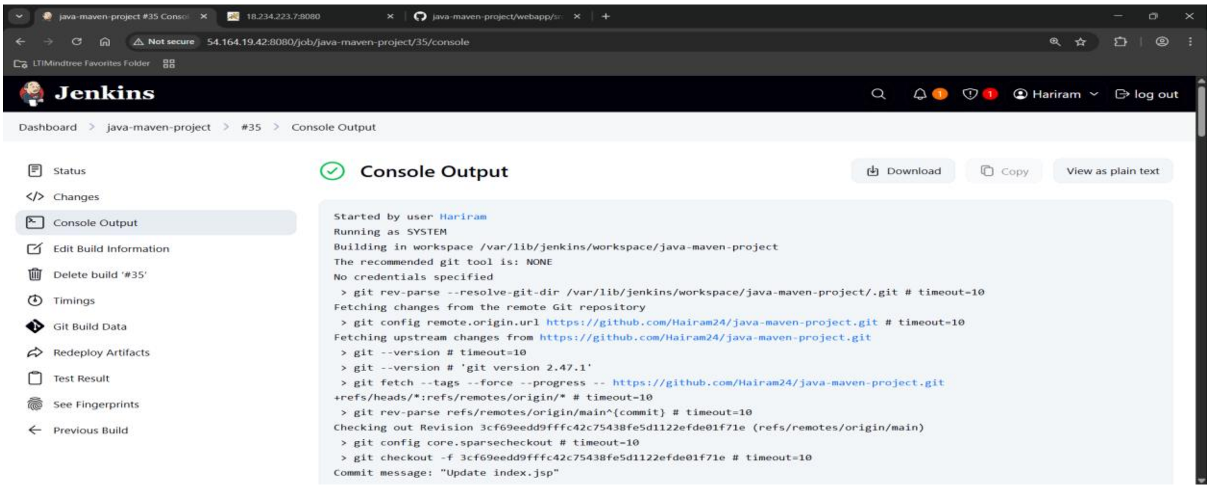


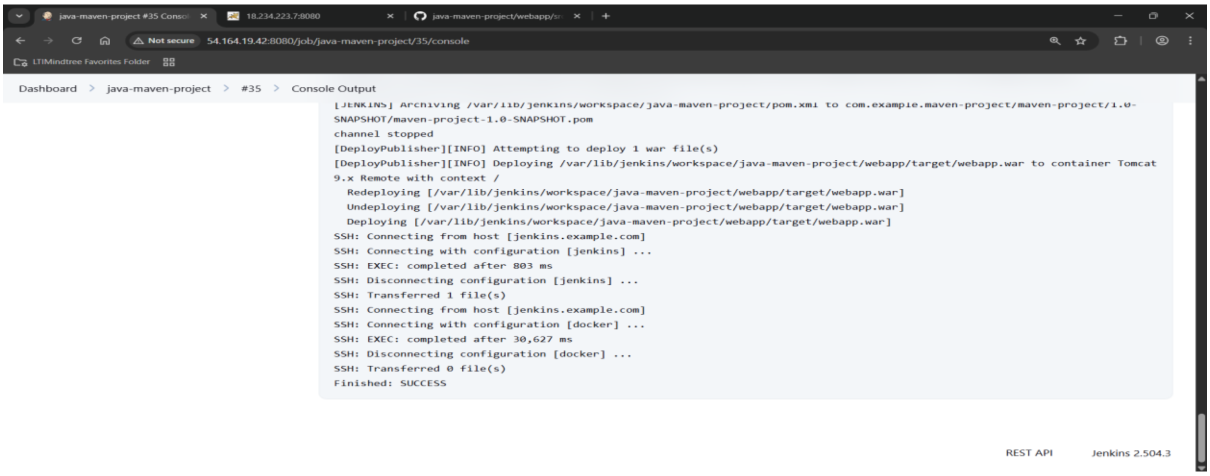
Tomcat sshd is edited and restarted.

Jenkins pub key is added to the tomcat authorized_keys.



Build successful.





Console output build successfully.

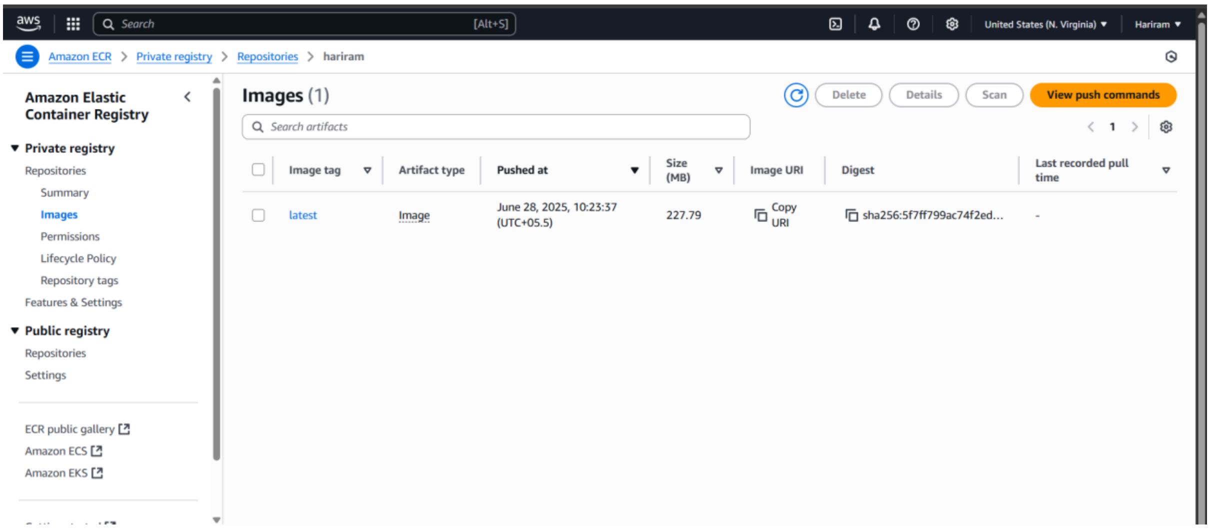
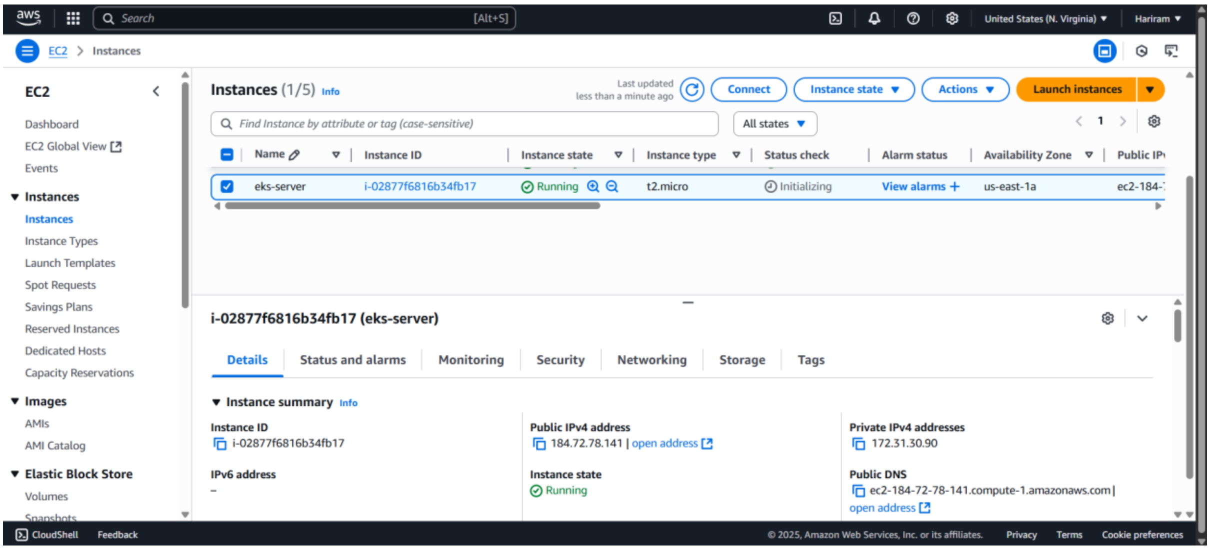
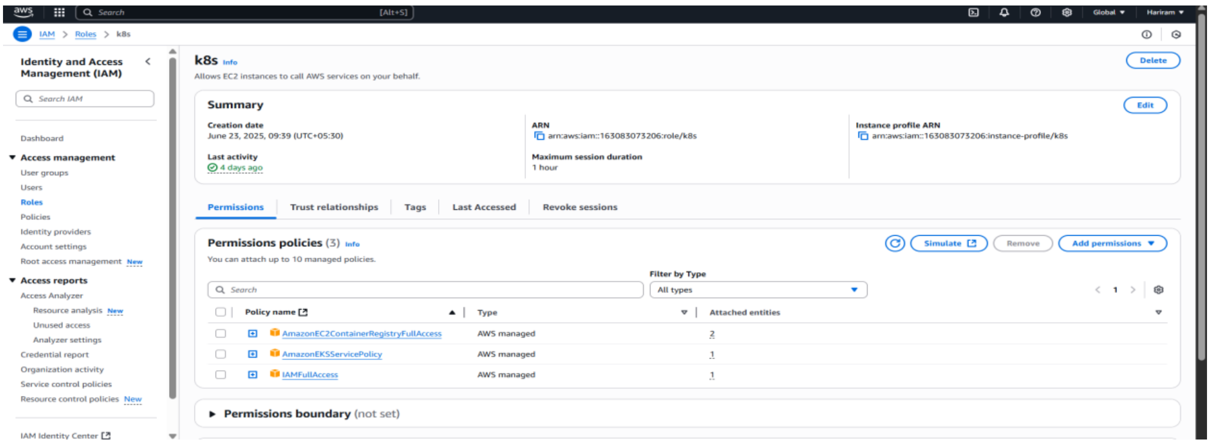


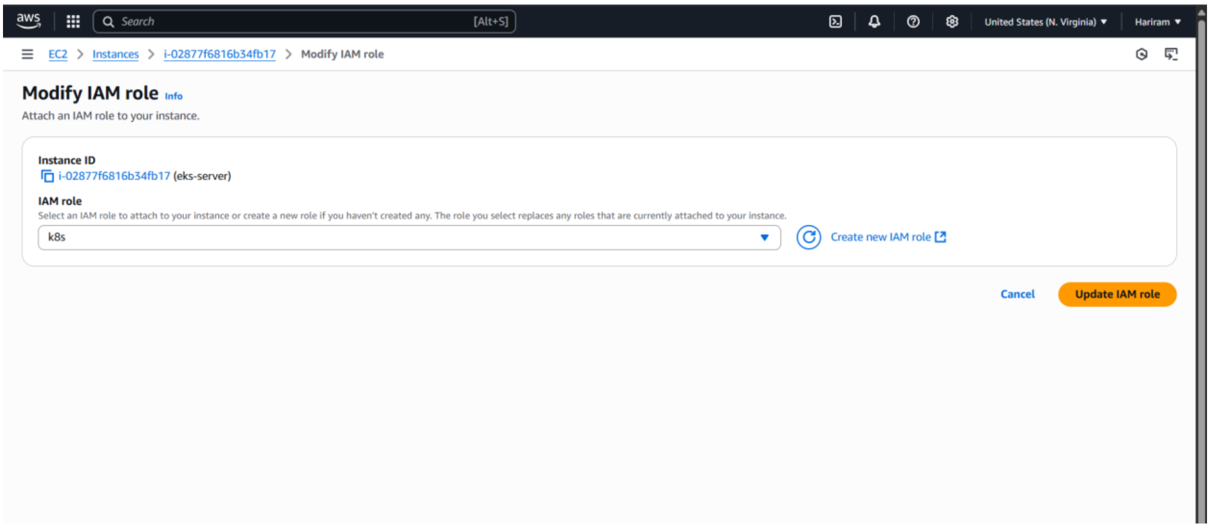
Image is created in ECR.



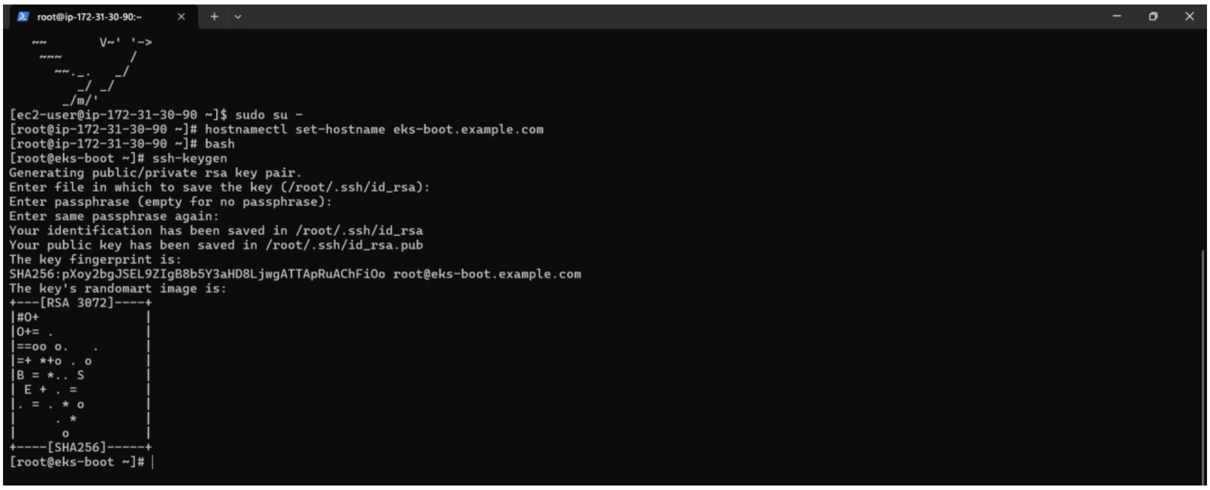
EKS server is created with amazon linux and existing security group k8s-sg with all k8s necessary ports allowed.



Roles is created with necessary permissions.



Roles is added to the eks-server.



EKS-server is started. Hostname is changed. Ssh key is generated.

Eks-server pub key is added to Jenkins.

```
no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ec2-user\" rather than the user \"root\".';echo;sleep 10;
exit 142" ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQDowCDZw93pIkZ5Zyr4FMsOjcgHTQnJdrf8gRXklMU5mZUBlAqJABcR5lsyzodN+N1iAX9Yee4zd3f+0meqX45ZxTXzDRNKGT50m7fbdB5Z4d
w/IUJsRcSOICVPvJQwKqB6Fcrm1Maa3yguwvt2pBWJcqgfvdOr1yisqKOGwztIUoHUDWCrlag8Lh+gXBLZx5FRF4mpglg67WRNP22dzE+hD5TiBe2DKQvSjU38FiiDcFRVm5X+EKa0VnqJBxVjqUspolzK/z
3W25cVLqh0/NrBQrdn+0J8YNPtYzo2Np5ghhKV3ie4z0tQcNY8Aa5ZWjFlptqmcIvg9Nzvdm25QDE/ aws_Ltimindtree

ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQCY5oIJGoTByF5rADOTAHB44WmJmloJctqnyUx2ua6oVcQRxJkVwXkhBbgNt5oVZnfgwR8sXSGsll5yBOOhdLAhdUcTLaWnV5pabz35wakqaT96hEYOLVFS
gjbPVrMcfoW77YZVN79U5zTs4+IxorA/EUB+0T81CeP8rbFnCdLNUGbxQBk1x7i7qAPUzTbU/O2XXhKFhfQ3eRU+ZApipOJzDLTuf6pqaamfWGSpFPL1BY4334qthfZnbA846stRL5zd5BDw5B7U5P27MA5q
h/nJkmSfv8pPHQIOWAXPDKFG2fPUTCgRiJqMz6tuzCBBFJlbfCgfBAn4izGd4ygVn0qXS9axUxpnJbYpoI+y08cNHpvfiuXXLIjbTkf+g7aHODGTeVhQ4aOjWuQJ3Flsbom/yisOChncRFM7hlb1UPnUZtRL
+HJRHczy9cwB4V5vf5n+S7PNEUi8Bj4YneqNgzf5WYfgiz9JGiZfe9Tn/hkkmB6n3LRSJPJ5VDqkWywNWeU= root@jenkins.example.com
~
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~
-- INSERT --                                                                                                                3,578          All
```
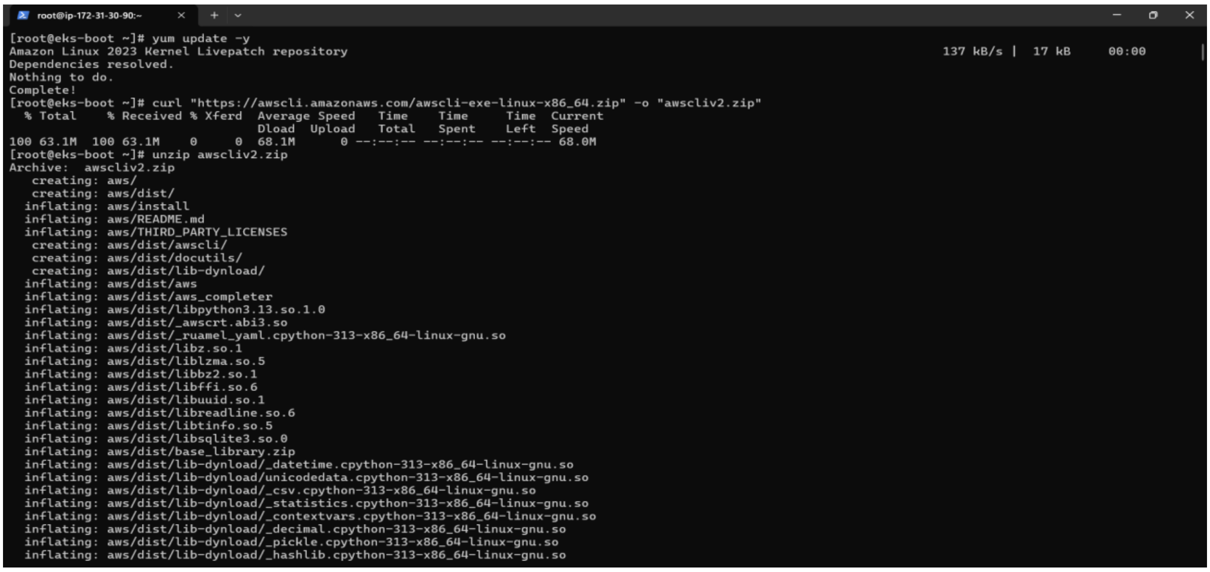

Jenkins pub key is added to eks-server.

```
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile      .ssh/authorized_keys

#AuthorizedPrincipalsFile none


# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no
-- INSERT --                                                                                                                33,10          29%
```

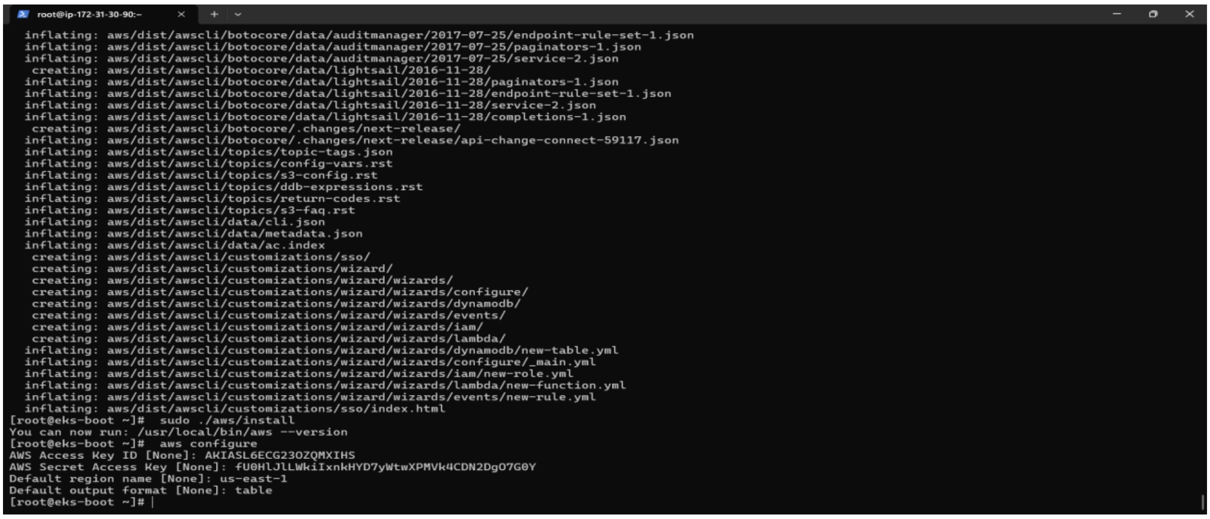
```
[root@eks-boot ~]# vim /etc/ssh/sshd_config
[root@eks-boot ~]# systemctl restart sshd
[root@eks-boot ~]# systemctl enable sshd
```

Eks-server sshd config is edited.



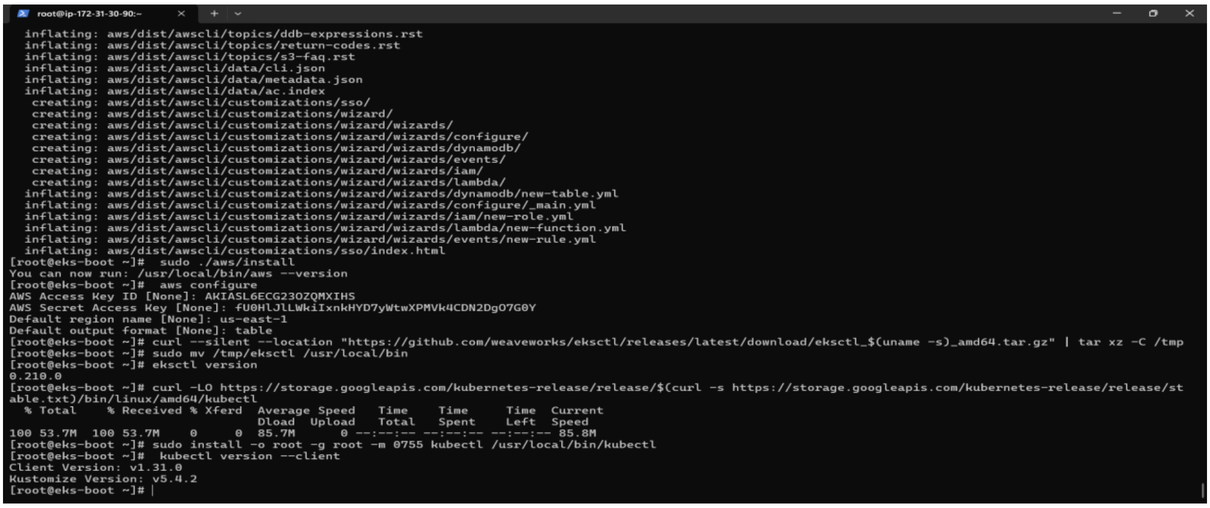
```
[root@eks-boot ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                                                    137 kB/s |  17 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@eks-boot ~]# curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 63.1M  100 63.1M    0     0  68.1M      0 --:--:-- --:--:-- --:--:-- 68.0M
[root@eks-boot ~]# unzip awscliv2.zip
Archive:  awscliv2.zip
   creating: aws/
   creating: aws/dist/
  inflating: aws/install
  inflating: aws/README.md
  inflating: aws/THIRD_PARTY_LICENSES
   creating: aws/dist/awscli/
   creating: aws/dist/docutils/
   creating: aws/dist/lib-dynload/
  inflating: aws/dist/aws
  inflating: aws/dist/aws_completer
  inflating: aws/dist/libpython3.13.so.1.0
  inflating: aws/dist/_awscrt.abi3.so
  inflating: aws/dist/_ruamel_yaml.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/libz.so.1
  inflating: aws/dist/liblzma.so.5
  inflating: aws/dist/libbz2.so.1
  inflating: aws/dist/libffi.so.6
  inflating: aws/dist/libuuid.so.1
  inflating: aws/dist/libreadline.so.6
  inflating: aws/dist/libtinfo.so.5
  inflating: aws/dist/libsqlite3.so.0
  inflating: aws/dist/base_library.zip
  inflating: aws/dist/lib-dynload/_datetime.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/unicodedata.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_csv.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_statistics.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_contextvars.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_decimal.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_pickle.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/lib-dynload/_hashlib.cpython-313-x86_64-linux-gnu.so
```

Amazon Cli package is installed.



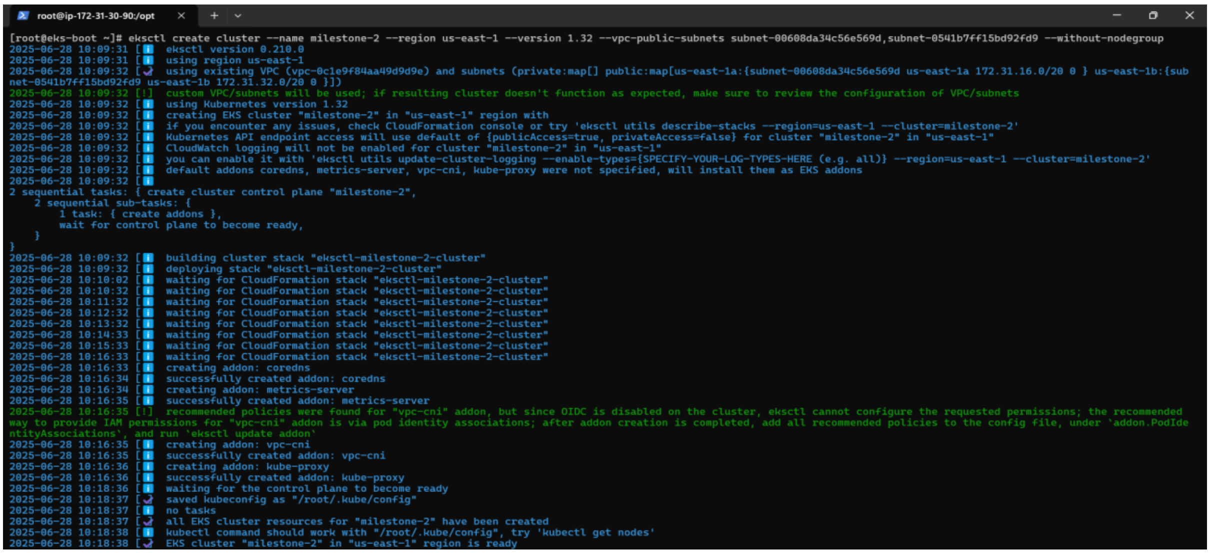
```
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/endpoint-rule-set-1.json
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/paginators-1.json
  inflating: aws/dist/awscli/botocore/data/auditmanager/2017-07-25/service-2.json
   creating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/paginators-1.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/endpoint-rule-set-1.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/service-2.json
  inflating: aws/dist/awscli/botocore/data/lightsail/2016-11-28/completions-1.json
   creating: aws/dist/awscli/botocore/.changes/next-release/
  inflating: aws/dist/awscli/botocore/.changes/next-release/api-change-connect-59117.json
  inflating: aws/dist/awscli/topics/topic-tags.json
  inflating: aws/dist/awscli/topics/config-vars.rst
  inflating: aws/dist/awscli/topics/s3-config.rst
  inflating: aws/dist/awscli/topics/ddb-expressions.rst
  inflating: aws/dist/awscli/topics/return-codes.rst
  inflating: aws/dist/awscli/topics/s3-faq.rst
  inflating: aws/dist/awscli/data/cli.json
  inflating: aws/dist/awscli/data/metadata.json
  inflating: aws/dist/awscli/data/ac.index
   creating: aws/dist/awscli/customizations/sso/
   creating: aws/dist/awscli/customizations/wizard/
   creating: aws/dist/awscli/customizations/wizard/wizards/
   creating: aws/dist/awscli/customizations/wizard/wizards/configure/
   creating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/
   creating: aws/dist/awscli/customizations/wizard/wizards/events/
   creating: aws/dist/awscli/customizations/wizard/wizards/iam/
   creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
  inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
  inflating: aws/dist/awscli/customizations/sso/index.html
[root@eks-boot ~]#  sudo ./aws/install
You can now run: /usr/local/bin/aws --version
[root@eks-boot ~]#  aws configure
AWS Access Key ID [None]: AKIASL6ECG23OZQMXIHS
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: table
[root@eks-boot ~]# |
```

Aws configuration credentials are given from the IAM user.

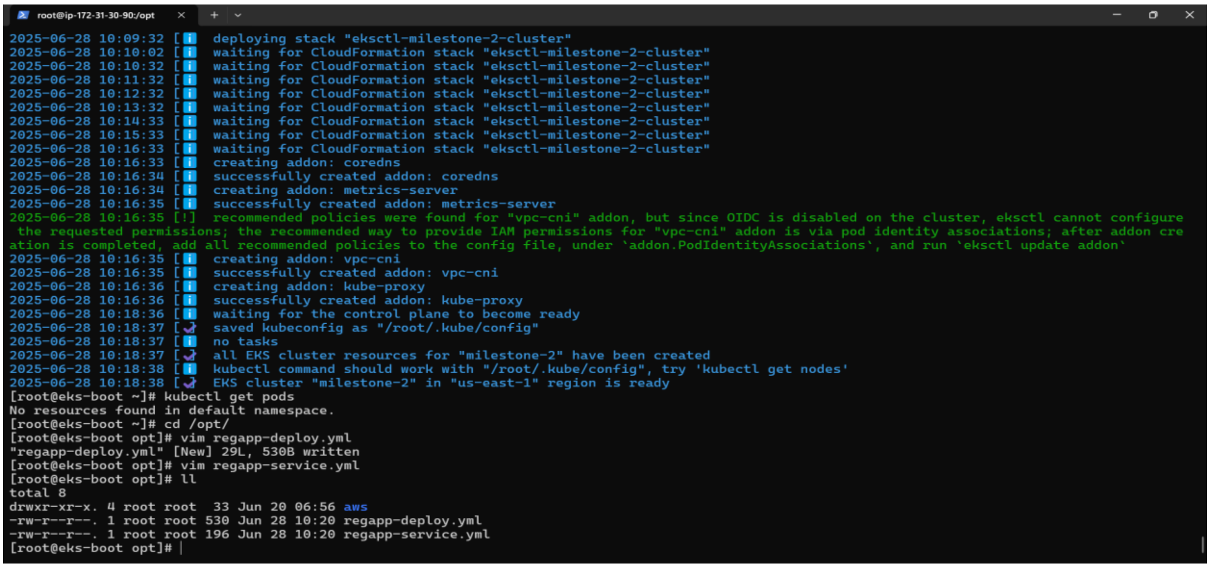


```
  inflating: aws/dist/awscli/topics/ddb-expressions.rst
  inflating: aws/dist/awscli/topics/return-codes.rst
  inflating: aws/dist/awscli/topics/s3-faq.rst
  inflating: aws/dist/awscli/data/cli.json
  inflating: aws/dist/awscli/data/metadata.json
  inflating: aws/dist/awscli/data/ac.index
   creating: aws/dist/awscli/customizations/sso/
   creating: aws/dist/awscli/customizations/wizard/
   creating: aws/dist/awscli/customizations/wizard/wizards/
   creating: aws/dist/awscli/customizations/wizard/wizards/configure/
   creating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/
   creating: aws/dist/awscli/customizations/wizard/wizards/events/
   creating: aws/dist/awscli/customizations/wizard/wizards/iam/
   creating: aws/dist/awscli/customizations/wizard/wizards/lambda/
  inflating: aws/dist/awscli/customizations/wizard/wizards/dynamodb/new-table.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/configure/_main.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/iam/new-role.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/lambda/new-function.yml
  inflating: aws/dist/awscli/customizations/wizard/wizards/events/new-rule.yml
  inflating: aws/dist/awscli/customizations/sso/index.html
[root@eks-boot ~]#  sudo ./aws/install
You can now run: /usr/local/bin/aws --version
[root@eks-boot ~]#  aws configure
AWS Access Key ID [None]: AKIASL6ECG23OZQMXIHS
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: table
[root@eks-boot ~]# curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp
[root@eks-boot ~]# sudo mv /tmp/eksctl /usr/local/bin
[root@eks-boot ~]# eksctl version
0.210.0
[root@eks-boot ~]# curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/st
able.txt)/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 53.7M  100 53.7M    0     0  85.7M      0 --:--:-- --:--:-- --:--:-- 85.8M
[root@eks-boot ~]# sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
[root@eks-boot ~]#  kubectl version --client
Client Version: v1.31.0
Kustomize Version: v5.4.2
[root@eks-boot ~]# |
```

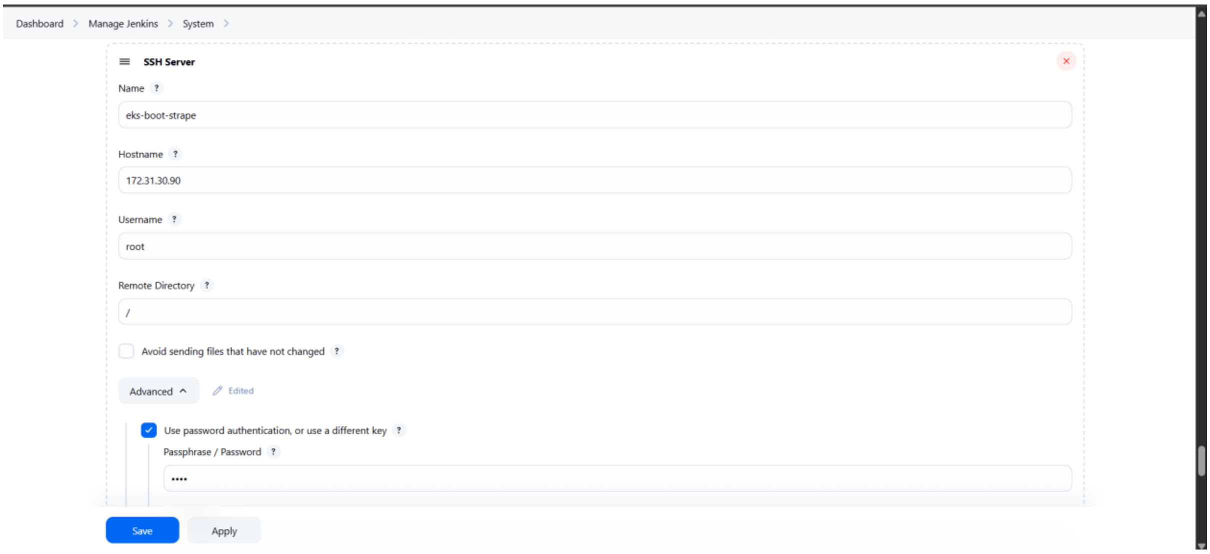
Kubernetes package is installed.

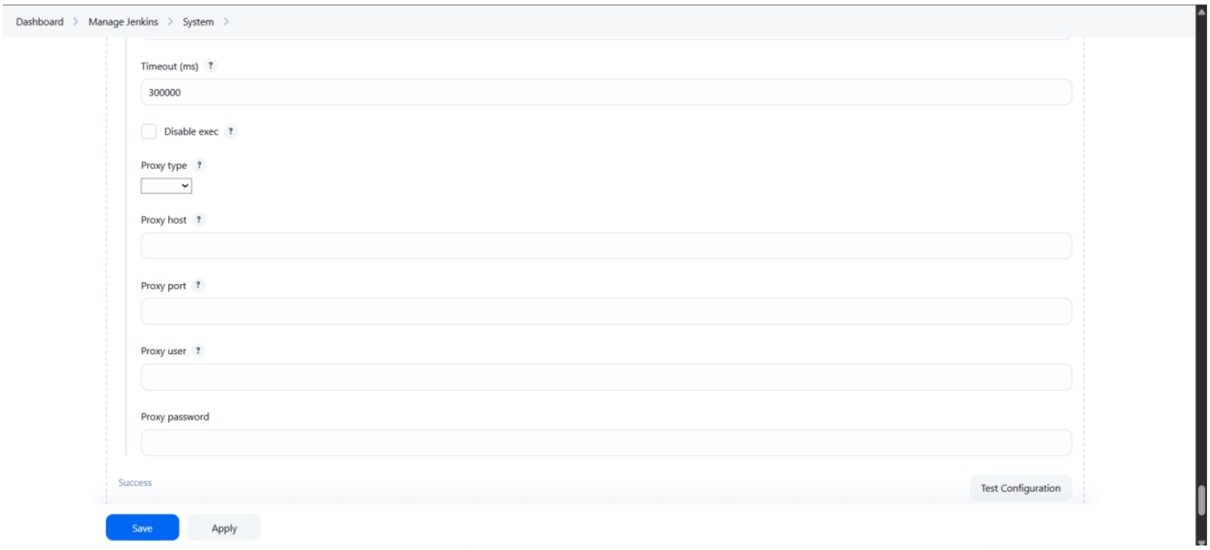


Cluster is created.

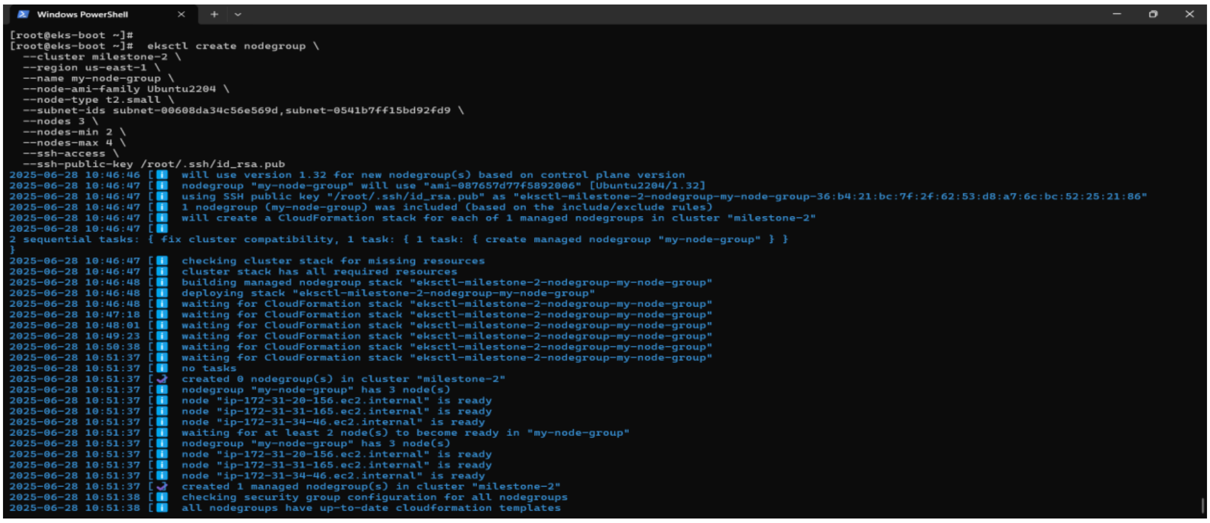


Deployment and service yaml file is added in /opt.

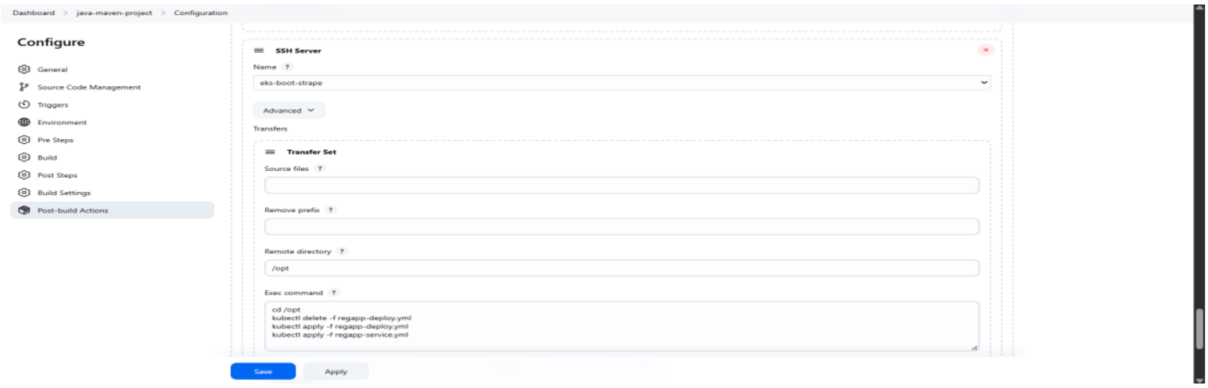




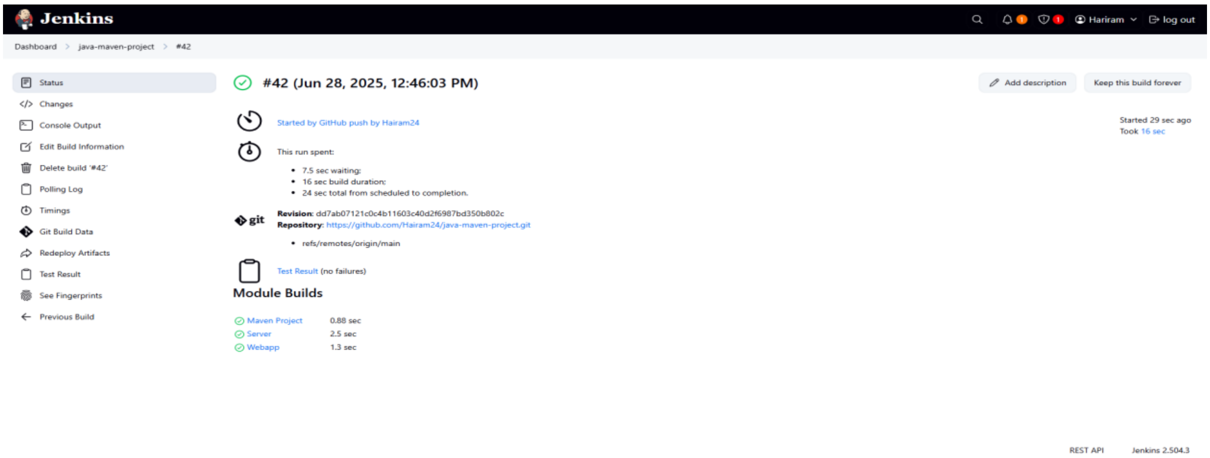
Eks-server ssh is added to Jenkins system.

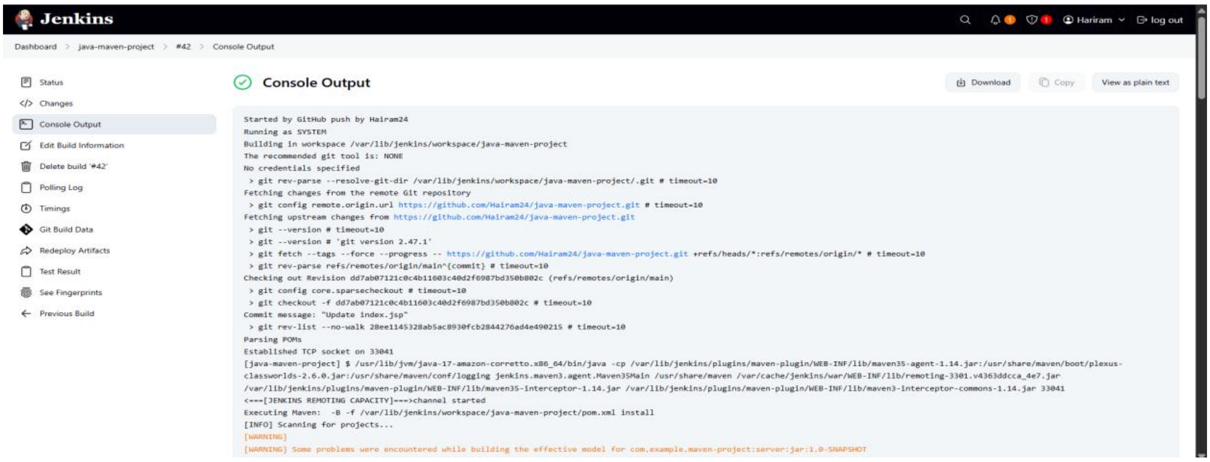


Node group is created.

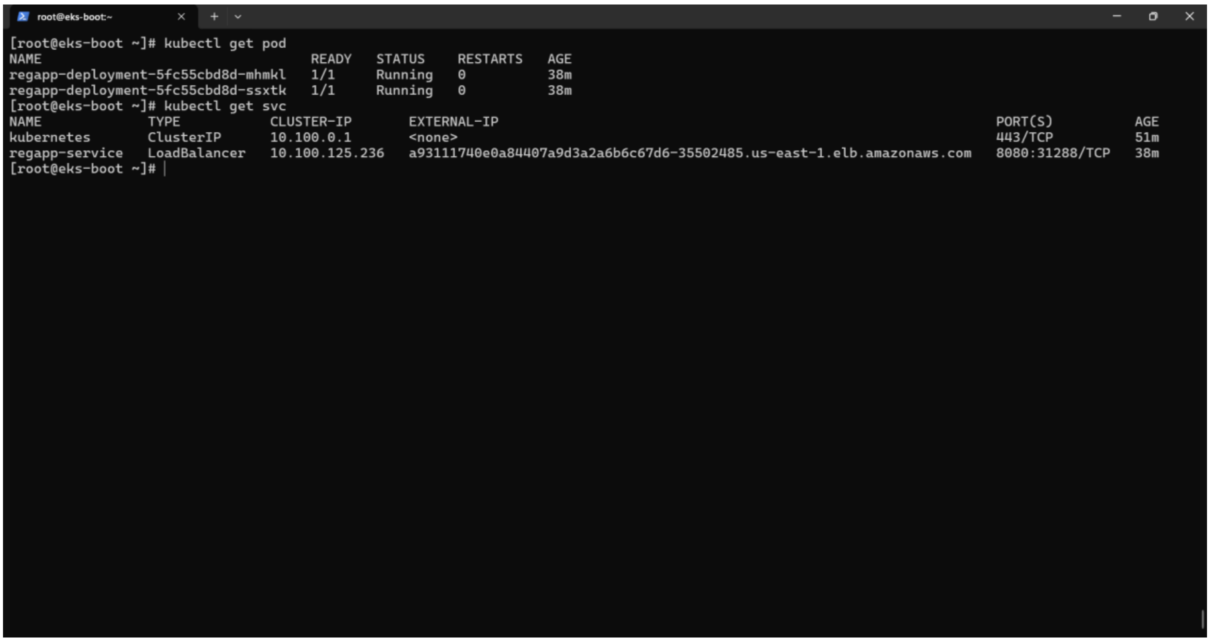


Post build eks-server is added with exec commands.

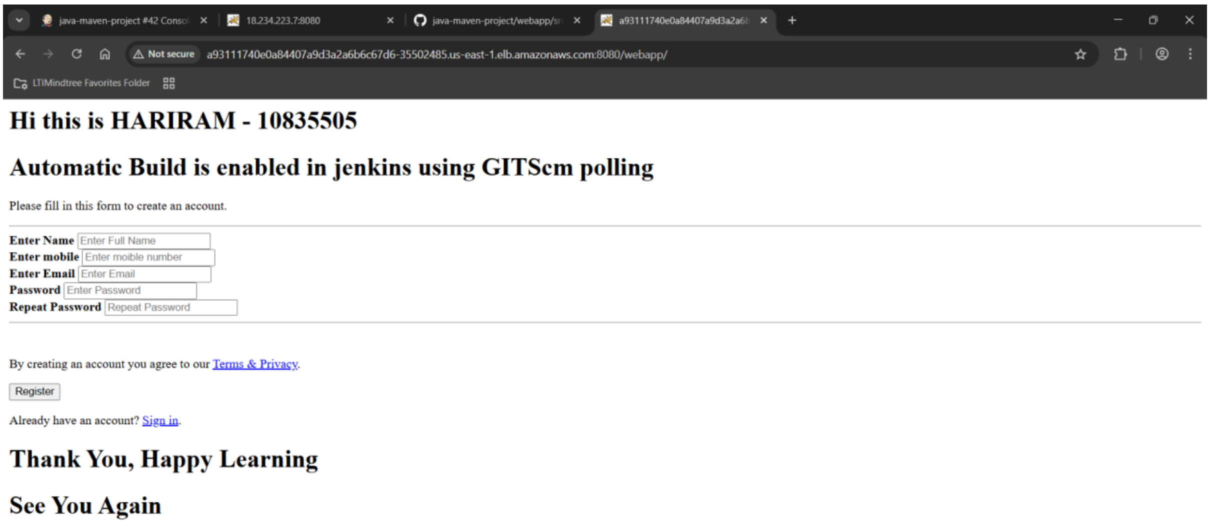




Build successfully.



Two pods are running after the build successful.



Final output is displayed in server ip:8080/webapp .